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UNIT 2: EV Traction Batteries, Battery Management Systems and Charging Systems

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Unit 2: EV Traction Batteries, Battery Management Systems and Charging Systems

The traction battery is the largest, heaviest and most expensive component of a battery electric vehicle (BEV). Its energy sets the range, its power sets the acceleration and regenerative capability, its voltage sets the architecture of the inverter and charger, and its ageing sets the service life of the vehicle. This unit develops the subject in the order an engineer meets it: selecting a storage technology, understanding how a cell works, sizing a pack, estimating its internal states, modelling it, managing it through a battery management system (BMS), and finally charging it.

Notation and sign convention. Throughout this unit the battery current I is taken as *positive during discharge* and negative during charge. State of charge is written as z (or SoC) and is dimensionless, $0 \le z \le 1$. Capacity Q is quoted in ampere-hours (Ah); wherever it appears with time in seconds it is converted to coulombs using 1 Ah = 3600 C. Specific quantities are per kilogram and density quantities are per litre.

2.1 Ragone Plot

2.1.1 Introduction to the Ragone Plot

Every energy storage device must answer two independent questions: *how much* energy can it hold, and *how fast* can that energy be delivered? A single number cannot answer both. The Ragone plot, named after D. V. Ragone, who used it in 1968 to compare battery systems for electrically powered vehicles [1], answers both at once by plotting the specific power of a device against its specific energy on logarithmic axes. Its theoretical basis was later formalised by Christen and Carlen [2].

2.1.2 Definition and Significance

A Ragone plot is a log–log chart with specific energy on one axis and specific power on the other. Each storage technology occupies a region, not a point, because a real device delivers less energy when it is discharged at higher power. The chart is significant because a single glance shows whether a technology can meet a vehicle's energy requirement and power requirement with an acceptable mass, and how long a discharge can be sustained.

2.1.3 Specific Energy and Specific Power

Specific energy. Specific energy is the energy delivered per unit mass:

$$e_s = \frac{E}{m} \qquad [\text{Wh/kg}], \tag{2.1}$$

where E is the delivered energy in Wh and m the mass in kg. It governs vehicle range for a given battery mass.

Specific power. Specific power is the power delivered per unit mass:

$$p_s = \frac{P}{m} \qquad [\text{W/kg}]. \tag{2.2}$$

It governs acceleration, hill climbing and regenerative capability.

Discharge time. Dividing (2.1) by (2.2) gives the time for which the power can be sustained:

$$t = \frac{e_s}{p_s} \quad [\text{h}] \qquad \Longleftrightarrow \qquad t\,[\text{s}] = 3600\,\frac{e_s}{p_s}. \tag{2.3}$$

On log–log axes $\log p_s = \log e_s - \log t$, so a line of constant discharge time is a straight line of unit slope. These are the dashed diagonals of Fig. 2.1.

2.1.4 Energy Density versus Power Density: Gravimetric and Volumetric Measures

Gravimetric measures are per unit mass (Wh/kg, W/kg); *volumetric* measures are per unit volume (Wh/L, W/L). The latter are called energy density and power density:

$$e_v = \frac{E}{\mathcal{V}}, \qquad p_v = \frac{P}{\mathcal{V}}, \tag{2.4}$$

where $\mathcal{V}$ is the volume in litres. A passenger-car floor pack is usually volume-limited, because the underfloor space is fixed, whereas a two-wheeler or an aircraft is more sensitive to mass. The terms are often loosely interchanged, so a careful engineer always checks the units. Values must also be compared at the same level (cell, module or pack), because housing, cooling and busbars typically reduce the cell-level specific energy by 25–40% at pack level.

2.1.5 Interpretation of the Axes

Figure 2.1 shows a Ragone plot with specific energy on the horizontal axis and specific power on the vertical axis. Both axes span several decades, which is the reason for the logarithmic scale: a supercapacitor and a fuel-cell system differ by more than two orders of magnitude in each coordinate. Moving right means more energy (more range); moving up means more power. Devices in the upper left deliver bursts lasting seconds, whereas devices in the lower

right sustain low power for hours. The reader should remember that the shaded regions are indicative envelopes drawn from typical published ranges [2–4]; they are not specifications.

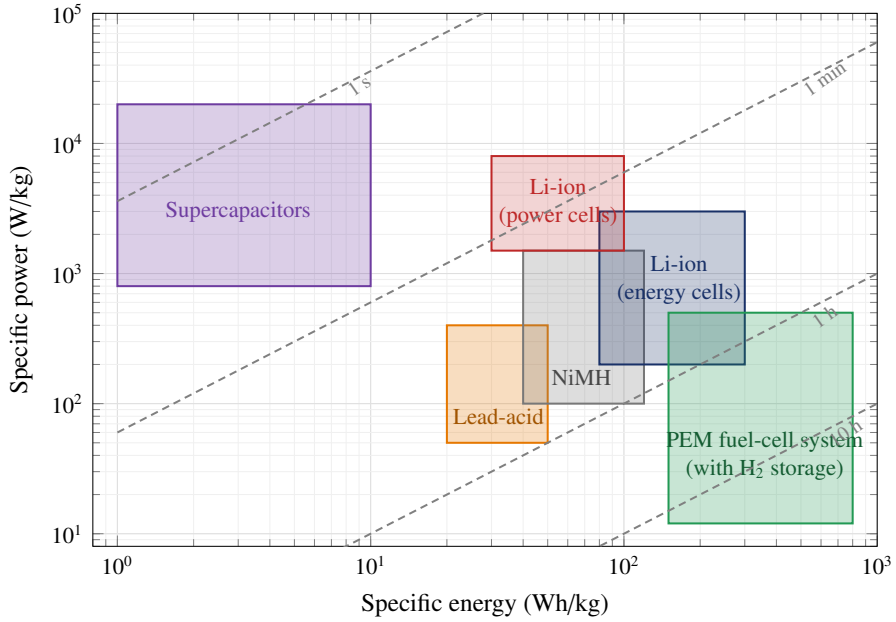


Figure 2.1: Indicative Ragone plot of energy storage technologies at cell or system level. Regions are representative ranges, not product specifications. Dashed diagonals are lines of constant discharge time.

2.1.6 Energy Storage Technologies on the Ragone Plot

Figure 2.1 places the technologies relevant to road vehicles:

- **Lead-acid** batteries lie at low specific energy and low specific power. They remain suitable for 12 V auxiliary service and for low-cost light vehicles.
- **Nickel-metal hydride (NiMH)** batteries roughly double the lead-acid energy and were the workhorse of early hybrid vehicles.
- **Lithium-ion** batteries span a wide region. Energy-optimised cells reach the right of the chart; power-optimised cells trade energy for power and move upward and left.
- **Supercapacitors** store charge electrostatically. They have very high specific power and cycle life but very little energy, so they discharge within seconds.
- **Fuel-cell systems** are energy converters with an external fuel tank. Their energy depends on the mass of stored hydrogen and is therefore high, but their power per kilogram is moderate, so they sit at the lower right. Fuel-cell vehicles normally need a battery buffer.

Figure 2.2 isolates the specific energy axis so that the technologies can be ranked directly. Note the factor of about ten between lead-acid and lithium-ion at the top of its range, and the further factor of about ten between lead-acid and a supercapacitor. This single comparison explains why long-range BEVs became practical only with lithium-ion cells.

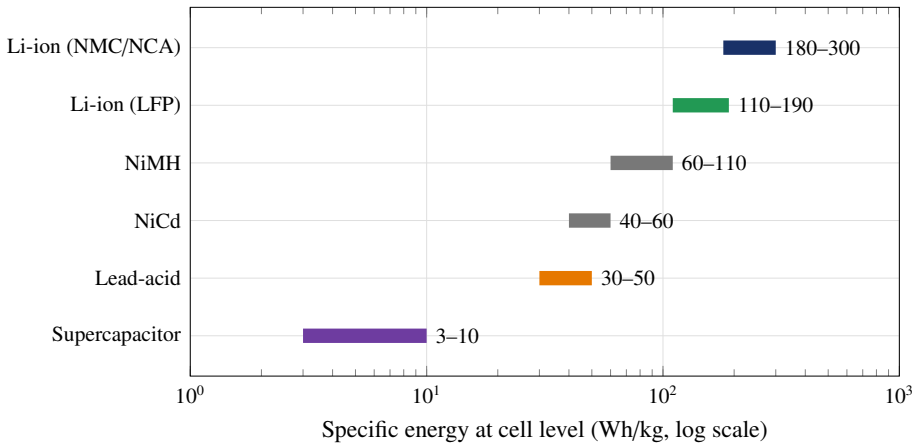


Figure 2.2: Indicative cell-level specific energy ranges (log scale). Representative values compiled from [3, 4]; individual products differ.

2.1.7 Battery and Supercapacitor Performance Characteristics

A battery stores energy in chemical bonds, so charge must cross an interface and diffuse through solids. These are slow, thermally activated steps, which limit power. A supercapacitor stores energy in the electric double layer, with no bulk chemical reaction, so its power is limited chiefly by its low series resistance. Its stored energy is $E = \frac{1}{2}CV^2$, which falls with the square of the voltage; consequently a supercapacitor bank needs a DC-DC converter to use its energy fully. It also has a high self-discharge rate compared with a lithium-ion battery.

2.1.8 Trade-off between Energy and Power Density

The curve of a real device bends downward at high power because internal losses grow with current. This can be derived from the simple V_{oc}–R model. Let a cell with open-circuit voltage V_{oc}, resistance R and capacity Q deliver a constant power P to a load. The terminal voltage $V_t = V_{oc} - IR$ and $P = V_t I$ give

$$RI^2 - V_{oc}I + P = 0 \quad \Rightarrow \quad I = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4RP}}{2R}, \tag{2.5}$$

the smaller root being the stable operating point. The discharge lasts $t = Q/I$, so the delivered energy is

$$E(P) = Pt = \frac{2RPQ}{V_{oc} - \sqrt{V_{oc}^2 - 4RP}}. \tag{2.6}$$

As $P \to 0$, $E \to QV_{oc}$, the full stored energy. A real solution exists only when $P \le P_{\max} = V_{oc}^2/(4R)$, which is the maximum power transfer condition; at that point half the energy is lost inside the cell. Equation (2.6) is precisely the knee of the Ragone curve.

The physical origin of the trade-off lies in electrode design, as illustrated in Fig. 2.3. A high-energy cell uses thick electrodes to pack in more active material, but ions must then travel a longer path, which raises resistance. A high-power cell uses many thin electrode layers with a large interfacial area and short ion paths, but more of its mass is inactive foil and separator.

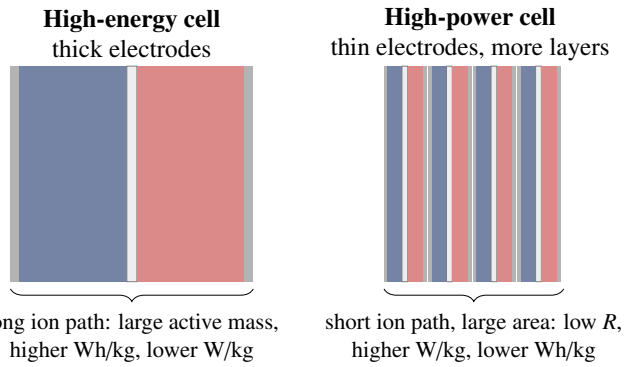


Figure 2.3: Illustration of high-energy and high-power storage devices: the electrode thickness sets the energy–power trade-off.

2.1.9 Applications of Ragone Plots in EV Battery Selection

The selection procedure uses both vehicle requirements together. For a required energy E_{req} and peak power P_{req} the minimum storage mass is

$$m_{\min} = \max\left(\frac{E_{\text{req}}}{e_s},\ \frac{P_{\text{req}}}{p_s}\right). \tag{2.7}$$

The first term dominates for long-range BEVs, so energy cells are chosen; the second dominates for hybrids, whose small battery must supply a large power, so power cells are chosen. The ratio $P_{\text{req}}/E_{\text{req}}$ (in h^{-1}) is the key figure: a BEV is typically around 1–3 h^{-1}, whereas a power-assist hybrid may need well above 10 h^{-1}. Any device whose region of Fig. 2.1 crosses that diagonal is a candidate.

Example 1: Ragone interpretation and energy–power comparison

Problem. A BEV needs 60 kWh of energy and 150 kW of peak power. Compare the minimum storage mass of three candidates with the assumed pack-level values below.

Given. Li-ion: 180 Wh/kg, 400 W/kg. Lead-acid: 35 Wh/kg, 150 W/kg. Supercapacitor: 6 Wh/kg, 5000 W/kg. (Illustrative values, not product data.)

Required. Mass needed for energy, mass needed for power, governing mass, and discharge time at the rated p_s.

Equations. (2.3) and (2.7).

Solution.

$$\begin{aligned}
\text{Li-ion:} \quad & m_E = \frac{60\,000}{180} = 333.3 \text{ kg}, \quad m_P = \frac{150\,000}{400} = 375 \text{ kg}, \quad t = \frac{180}{400} = 0.45 \text{ h} \\
\text{Lead-acid:} \quad & m_E = \frac{60\,000}{35} = 1714 \text{ kg}, \quad m_P = \frac{150\,000}{150} = 1000 \text{ kg}, \quad t = \frac{35}{150} = 0.233 \text{ h} \\
\text{Supercap:} \quad & m_E = \frac{60\,000}{6} = 10\,000 \text{ kg}, \quad m_P = \frac{150\,000}{5000} = 30 \text{ kg}, \quad t = \frac{6}{5000} \text{ h} = 4.3 \text{ s}
\end{aligned}$$

Answer. Governing masses are 375 kg (Li-ion, power-limited), 1714 kg (lead-acid, energy-limited) and 10 000 kg (supercapacitor, energy-limited).

Interpretation. Only lithium-ion is feasible. The small margin between 333 kg and 375 kg shows why a slightly higher power capability of the cell, rather than more energy, would reduce the mass here. The supercapacitor needs only 30 kg for power, which is why it is attractive as a short-term power buffer alongside an energy battery.

Example 2: Battery specific energy and specific power

Problem. A pack stores 75 kWh, has a mass of 480 kg and a volume of 300 L, and delivers 200 kW peak. Find its gravimetric and volumetric figures.

Given. $E = 75$ kWh, $m = 480$ kg, $\mathcal{V} = 300$ L, $P = 200$ kW (hypothetical pack).

Required. e_s, p_s, e_v, p_v.

Equations. (2.1), (2.2) and their volumetric forms.

Solution.

$$\begin{aligned}
e_s &= \frac{75\,000}{480} = 156.3 \text{ Wh/kg}, & p_s &= \frac{200\,000}{480} = 416.7 \text{ W/kg}, \\
e_v &= \frac{75\,000}{300} = 250 \text{ Wh/L}, & p_v &= \frac{200\,000}{300} = 666.7 \text{ W/L}.
\end{aligned}$$

Answer. 156.3 Wh/kg, 416.7 W/kg, 250 Wh/L, 666.7 W/L.

Interpretation. The peak power-to-energy ratio is $200/75 = 2.67\ \text{h}^{-1}$, so the peak discharge lasts about 22 minutes and corresponds to about 2.7C.

2.1.10 Limitations of Ragone Plot-Based Comparisons

The Ragone plot is a screening tool. Its limitations should be stated explicitly:

- It says nothing about cost, cycle life, calendar life, safety or charging capability.

- Published regions mix cell, module and system-level data; comparing them directly is misleading.
- The curve assumes constant-power discharge at one temperature, whereas a vehicle sees a dynamic profile over a wide temperature range.
- The curves shift with state of charge and ageing, as Section 2.5 shows.
- It ignores volume unless a separate volumetric chart is drawn.
- It ignores charge acceptance. Regenerative braking and fast charging are limited by different mechanisms from discharge.

2.2 Battery Types and Battery Pack

2.2.1 Introduction to Electrochemical Batteries

An electrochemical battery converts chemical energy directly into electrical energy through spatially separated oxidation and reduction reactions. Strictly, a *cell* is the basic electrochemical unit and a *battery* is a connected group of cells, although common usage calls either a battery.

2.2.2 Primary and Secondary Batteries

Primary batteries are designed for a single discharge because their reactions are not practically reversible. *Secondary* batteries can be recharged many times because the reaction can be reversed by an external current. Traction batteries are always secondary. Figure 2.4 classifies the technologies considered here.

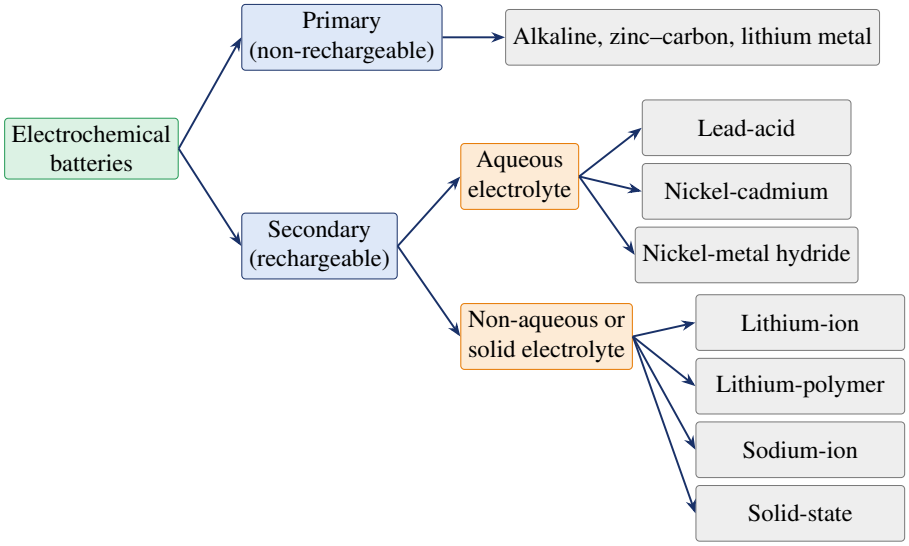


Figure 2.4: Classification of battery technologies by rechargeability and electrolyte family.

The first branch in Fig. 2.4 separates single-use from rechargeable cells. The second branch reflects an engineering boundary: water decomposes at about 1.23 V (thermodynamically),

which limits aqueous cells to roughly 2 V, whereas organic and solid electrolytes allow 3–4 V cells. Because energy scales with voltage, this is the main reason why non-aqueous systems lead in specific energy.

2.2.3 Classification of Rechargeable Batteries

The following subsections review each rechargeable family in terms of its construction, reactions, nominal voltage, strengths and limitations [3].

2.2.3.1 Lead-Acid Batteries

A lead-acid cell has a lead dioxide positive plate, a sponge lead negative plate and a sulphuric acid electrolyte. On discharge both plates convert to lead sulphate:

$$\mathrm{Pb} + \mathrm{PbO_2} + 2\mathrm{H_2SO_4} \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftharpoons}} 2\mathrm{PbSO_4} + 2\mathrm{H_2O}, \qquad V_{\text{nom}} \approx 2.0\ \text{V}. \tag{2.8}$$

The technology is low in cost and highly recyclable, and it has high surge capability for its cost, which suits engine starting. However, it has low specific energy, poor deep-cycle life, and suffers sulphation if left discharged. It survives in EVs mainly as the 12 V auxiliary battery and in low-speed vehicles.

2.2.3.2 Nickel-Cadmium Batteries

A NiCd cell uses nickel oxyhydroxide and cadmium in an alkaline potassium hydroxide electrolyte, with a nominal voltage of 1.2 V. It is robust, tolerates abuse and works at low temperatures, but cadmium is toxic and the cell shows the memory effect. Its use in vehicles has been effectively abandoned, largely for environmental reasons.

2.2.3.3 Nickel-Metal Hydride Batteries

NiMH replaces cadmium with a hydrogen-absorbing metal alloy. It keeps the 1.2 V nominal voltage but offers higher capacity and better environmental acceptability. It was used widely in hybrid vehicles due to its long shallow-cycle life and tolerance of overcharge. Its drawbacks are high self-discharge, a low coulombic efficiency at high SoC, and heat generation at the end of charge.

2.2.3.4 Lithium-Ion Batteries

Lithium-ion cells shuttle lithium ions between two host materials (Section 2.4). With a nominal voltage of about 3.2–3.7 V for graphite-based cells, they lead in specific energy and offer high coulombic efficiency and long cycle life. They need a BMS for safe operation because they tolerate no overcharge and can enter thermal runaway.

2.2.3.5 Lithium-Polymer Batteries

The term “lithium-polymer” is used loosely. Commercial cells so described are normally lithium-ion cells with a gelled polymer electrolyte or simply a pouch format. The chemistry

is otherwise the same; the benefit is packaging flexibility and low mass. Genuine solid polymer electrolyte cells, which usually run at elevated temperature, are a different and much rarer technology.

2.2.3.6 Sodium-Ion Batteries

Sodium-ion cells follow the same rocking-chair principle, replacing lithium with sodium. Sodium is abundant and the cathode can avoid nickel and cobalt, and aluminium can replace copper as the negative collector. Hard carbon replaces graphite since sodium does not intercalate into graphite adequately. Specific energy is currently below that of LFP, although the cells show good low-temperature performance, so they are attractive for low-cost, shorter-range vehicles and stationary storage [5].

2.2.3.7 Solid-State Batteries

Solid-state batteries replace the flammable liquid electrolyte with a solid ionic conductor (sulphide, oxide or polymer), which may enable a lithium metal negative electrode and therefore higher specific energy. Open challenges include interfacial resistance, dendrite growth, stack pressure and manufacturing scale [6]. They should be treated as an emerging technology; claimed values for pre-commercial cells are not comparable with mass-production data.

2.2.4 Comparison of Battery Technologies

Table 2.1 collects the main characteristics. The table shows why lead-acid and nickel systems have been displaced: the specific energy and cycle life of lithium-ion more than offset its higher cost per kWh, particularly as that cost has fallen with scale.

Table 2.1: Indicative comparison of rechargeable battery technologies at cell level [3, 4]. Values are representative ranges; manufacturer datasheets govern any actual design.

Technology	Nominal voltage (V)	Specific energy (Wh/kg)	Cycle life (typ.)	Self-discharge	Main role
Lead-acid	2.0	30–50	200–800	Low to moderate	12 V auxiliary, low-speed EV
NiCd	1.2	40–60	1000+	Moderate	Legacy
NiMH	1.2	60–110	500–1000+	High	HEV
Li-ion (NMC/NCA)	3.6–3.7	180–300	1000–2500	Low	Long-range BEV
Li-ion (LFP)	3.2	110–190	2000–6000	Low	BEV, bus, storage
Sodium-ion	2.8–3.2	100–170	1000–4000	Low	Emerging low-cost EV
Solid-state	Design dependent	Target >300	Under development	–	Emerging

2.2.5 Requirements of EV Traction Batteries

A traction battery must satisfy a combination of requirements:

- high specific energy and energy density, for range and packaging;
- high specific power, for acceleration and regenerative braking;
- long cycle and calendar life, ideally matching the vehicle life;
- a wide operating temperature range;
- intrinsic and system-level safety under abuse, crash and fault conditions;
- fast charge acceptance without lithium plating;
- low cost per kWh, and recyclable, responsibly sourced materials.

No chemistry maximises all of these, and the art of battery engineering lies in the compromise.

2.2.6 Battery Cell, Module and Pack Architecture

2.2.6.1 Cell Formats: Cylindrical, Prismatic and Pouch

Figure 2.5 shows the three commercial formats. The internal electrochemistry can be identical; what differs is the mechanical package and hence the thermal and manufacturing behaviour.

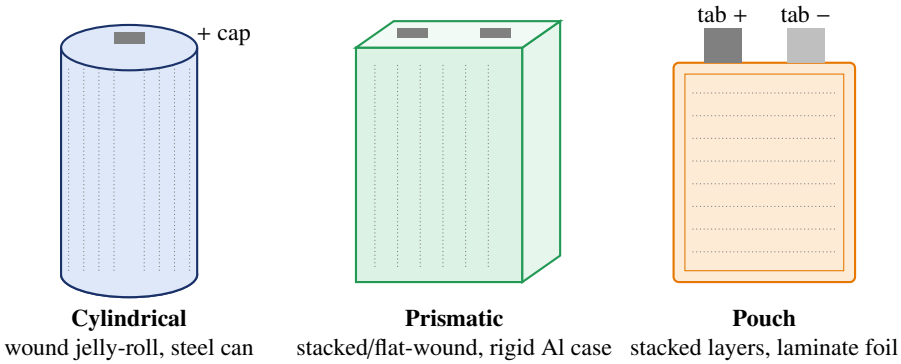


Figure 2.5: Cylindrical, prismatic and pouch cell structures.

Cylindrical cells. Long electrode strips are wound into a jelly-roll and sealed in a steel can, as on the left of Fig. 2.5. The can resists internal pressure and the format is highly automated. Formats are named by size: 18650 (18 mm diameter, 65 mm long), 21700 and 4680. The small capacity per cell means that a pack needs thousands of cells, with many joints.

Prismatic cells. A flat-wound or stacked electrode assembly sits in a rigid aluminium case. Prismatic cells pack with little wasted space and offer large capacities, and have become common in LFP packs. The rigid case carries part of the structural load. Their disadvantages are swelling forces and a lower heat transfer area than the pouch per unit energy.

Pouch cells. Stacked electrode sheets are sealed in a polymer-aluminium laminate. The pouch gives the highest cell-level specific energy and a large cooling surface, but it needs external mechanical support and a compression arrangement, and has no inherent vent.

2.2.6.2 Cell Interconnection and Pack Hierarchy

Figure 2.6 shows how cells become a module and modules become a pack. The module is a mechanically and electrically manageable subassembly. The pack adds the enclosure, BMS, contactors, fuses and thermal system. In *cell-to-pack* designs, which are growing in popularity, the module level is eliminated to raise volumetric utilisation.

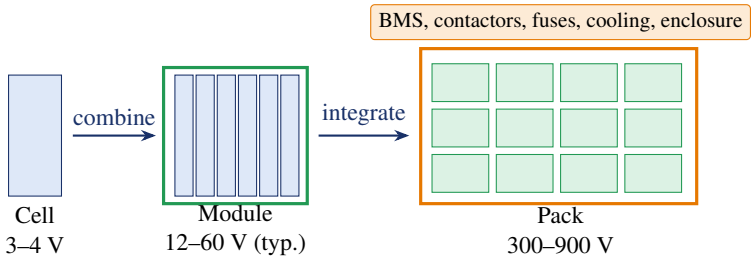


Figure 2.6: Battery cell-to-module-to-pack architecture. Voltage ranges are typical, not limits.

2.2.6.3 Series Configuration

Cells in series add their voltages while carrying the same current, as Fig. 2.7 shows:

$$V_{\text{string}} = N_s V_c, \qquad Q_{\text{string}} = Q_c, \qquad R_{\text{string}} = N_s R_c. \tag{2.9}$$

Series connection raises voltage, which lowers current for a given power and reduces conductor mass and I^2R loss. Its weakness is that the same current flows through all cells, so the weakest cell limits the whole string; a single open cell interrupts the string.

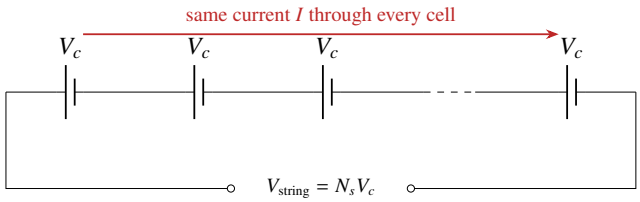


Figure 2.7: Series-connected battery cells.

2.2.6.4 Parallel Configuration

Cells in parallel share the same voltage and add their capacities, as Fig. 2.8 shows:

$$V_{\text{group}} = V_c, \qquad Q_{\text{group}} = N_p Q_c, \qquad R_{\text{group}} = \frac{R_c}{N_p}. \tag{2.10}$$

Parallel cells self-equalise their voltage, but differences in resistance cause unequal current sharing, so interconnect design matters. One weak cell in a parallel group will also take current

from its neighbours if it develops an internal short, hence the common practice of fusing each cell or using a current-limiting wire bond.

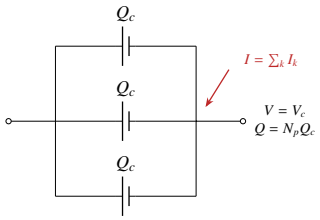


Figure 2.8: Parallel-connected battery cells.

2.2.6.5 Series-Parallel Battery Packs

Practical packs combine both. In the *parallel-first* (or N_ppN_ss) arrangement of Fig. 2.9, cells are paralleled into groups and the groups are connected in series. The pack parameters follow from (2.9) and (2.10):

$$V_{\text{pack}} = N_s V_c, \qquad Q_{\text{pack}} = N_p Q_c, \tag{2.11}$$

$$N_{\text{cells}} = N_s N_p, \qquad E_{\text{pack}} = N_s N_p V_c Q_c, \qquad R_{\text{pack}} = \frac{N_s}{N_p} R_c. \tag{2.12}$$

Parallel-first needs only one voltage sense wire per group. It is the dominant approach, because the BMS monitors each group as one equivalent cell. The string-first alternative isolates a failed string but requires monitoring of every cell.

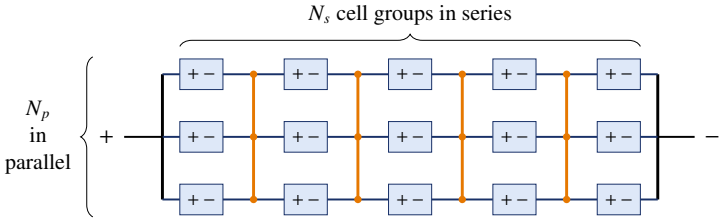


Figure 2.9: Series-parallel battery pack, parallel-first arrangement. Vertical links (orange) tie cells of the same series position.

Example 3: Battery pack series and parallel configuration

Problem. A pack is built from hypothetical 21700-size cells in a 96s46p arrangement. Find its main parameters.

Given. Cell: $V_c = 3.6$ V, $Q_c = 5$ Ah, $R_c = 20$ mΩ, $V_{\text{max}} = 4.2$ V, $V_{\text{min}} = 2.5$ V.

Required. Nominal voltage, voltage window, capacity, energy, number of cells and resistance.

Equations. (2.11) and (2.12).

Solution. $V_{\text{pack}} = 96 \times 3.6 = 345.6$ V; window $96 \times 2.5 = 240$ V to $96 \times 4.2 = 403.2$ V. $Q_{\text{pack}} = 46 \times 5 = 230$ Ah. $E = 345.6 \times 230 = 79.49$ kWh. $N_{\text{cells}} = 96 \times 46 = 4416$. $R_{\text{pack}} = (96/46) \times 0.020 = 41.7$ mΩ.

Answer. 345.6 V (240–403.2 V), 230 Ah, 79.5 kWh, 4416 cells, 41.7 mΩ.

Interpretation. The voltage swings by about ±20% around nominal. The inverter and charger must operate across this entire window, not merely at 345.6 V.

2.2.7 Typical EV Battery Pack Architecture

Figure 2.10 gives the layout of a typical pack. The modules are connected in series by busbars; a manual service disconnect (MSD) or fuse sits at the electrical midpoint so that opening it halves the exposed voltage during service. Current then passes through a sensor and the main contactors to the high-voltage terminals. The BMS master coordinates all of these through signal wiring.

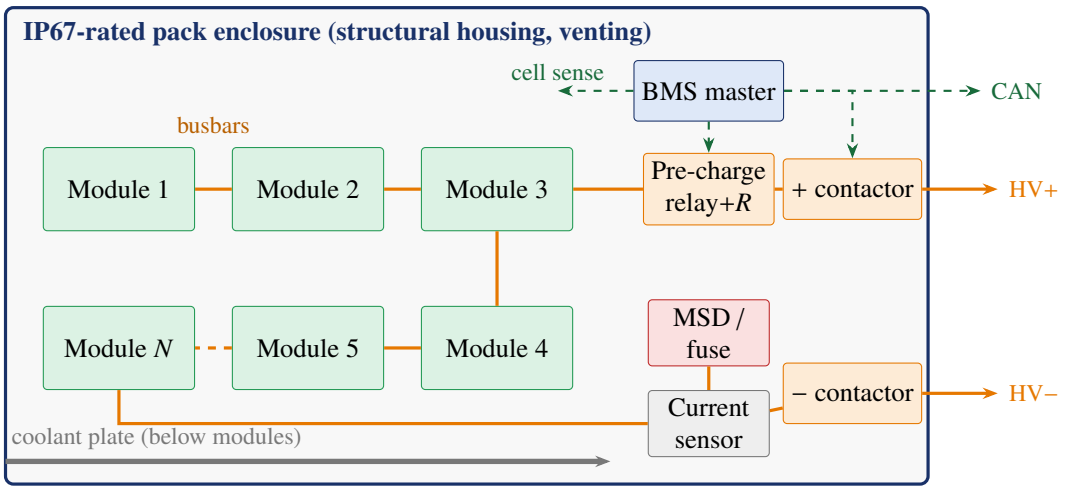


Figure 2.10: Typical EV battery pack architecture. Orange: high-voltage path; green dashed: signals.

2.2.7.1 Battery Enclosure and Mechanical Protection

The enclosure protects the cells from crash loads, road debris, water and dust, and contains fire or venting events. It is commonly an aluminium or steel tray with a sealed lid, and is frequently rated around IP67. Modern designs make it part of the vehicle body structure.

2.2.7.2 Electrical Interconnections and Busbars

Busbars are copper or aluminium conductors joining cells and modules. Their cross-section is designed for peak current and temperature rise, and their joints (laser welded or bolted) must retain low resistance over the vehicle life, because a poor joint is a concentrated heat source.

2.2.7.3 Thermal Management Provisions

Cell ageing and safety depend strongly on temperature, so packs normally have cold plates, a coolant loop and sometimes a heater. Air cooling suits only low-power small packs, while liquid cooling is the norm for passenger-car packs. The aim is both a target mean temperature, typically 20–40°C, and a low temperature spread across cells, typically a few degrees.

2.2.7.4 High-Voltage Battery Safety Components

These include main positive and negative contactors, a pre-charge circuit, a fuse or pyrotechnic disconnect, the MSD, the current sensor, an insulation monitor, a high-voltage interlock loop (HVIL), and vent paths. Their operation is explained in Section 2.13.

2.3 Basic Battery Operation

2.3.1 Electrochemical Energy Conversion

A cell produces electricity because the two electrode materials have different electrochemical potentials. When an external circuit is closed, a spontaneous reaction proceeds, but it is divided into two half-reactions at separate electrodes. Electrons cannot travel through the electrolyte, so they are forced around the external circuit to do work, whereas the ions complete the circuit inside the cell.

2.3.2 Construction: Electrodes, Electrolyte, Separator and Collectors

Figure 2.11 shows the universal structure of a cell.

- The **anode** is the electrode where oxidation occurs and the **cathode** the one where reduction occurs. These labels strictly interchange during charge, so in rechargeable cells it is safest to speak of the *negative* and *positive* electrodes. By industry convention, “anode” means the negative electrode and “cathode” the positive, named for discharge.
- The **electrolyte** is an ionic conductor and electronic insulator.
- The **separator** is a porous, electronically insulating membrane that prevents electrode contact while letting ions pass.
- The **current collectors** are metal foils that carry electrons between the electrode material and the terminals.

2.3.3 Electrochemical Potential

Each electrode has an electrode potential that reflects the free energy of its reaction. The cell’s open-circuit voltage is the difference between them, and it is linked to the reaction Gibbs free energy ΔG by

$$V_{oc} = -\frac{\Delta G}{nF}, \tag{2.13}$$

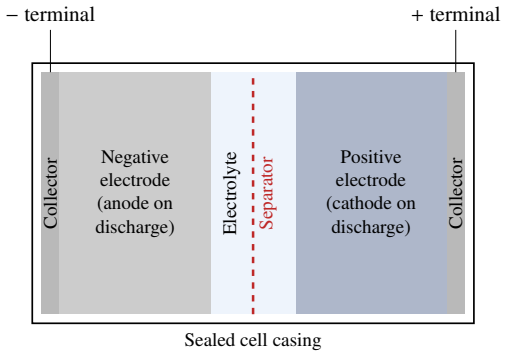


Figure 2.11: Battery cell construction diagram.

in which n is the number of electrons transferred per reaction and $F = 96\,485$ C/mol is Faraday's constant. The dependence of voltage on concentration is captured by the Nernst equation

$$V_{oc} = V^0 - \frac{\mathcal{R}T}{nF} \ln \frac{a_{\text{products}}}{a_{\text{reactants}}}, \tag{2.14}$$

where V^0 is the standard potential, $\mathcal{R} = 8.314$ J/(mol K) the gas constant, T the absolute temperature, and a the activities. Thus voltage varies with state of charge, which is the physical basis of voltage-based SoC estimation.

2.3.4 Oxidation and Reduction; Electron and Ion Movement

On discharge the negative electrode is oxidised, releasing electrons and positive ions. Those electrons pass through the external load to the positive electrode where reduction consumes them, as shown in the left half of Fig. 2.12. The positive ions move through the electrolyte in the same direction, from negative to positive. Note carefully that conventional current flows from the positive terminal through the load to the negative terminal outside the cell, opposite to the electron direction, and from negative to positive inside the cell, which agrees with the direction of the cations.

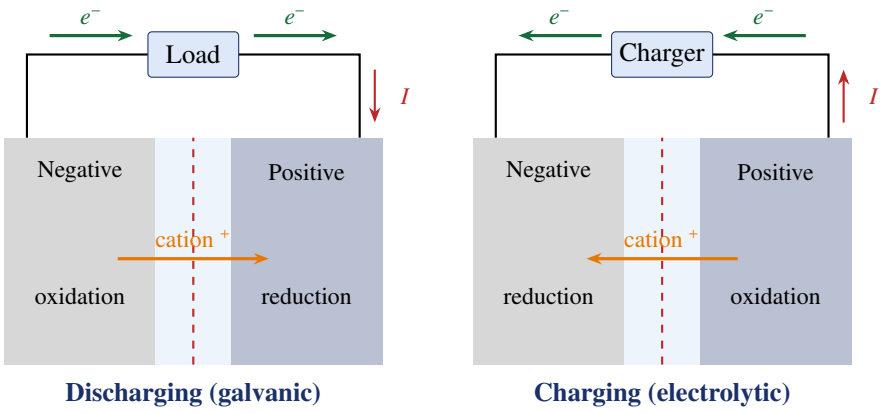


Figure 2.12: Charging and discharging operation of a cell. Electrons travel in the external circuit, cations in the electrolyte.

2.3.5 Charging and Discharging Processes

On charge, shown on the right of Fig. 2.12, an external source with a voltage above V_{oc} forces the reactions backward: the positive electrode is oxidised, the negative is reduced, and cations travel from positive to negative. The discharge reaction is spontaneous (galvanic), whereas the charge reaction requires energy input (electrolytic). A closer look at the transport path is given in Fig. 2.13, for a lithium-ion cell on discharge. Three serial processes appear: solid-phase diffusion out of the negative particles, migration and diffusion through the electrolyte, and charge transfer at the positive particles. The slowest controls the rate capability.

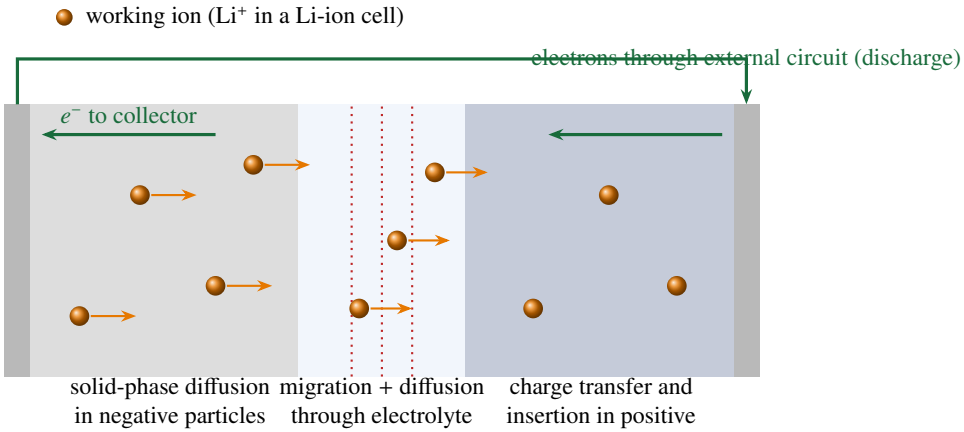


Figure 2.13: Ion and electron movement in a cell during discharge: ions and electrons both travel from the negative to the positive electrode, by different paths.

2.3.6 Open-Circuit Voltage, Terminal Voltage and Internal Resistance

Open-circuit voltage V_{oc} (OCV) is the voltage at the terminals with no current, after the cell has rested until it reaches equilibrium. *Terminal voltage* V_t is the voltage measured under load. For the sign convention used here,

$$V_t = V_{oc}(z) - IR_{\text{int}}. \tag{2.15}$$

Thus the terminal voltage is below OCV on discharge ($I > 0$) and above OCV on charge ($I < 0$). *Internal resistance* R_{int} lumps the electronic resistance of the foils and tabs, the ionic resistance of the electrolyte, and the charge-transfer resistance. It rises at low temperature and low SoC and with ageing.

2.3.7 Polarisation Effects

The total voltage departure from equilibrium is called polarisation or overpotential. It has three components:

- **Ohmic polarisation** appears instantly with current and disappears instantly with it.

- **Activation polarisation** arises from the kinetics of charge transfer and develops in milliseconds to seconds. It is described by the Butler–Volmer equation.
- **Concentration polarisation** arises from ion concentration gradients and develops over tens of seconds to minutes.

This separation of time scales is the justification for using resistor-capacitor (RC) branches in the equivalent circuit models of Section 2.11.

2.3.8 Capacity and Energy

Capacity is the charge that a cell can deliver between its fully charged and cut-off states:

$$Q = \int_0^{t_d} I(t)\,dt \qquad [\text{A s or, dividing by 3600, Ah}]. \tag{2.16}$$

Energy delivered is the time integral of power,

$$E = \int_0^{t_d} V_t(t) I(t)\,dt \approx \bar{V}\,Q \qquad [\text{Wh when } t \text{ in h}], \tag{2.17}$$

where $\bar{V}$ is the mean discharge voltage. Rated capacity depends on the discharge rate and temperature at which it is measured, so a capacity statement is meaningless without those test conditions. The discharge rate is commonly expressed as a *C-rate*:

$$C_{\text{rate}} = \frac{|I|}{Q_n} \qquad [\text{h}^{-1}], \tag{2.18}$$

where Q_n is the nominal capacity in Ah. A 50 Ah cell discharged at 100 A runs at 2C and an ideal discharge would last $1/2$ h.

2.3.9 Coulombic Efficiency and Energy Efficiency

Charge can be lost to side reactions. The *coulombic efficiency* over a cycle is

$$\eta_C = \frac{Q_{\text{dis}}}{Q_{\text{ch}}} = \frac{\int_{\text{dis}} |I|\,dt}{\int_{\text{ch}} |I|\,dt}. \tag{2.19}$$

For lithium-ion this exceeds 99% and is usually much closer to unity. The *energy efficiency* includes voltage losses as well:

$$\eta_E = \frac{E_{\text{dis}}}{E_{\text{ch}}} = \frac{\int_{\text{dis}} V_t |I|\,dt}{\int_{\text{ch}} V_t |I|\,dt}. \tag{2.20}$$

These definitions are developed in Section 2.5.

2.3.10 Battery Voltage Characteristics

Figure 2.14 shows a representative voltage against SoC. The OCV rises monotonically with SoC. Loaded curves are offset from it by IR: below on discharge, above on charge. The gap between charge and discharge curves is the voltage penalty of a cycle, which becomes heat.

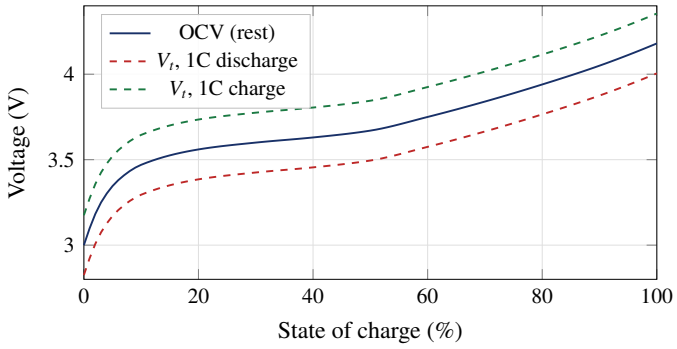


Figure 2.14: Typical battery voltage versus SoC (illustrative curves from a hypothetical 50 Ah NMC/graphite cell).

2.3.11 Charging and Discharging Voltage Profiles

Figure 2.15 plots terminal voltage against capacity throughput. On discharge the voltage falls gently through the plateau and then drops sharply near the end as the electrode runs out of available sites; the V_{min} cut-off protects the cell from this. On charge the voltage climbs to V_{max}, at which point control must transfer to constant-voltage mode (Section 3.3). The horizontal extent of each curve is the usable capacity at that rate.

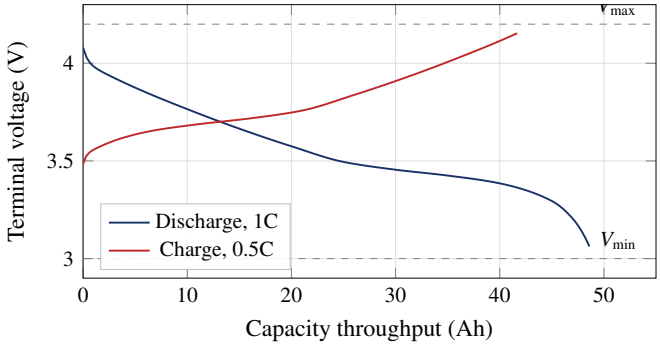


Figure 2.15: Charging and discharging characteristic curves (illustrative simulation of a 50 Ah cell).

Example 4: Battery terminal voltage and internal resistance losses

Problem. A cell with OCV 3.7 V and internal resistance 2 mΩ discharges at 150 A and later charges at 100 A at the same SoC. Find the terminal voltage and heat in each case.

Given. V_{oc} = 3.7 V, R = 2 mΩ; I = +150 A, then I = −100 A.

Required. V_t, internal loss, output power, and voltage efficiency on discharge.

Equations. (2.15), and $P_{\text{loss}} = I^2R$.

Solution. Discharge: $V_t = 3.7 - 150(0.002) = 3.40$ V; $P_{\text{loss}} = 150^2(0.002) = 45$ W; chemical power $V_{oc}I = 555$ W; output $V_tI = 510$ W; $\eta_V = 3.40/3.70 = 91.9\%$.

Charge: $V_t = 3.7 - (-100)(0.002) = 3.90$ V; $P_{\text{loss}} = 100^2(0.002) = 20$ W.

Answer. 3.40 V with 45 W loss on discharge; 3.90 V with 20 W loss on charge.

Interpretation. Loss grows with the square of current: 50% more current gives 2.25 times the heat. A sign error in (2.15) would give the wrong side of the OCV, so the convention must be fixed before any calculation.

2.4 Battery Chemistry: Lithium-Ion Batteries

2.4.1 Introduction to Lithium-Ion Technology

Lithium has the lowest electrochemical potential of the metals and low atomic mass, which makes it attractive for high voltage and high specific energy. Using lithium metal directly proved unsafe due to dendrites, so the lithium-ion cell instead keeps lithium in ionic form within host materials [7, 8].

2.4.2 Working Principle, Intercalation and Deintercalation

A lithium-ion cell is a "rocking-chair" device: lithium ions shuttle between two hosts and no bulk metal plating occurs in normal operation. *Intercalation* is the reversible insertion of ions into a host lattice; *deintercalation* is their extraction. Figure 2.16 shows the concept: the graphite layers keep their structure while lithium enters and leaves the gaps between them. Because the host is not consumed or rebuilt, as lead plates are, cycle life can be long.

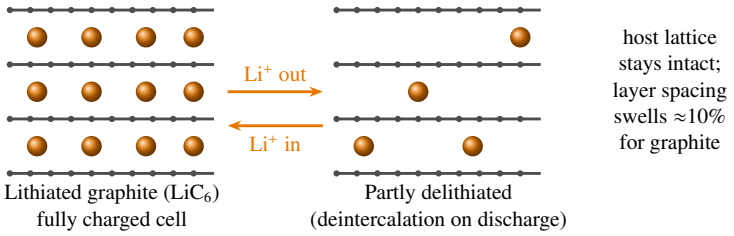


Figure 2.16: Lithium-ion intercalation process in a layered graphite host (schematic, not to scale).

The construction of a cell is shown in Fig. 2.17, and the mechanism in Fig. 2.18. On discharge, Li^+ leaves the graphite, crosses the separator, and enters the metal oxide, while electrons travel through the load. On charge the charger reverses both flows.

2.4.3 Cell Components

Anode materials. Graphite is dominant, holding one lithium atom per six carbon atoms (LiC_6) for a theoretical 372 mAh/g, at a low potential near that of lithium metal. Small ad-

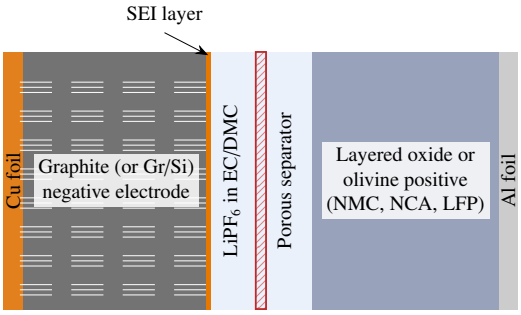


Figure 2.17: Lithium-ion cell construction (schematic, thicknesses exaggerated).

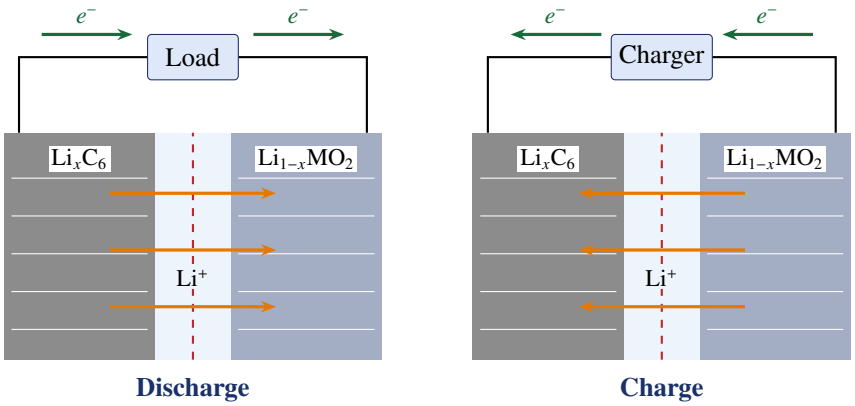


Figure 2.18: Lithium-ion charging and discharging mechanism.

ditions of silicon raise capacity since silicon has a far higher theoretical capacity, but it swells greatly on lithiation. Lithium titanate ($Li_4Ti_5O_{12}$, LTO) gives a safe, stable, long-life alternative at a lower cell voltage.

Cathode materials. These include the layered oxides (LCO, NMC, NCA), the olivine phosphates (LFP, LMFP) and the spinels (LMO). The cathode generally limits the capacity and sets much of the cost and safety behaviour.

Electrolyte composition and function. A typical liquid electrolyte is a lithium salt, $LiPF_6$, dissolved in a blend of organic carbonates such as ethylene carbonate (EC) and dimethyl carbonate (DMC), plus additives [9]. It must conduct ions, remain stable across the cell voltage, and form a protective film. It is flammable, which is the central safety concern.

Separator characteristics. The separator is a thin (typically some 10–25 µm) microporous polyolefin film, frequently ceramic-coated. It must be thin for low resistance but robust against puncture. Some have a shutdown feature, in which pores close at elevated temperature.

Current collectors. Copper is used at the negative because aluminium alloys with lithium at low potential, and aluminium is used at the positive because copper would dissolve at high

potential.

2.4.4 Charge and Discharge Electrochemical Reactions

For a layered oxide cathode of general formula $LiMO_2$ (M = transition metals), the half-reactions on discharge are

$$\text{Negative (oxidation):} \quad Li_xC_6 \rightarrow 6C + xLi^+ + xe^- \tag{2.21}$$

$$\text{Positive (reduction):} \quad Li_{1-x}MO_2 + xLi^+ + xe^- \rightarrow LiMO_2 \tag{2.22}$$

$$\text{Overall:} \quad Li_xC_6 + Li_{1-x}MO_2 \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftharpoons}} 6C + LiMO_2. \tag{2.23}$$

For LFP the positive reaction is $FePO_4 + Li^+ + e^- \rightarrow LiFePO_4$. The charge is carried by the working lithium ion; the amount of cyclable lithium is therefore a key inventory which decreases with ageing.

2.4.5 Lithium-Ion Transport

Ion transport comprises solid diffusion in the active particles, charge transfer at the particle surface, and migration and diffusion in the electrolyte. At high rates, the electrolyte becomes depleted near the positive electrode on discharge, and at low temperatures diffusion slows dramatically. These limits explain the falling capacity at high C-rate and low temperature.

2.4.6 SEI Layer and its Importance

During the first charge, the electrolyte reduces on the graphite surface at low potential, forming the *solid electrolyte interphase* (SEI) [10]. A good SEI passes Li^+ but blocks electrons and solvent, stopping further electrolyte decomposition, so it is essential for stability. Its formation consumes some cyclable lithium, which is the origin of the first-cycle capacity loss. Over life the SEI thickens slowly, which is the main mechanism of calendar ageing and resistance growth [11].

2.4.7 Cell Voltage and Operating Voltage Range

Voltage limits are chemistry-specific. NMC/graphite cells commonly operate between about 2.5–3.0 V and 4.2 V with a nominal around 3.6–3.7 V, whereas LFP/graphite operates about 2.5–3.65 V with a very flat 3.2–3.3 V plateau. Exceeding $V_{\max}$ degrades the cathode and electrolyte, and discharging beneath $V_{\min}$ can dissolve the copper collector. Figure 2.19 compares representative curves. The flat LFP plateau gives steady power but makes voltage-based SoC estimation difficult, a point revisited in Section 2.7.

2.4.8 Capacity, Specific Energy and Specific Power

Practical capacity depends on the active material loading, the voltage window, the rate and the temperature. Cell specific energy is the product of voltage and capacity divided by *total*

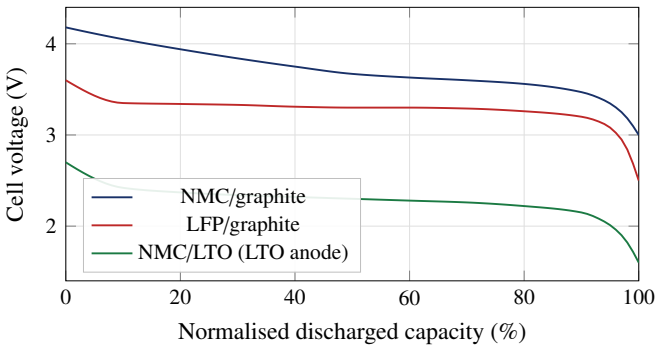


Figure 2.19: Representative discharge voltage curves of NMC, LFP and LTO-anode cells (illustrative shapes, low rate).

cell mass, including foil, separator, electrolyte and housing. So the high capacity of a single material does not translate directly to the cell. Specific power depends on $V_{oc}^2/(4R)$, but in practice is limited by the voltage and thermal limits of the cell.

2.4.9 Cycle Life and Calendar Ageing

Cycle life is the number of cycles to a defined end of life, often 80% of initial capacity, under stated DoD, C-rate and temperature. *Calendar ageing* occurs with time, even at rest, and is accelerated by high SoC and high temperature. Both are treated quantitatively in Section 2.7.2.

2.4.10 Temperature Dependence

Reaction rates follow an Arrhenius law

$$k(T) = A \exp\!\left(-\frac{E_a}{\mathcal{R}T}\right), \tag{2.24}$$

where A is a pre-exponential factor and E_a the activation energy (J/mol). Low temperature therefore raises resistance and reduces available capacity, whereas high temperature improves power but accelerates ageing. A commonly quoted rule of thumb is that ageing rate roughly doubles for each 10°C rise, but the factor depends on the mechanism.

2.4.11 Charging Rate and C-Rate

The C-rate of (2.18) is the natural normalisation of current, since it allows cells of different capacity to be compared. Note that a 1C discharge ideally lasts one hour regardless of cell size. High charge rates give convenience, but they raise heat and risk lithium plating.

2.4.12 Lithium Plating

On charge, the graphite potential falls. If it reaches that of metallic lithium, Li^+ deposits as metal on the particle surface instead of intercalating [12]. This occurs at high charge rate, at low temperature, or at high SoC. Plating consumes cyclable lithium, can grow dendrites

through the separator, and is the main reason fast charging is restricted at low temperature and tapered at high SoC.

2.4.13 Thermal Runaway Fundamentals

Thermal runaway is a self-accelerating exothermic failure [13]. Triggers include mechanical damage, an internal short, overcharge or external heating. As temperature rises, a sequence follows: the SEI decomposes, the anode reacts with electrolyte, the separator melts allowing an internal short, the cathode releases oxygen, and the electrolyte burns. Each stage feeds the next because heat generation grows exponentially with temperature (Arrhenius behaviour), whereas heat rejection grows only about linearly. The danger at pack level is propagation between cells. Hence the BMS must detect the early signs and the pack must be designed to delay propagation.

2.4.14 Important Lithium-Ion Chemistries

Lithium iron phosphate (LFP). $LiFePO_4$ cathode, graphite anode, nominal 3.2 V. The strong P–O bond resists oxygen release, giving excellent thermal stability, long cycle life and low cost, with no cobalt or nickel. Its drawbacks are a lower specific energy and voltage, a flat OCV, and poor low-temperature performance. It is used widely in buses, entry-level cars and, increasingly, mainstream cars.

Lithium nickel manganese cobalt oxide (NMC). $LiNi_xMn_yCo_zO_2$, graphite anode, nominal 3.6–3.7 V. The ratio is denoted NMC111, NMC622, NMC811 by proportion; more nickel gives higher capacity but less thermal stability, and less cobalt reduces cost. It is the mainstream of long-range BEVs.

Lithium nickel cobalt aluminium oxide (NCA). $LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$-type cathode, nominal about 3.6 V. Offers very high specific energy, but it needs careful thermal control, and is less tolerant of abuse.

Lithium cobalt oxide (LCO). $LiCoO_2$, nominal about 3.7 V. It has high volumetric energy density but cobalt-related cost and limited thermal stability, so it is used in consumer electronics rather than traction.

Lithium titanate (LTO). $Li_4Ti_5O_{12}$ anode with NMC or LMO cathode, nominal about 2.3–2.4 V. The anode operates at about 1.55 V versus Li, which avoids plating and SEI growth, so it has very long life and high charge rates, even in the cold. Its low voltage gives low specific energy and high cost per kWh, restricting it to specialised buses and high-cycle applications.

Other emerging chemistries. These include lithium manganese iron phosphate (LMFP), which raises the LFP voltage; silicon-rich anodes; lithium-rich and high-manganese cath-

odes; and lithium-sulphur, which offers high theoretical specific energy but a short life so far. Sodium-ion and solid-state cells were covered in Section 2.2.

Table 2.2: Comparative lithium-ion chemistry table. Approximate cell-level ranges for orientation only [3, 4, 7]; values depend on design and test conditions and are not universal.

Property	LFP	NMC	NCA	LCO	LTO (anode)
Cathode / anode	$LiFePO_4$ / graphite	NMC / graphite	NCA / graphite	$LiCoO_2$ / graphite	NMC or LMO / $Li_4Ti_5O_{12}$
Nominal voltage (V)	3.2	3.6–3.7	3.6	3.7	2.3–2.4
Specific energy (Wh/kg)	110–190	180–280	200–300	150–240	60–90
Power capability	Good	Good to high	Good	Moderate	Very high
Cycle life (typ.)	2000–6000	1000–2500	800–1500	500–1000	5000–20 000
Thermal stability	Very good	Moderate, falls with Ni	Lower	Lower	Very good
Cost per kWh	Low	Medium	Medium to high	High	High
Safety	High	Moderate	Needs close control	Needs close control	High
EV application	Buses, 2W/3W, mass-market cars	Long-range cars	Long-range cars	Not used for traction	Fast-charge buses
Main limitation	Lower energy; flat OCV	Ni, Co supply; heat	Thermal sensitivity	Cost; stability	Low energy, cost

Table 2.2 brings these properties together. Choosing between chemistries is a question of which constraint is dominant: a cost-driven city vehicle in hot Indian conditions points to LFP, while a premium long-range car points to high-nickel NMC or NCA. Advantages and limitations of the main candidates are examined further in Table 2.7.

2.5 Electric Vehicle Battery Efficiency and Effects on Performance

2.5.1 Battery Input Power, Output Power and Internal Losses

Applying (2.15), the chemical power, the terminal power and the loss are related as follows. On discharge ($I > 0$),

$$P_{\text{out}} = V_t I = V_{oc} I - I^2 R, \tag{2.25}$$

and on charge, with $|I|$ entering the cell,

$$P_{\text{in}} = V_t |I| = V_{oc} |I| + I^2 R. \tag{2.26}$$

In both cases the internal power loss, or Joule heating, is

$$P_{\text{loss}} = I^2 R. \tag{2.27}$$

Dimensional check: $\text{A}^2\,\Omega = \text{W}$. A more complete heat balance [14] adds a reversible term:

$$\dot{Q}_{\text{gen}} = I\,(V_{oc} - V_t) - IT\frac{\partial V_{oc}}{\partial T}, \tag{2.28}$$

where the first term is the irreversible heat (equal to I^2R for a purely resistive model) and the second the reversible entropic heat, which changes sign with current direction.

2.5.2 Battery Energy, Power, Coulombic, Voltage and Round-Trip Efficiency

Battery power efficiency. The instantaneous power efficiency on discharge follows from (2.25):

$$\eta_{P,\text{dis}} = \frac{V_t I}{V_{oc} I} = \frac{V_t}{V_{oc}} = 1 - \frac{IR}{V_{oc}}, \tag{2.29}$$

and on charge,

$$\eta_{P,\text{ch}} = \frac{V_{oc}}{V_t} = \frac{V_{oc}}{V_{oc} + |I|R}. \tag{2.30}$$

Efficiency falls linearly with current on discharge, which is the instantaneous penalty of high power.

Voltage efficiency. Over a full cycle the *voltage efficiency* is the ratio of mean discharge voltage to mean charge voltage:

$$\eta_V = \frac{\bar{V}_{\text{dis}}}{\bar{V}_{\text{ch}}}. \tag{2.31}$$

Round-trip energy efficiency. Substituting $E = \bar{V}Q$ in (2.20) shows that round-trip energy efficiency factorises into coulombic and voltage parts:

$$\eta_E = \frac{\bar{V}_{\text{dis}} Q_{\text{dis}}}{\bar{V}_{\text{ch}} Q_{\text{ch}}} = \eta_V\,\eta_C. \tag{2.32}$$

For a symmetric cycle at current I around a mean OCV $\bar{V}_{oc}$,

$$\eta_E \approx \frac{\bar{V}_{oc} - IR}{\bar{V}_{oc} + IR}\,\eta_C. \tag{2.33}$$

Equation (2.33) produces the curves of Fig. 2.20. Doubling resistance or C-rate has the same effect, since only the product IR matters.

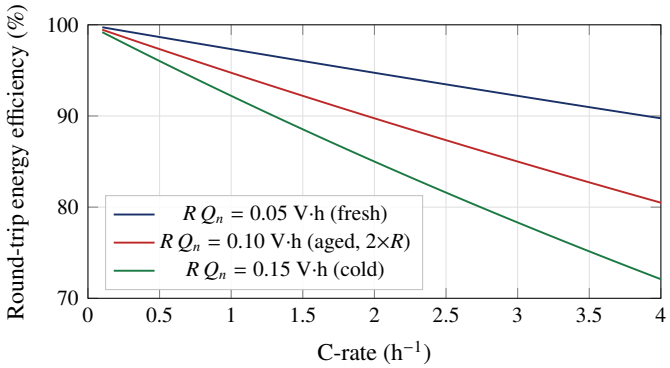


Figure 2.20: Round-trip energy efficiency versus C-rate from (2.33) (with $\bar{V}_{oc} = 3.7$ V and $\eta_C = 1$, illustrative).

Example 5: Battery charging and discharging efficiency

Problem. In a laboratory cycle a cell takes in 100 Ah at a mean voltage of 3.80 V and delivers 99 Ah at a mean voltage of 3.60 V. Find its efficiencies.

Given. $Q_{\text{ch}} = 100$ Ah, $\bar{V}_{\text{ch}} = 3.80$ V, $Q_{\text{dis}} = 99$ Ah, $\bar{V}_{\text{dis}} = 3.60$ V.

Required. η_C, η_V, η_E and energy lost per cycle.

Equations. (2.19), (2.31), (2.32).

Solution. $\eta_C = 99/100 = 0.990$. $\eta_V = 3.60/3.80 = 0.9474$. $\eta_E = 0.990 \times 0.9474 = 0.9379$. Check directly: $E_{\text{ch}} = 380$ Wh, $E_{\text{dis}} = 356.4$ Wh, ratio 0.9379. Energy lost = 23.6 Wh.

Answer. $\eta_C = 99.0\%$, $\eta_V = 94.7\%$, $\eta_E = 93.8\%$; loss 23.6 Wh per cycle.

Interpretation. Most of the loss is voltage loss, and it becomes heat that the thermal system must remove.

2.5.3 Self-Discharge

Self-discharge is the slow loss of stored charge at open circuit due to internal side reactions. It is small for lithium-ion, on the order of a few per cent per month, and larger when hot. It matters mostly for long parked periods and for the imbalance it creates between cells.

2.5.4 Temperature and High-Rate Effects

Figure 2.21 shows the indicative influence of temperature. At −20°C resistance may be several times the 25°C value and the accessible capacity is substantially reduced. High-rate charge and discharge raise the IR drop, which shortens the time to the voltage limits so the delivered capacity falls, and raise cell temperature.

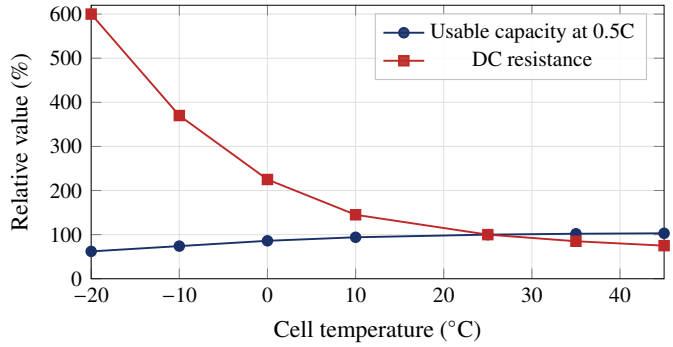


Figure 2.21: Effect of cell temperature on capacity and resistance (illustrative trends; both normalised to 25°C).

2.5.5 Battery Ageing Effects: Capacity Fade, Power Fade and Voltage Sag

Capacity fade is the loss of deliverable charge, which shortens range. *Power fade* is the fall in deliverable power, which arises from resistance growth. *Voltage sag* is the short-term fall in terminal voltage under a load step, given by $\Delta V = \Delta I\, R$, and it scales with resistance. Figure 2.22 shows a hard acceleration: for the same current step, doubling the resistance doubles the sag.

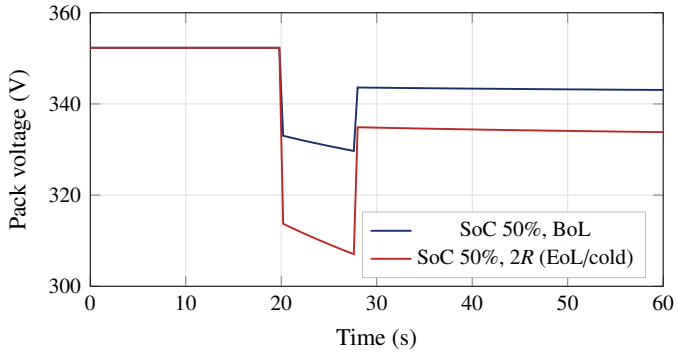


Figure 2.22: Voltage sag of a hypothetical 96s pack during a load step (illustrative simulation).

The maximum deliverable power is limited both by the matched-load condition and, more stringently, by the minimum voltage limit:

$$P_{\max} = \min\left(\frac{V_{oc}^2}{4R},\ \frac{V_{\min}(V_{oc} - V_{\min})}{R}\right). \tag{2.34}$$

Since V_{oc} falls with SoC and R rises with age, the limit falls as Fig. 2.23 shows. This is why an EV feels slower at low charge and in the cold, and why its dashboard may warn of limited power.

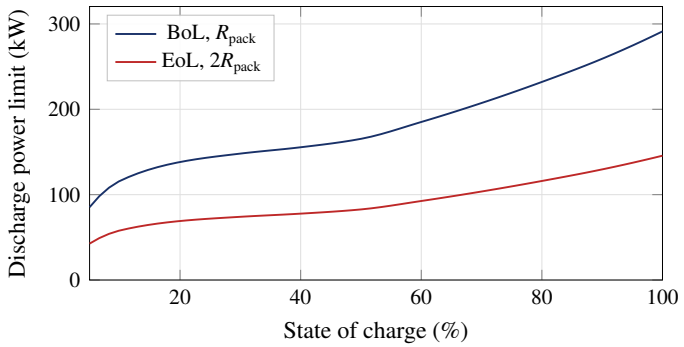


Figure 2.23: Discharge power limit (2.34) versus SoC of a hypothetical 96s3p pack at beginning (BoL) and end of life (EoL).

2.5.6 Effects on Vehicle Performance

Available energy and driving range. The usable energy at a given state is

$$E_{\text{avail}} = E_{\text{nom}}\,\text{SoH}\,(z - z_{\min})\,\eta_{\text{dis}}, \tag{2.35}$$

and the expected range is

$$d = \frac{E_{\text{avail}}}{e_v}, \tag{2.36}$$

where e_v is the consumption in Wh/km at the battery terminals.

Acceleration performance and maximum speed. Wheel power is $P_w = P_{\text{bat}}\,\eta_{\text{dt}}$, so the power limit (2.34) feeds directly into acceleration. Also, the inverter's maximum AC voltage is proportional to DC voltage, so when pack voltage sags, the motor's base speed falls, which reduces torque at high speed, and can limit maximum speed.

Regenerative braking performance. A battery near full charge or cold has a low charge acceptance, because the terminal voltage would exceed $V_{\max}$ or plating would be risked. The regen power limit is

$$P_{\text{regen,max}} = V_{\max}\,\frac{V_{\max} - V_{oc}}{R}, \tag{2.37}$$

which vanishes at $V_{oc} = V_{\max}$. So regenerative braking is weak straight after a full charge, and friction brakes must take over.

Battery thermal performance. A lumped energy balance links the loss to the temperature rise:

$$mc_p\frac{dT}{dt} = \dot{Q}_{\text{gen}} - hA\,(T - T_{\text{amb}}), \tag{2.38}$$

where m is mass (kg), c_p specific heat (J/kg K), hA the heat transfer conductance (W/K) and T_{amb} the coolant or ambient temperature. The steady temperature rise is $\Delta T = \dot{Q}_{\text{gen}}/(hA)$ [15].

Ambient and operating temperature. Ambient temperature acts through two routes: the cell temperature affects resistance and capacity, and the cabin HVAC draws power from the pack. Figure 2.24 shows that range falls substantially in cold conditions, particularly with resistive heating, and modestly in strong heat because of air conditioning. In a hot Indian summer, a cooled pack also pays a parasitic load to hold the cell temperature in its preferred window.

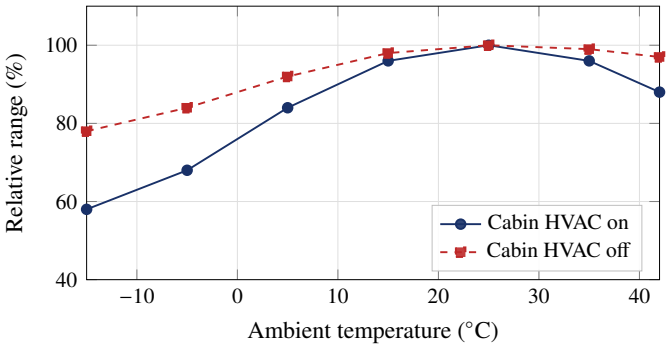


Figure 2.24: Indicative effect of ambient temperature on range (illustrative trend; actual behaviour depends on vehicle, heat pump and cycle).

Effects of SoC and SoH. SoC changes available energy and power via OCV and resistance. SoH reduces capacity, hence range, and increases resistance, hence reduces power and efficiency.

Example 6: Effect of resistance, temperature, SoC and SoH on EV performance

Problem. A hypothetical 70 kWh pack with $V_{oc} = 400$ V, $R = 0.10\ \Omega$, $V_{\min} = 320$ V, DoD 0.90, $\eta_{\text{dis}} = 0.95$ and $e_v = 160$ Wh/km supplies a 100 kW demand at the terminals. Evaluate: (a) the effect of doubling R; (b) the range when SoH is 0.88; (c) the range in the cold, with capacity at 75% and consumption 210 Wh/km; (d) range from 90% of the usable window.

Given. As stated (hypothetical).

Required. Current, terminal voltage, loss and power limit for (a); ranges for (b)–(d).

Equations. (2.5), (2.27), (2.34), (2.35), (2.36).

Solution. (a) For $R = 0.10\ \Omega$,

$$I = \frac{400 - \sqrt{400^2 - 4(0.10)(100\,000)}}{2(0.10)} = 267.9\ \text{A},$$

so $V_t = 373.2$ V and $P_{\text{loss}} = 7.18$ kW. For $R = 0.20\ \Omega$, $I = 292.9$ A, $V_t = 341.4$ V, and $P_{\text{loss}} = 17.16$ kW. The voltage-limited power is $320(80)/R$, which is 256 kW for 0.10 Ω and 128 kW for 0.20 Ω.

Baseline range $= 70 \times 0.90 \times 0.95/0.160 = 374$ km.

(b) $d = 70 \times 0.88 \times 0.90 \times 0.95/0.160 = 329$ km.

(c) $d = 70 \times 0.75 \times 0.90 \times 0.95/0.210 = 214$ km.

(d) $d = 0.9 \times 374 = 337$ km.

Answer. Doubling R more than doubles the loss (7.18 to 17.16 kW) and halves peak power (256 to 128 kW). Range: 374 km baseline, 329 km at SoH 0.88, 214 km in the cold, 337 km from 90% of the window.

Interpretation. Loss rises by more than the resistance ratio because the current must also rise to hold terminal power as voltage sags. Cold conditions are the worst case, since capacity and consumption both move in the wrong direction.

2.6 Calculation of Electric Vehicle Battery Capacity

2.6.1 Design Logic

A designer starts from what the customer wants, namely a range, and works backward to a number of cells. The chain has four links: energy needed at the battery terminals, energy that must be installed to provide it, charge and voltage of the pack, and finally the cell count. Figure 2.25 summarises it, and each link is derived below.

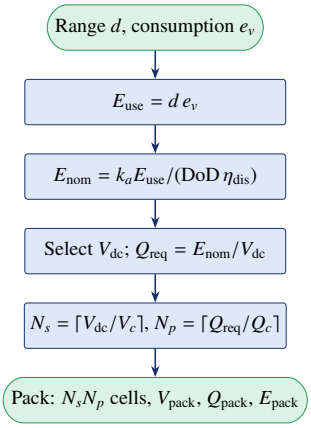


Figure 2.25: Battery capacity calculation sequence.

2.6.2 Energy Requirement Based on Driving Range

Vehicle energy consumption e_v (Wh/km) is the energy drawn from the battery terminals per kilometre. It comes from a road-load calculation or a measured drive cycle and already includes drivetrain losses and auxiliary loads if defined at the battery. If only wheel energy e_w is known then

$$e_v = \frac{e_w}{\eta_{dt}} + \frac{P_{aux}}{\bar{v}}, \tag{2.39}$$

where η_{dt} is the drivetrain efficiency (inverter, motor and gear), P_{aux} the mean auxiliary load in W, and $\bar{v}$ the mean speed in km/h. The second term shows that auxiliaries matter most at low speed: a 2 kW air conditioner at 20 km/h adds 100 Wh/km.

The first relation in the chain is the **battery energy requirement**, which is the usable energy for design range d (km):

$$E_{use} = d\, e_v. \tag{2.40}$$

Environmental and operating conditions enter via e_v: one must size for the design case (for example hot-climate air conditioning or a loaded vehicle), not the laboratory best case.

2.6.3 Usable and Nominal Energy

Nominal energy E_{nom} is the rated energy of the installed pack, $V_{nom} Q_{nom}$. *Usable energy* is what the vehicle can actually deliver. They differ for three reasons:

- **Depth of discharge and SoC window.** The BMS keeps the pack within a window, for example $z_{min} = 0.05$ to $z_{max} = 0.95$, so $\mathrm{DoD} = z_{max} - z_{min} = 0.90$. Avoiding the extremes protects life.
- **Battery efficiency.** The internal loss consumes a fraction $1 - \eta_{dis}$ of energy.
- **Ageing margin.** The end-of-life pack must still meet the range requirement. A multiplier $k_a \geq 1$ provides this extra installed energy. A design safety factor can be included within it.

Hence the **usable battery energy** of an installed pack, and the **nominal battery capacity** needed, are

$$E_{use} = \frac{E_{nom}\, \mathrm{DoD}\, \eta_{dis}}{k_a}, \tag{2.41}$$

$$E_{nom} = \frac{k_a\, E_{use}}{\mathrm{DoD}\, \eta_{dis}} = \frac{k_a\, d\, e_v}{\mathrm{DoD}\, \eta_{dis}}. \tag{2.42}$$

2.6.4 Pack Voltage Selection and Ah Capacity

Pack voltage is a system choice. Since $P = VI$, a higher voltage lowers current, cable mass and I^2R loss, and supports faster DC charging. Passenger cars generally use nominal 350–400 V or 800 V systems; two-wheelers use 48–72 V. Higher voltage raises insulation and semiconductor cost. With the DC bus chosen, the **ampere-hour capacity** is

$$Q_{req} = \frac{1000\, E_{nom}}{V_{dc}} \quad [\mathrm{Ah}], \tag{2.43}$$

with E_{nom} in kWh and V_{dc} in V.

2.6.5 Series and Parallel Cell Selection

For a cell of voltage V_c and capacity Q_c, the **series cell count** is the smallest integer that reaches the bus voltage,

$$N_s = \left\lceil \frac{V_{\mathrm{dc}}}{V_c} \right\rceil. \tag{2.44}$$

Because N_s is rounded, the actual pack voltage differs, so the **parallel cell count** is taken from the energy using that actual voltage:

$$N_p = \left\lceil \frac{1000\, E_{\mathrm{nom}}}{N_s V_c Q_c} \right\rceil. \tag{2.45}$$

The remaining relations are the **total cell count**, **pack voltage**, **pack capacity** and **pack energy**, in line with (2.11)–(2.12):

$$N_{\mathrm{cells}} = N_s N_p, \qquad V_{\mathrm{pack}} = N_s V_c, \qquad Q_{\mathrm{pack}} = N_p Q_c, \tag{2.46}$$

$$E_{\mathrm{pack}} = \frac{V_{\mathrm{pack}} Q_{\mathrm{pack}}}{1000} \quad [\mathrm{kWh}]. \tag{2.47}$$

The **pack current** for a power demand P and the **battery power rating** for a permitted current $I_{\max}$ are

$$I_{\mathrm{pack}} = \frac{P}{V_{\mathrm{pack}}}, \qquad P_{\mathrm{rated}} = V_{\mathrm{pack}} I_{\max} = V_{\mathrm{pack}}\, C_{\max}\, Q_{\mathrm{pack}}. \tag{2.48}$$

The expected range follows by inverting (2.42) with the actual pack energy.

Algorithm 1 lists these operations. It checks the end-of-life range after integer rounding, because the rounding can only add energy.

Algorithm 1 Battery capacity calculation

Require: Range d, consumption e_v, bus voltage V_{dc}, DoD, η_{dis}, k_a, cell (V_c, Q_c)
Ensure: N_s, N_p, V_{pack}, Q_{pack}, E_{pack}, expected range

1: $E_{\mathrm{use}} \leftarrow d\, e_v$
2: $E_{\mathrm{nom}} \leftarrow k_a E_{\mathrm{use}} / (\mathrm{DoD}\, \eta_{\mathrm{dis}})$
3: $N_s \leftarrow \lceil V_{\mathrm{dc}} / V_c \rceil$; $V_{\mathrm{pack}} \leftarrow N_s V_c$
4: $N_p \leftarrow \lceil E_{\mathrm{nom}} / (V_{\mathrm{pack}} Q_c) \rceil$; $Q_{\mathrm{pack}} \leftarrow N_p Q_c$
5: $E_{\mathrm{pack}} \leftarrow V_{\mathrm{pack}} Q_{\mathrm{pack}}$
6: $d_{\mathrm{EoL}} \leftarrow E_{\mathrm{pack}} \mathrm{DoD}\, \eta_{\mathrm{dis}} / (k_a e_v)$
7: **if** $d_{\mathrm{EoL}} < d$ **then**
8: $\quad N_p \leftarrow N_p + 1$ and repeat from step 4
9: **end if**
10: **return** pack configuration

Example 7: EV battery capacity for a specified driving range (mandatory case study)

Problem. Design a pack for a hypothetical BEV. All parameters below are design assumptions, not manufacturer data.

Given. Range $d = 350$ km; consumption $e_v = 160$ Wh/km at the battery terminals; nominal bus $V_{\mathrm{dc}} = 400$ V; DoD = 0.90; $\eta_{\mathrm{dis}} = 0.96$; ageing margin $k_a = 1.10$; cell 3.6 V, 60 Ah.

Required. Usable energy, nominal energy, Ah capacity, N_s, N_p, cell total, pack voltage, capacity and energy, and expected range.

Equations. (2.40), (2.42)–(2.47).

Solution.

$$
\begin{aligned}
E_{\mathrm{use}} &= 350 \times 160 = 56\,000 \text{ Wh} = 56.0 \text{ kWh} \\
E_{\mathrm{nom}} &= \frac{1.10 \times 56.0}{0.90 \times 0.96} = 71.30 \text{ kWh} \\
Q_{\mathrm{req}} &= \frac{71\,296}{400} = 178.2 \text{ Ah} \\
N_s &= \lceil 400/3.6 \rceil = \lceil 111.1 \rceil = 112 \\
N_p &= \left\lceil \frac{71\,296}{112 \times 3.6 \times 60} \right\rceil = \lceil 2.95 \rceil = 3 \\
N_{\mathrm{cells}} &= 112 \times 3 = 336 \\
V_{\mathrm{pack}} &= 112 \times 3.6 = 403.2 \text{ V}, \qquad Q_{\mathrm{pack}} = 3 \times 60 = 180 \text{ Ah} \\
E_{\mathrm{pack}} &= 403.2 \times 180 = 72\,576 \text{ Wh} = 72.58 \text{ kWh} \\
d_{\mathrm{BoL}} &= \frac{72\,576 \times 0.90 \times 0.96}{160} = 392 \text{ km} \\
d_{\mathrm{EoL}} &= \frac{392}{1.10} = 356 \text{ km}
\end{aligned}
$$

Answer. A 112s3p pack of 336 cells: 403.2 V, 180 Ah, 72.58 kWh, with an expected range of 392 km when new and 356 km at end of life.

Interpretation. Nominal energy (72.6 kWh) exceeds usable energy (56 kWh) by about 30% because of the SoC window, efficiency and ageing allowance. The installed energy above the requirement is not waste; it is what guarantees the 350 km target at end of life.

2.7 SoC and SoH Estimation

Neither state of charge nor state of health can be measured directly by a sensor: the BMS can only measure voltage, current and temperature. They must therefore be *estimated*. Figure 2.26 gives the overall architecture. The estimator combines measurements with stored maps, and supplies the rest of the vehicle with the SoC, state of power (SoP) and range.

2.7.1 State of Charge

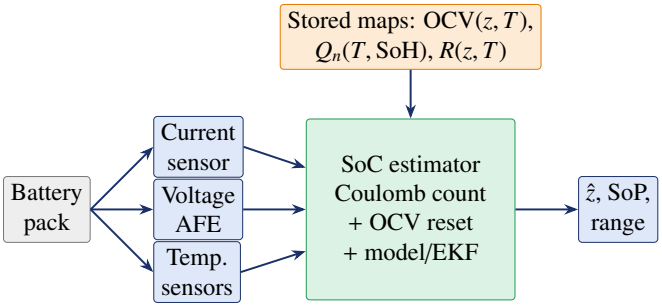


Figure 2.26: SoC estimation architecture.

2.7.1.1 Definition and Physical Interpretation

SoC is the ratio of the charge remaining to the charge available at full, on the present capacity:

$$z = \frac{Q_{\text{rem}}}{Q_n}. \tag{2.49}$$

Physically, it measures the lithium concentration in the electrodes relative to the limits of use. It is the battery analogue of a fuel gauge, with one crucial difference: the capacity of the "tank" shrinks with age and varies with temperature.

2.7.1.2 Importance in EV Operation and Measurement Challenges

SoC is needed for the range display, for preventing overcharge and over-discharge, for setting regen and power limits, and for balancing. It is difficult to estimate accurately because of the following:

- the OCV curve is flat for some chemistries, especially LFP;
- OCV can be measured only after long rest, and it shows hysteresis;
- current sensors have offset, gain error and drift;
- capacity and resistance change with temperature and ageing;
- a pack comprises many cells that are not identical.

2.7.1.3 Coulomb Counting Method

Coulomb counting integrates current over time. Since the discharge current ($I > 0$) reduces stored charge, charge conservation gives

$$\frac{dz}{dt} = -\frac{\eta(t) I(t)}{Q_n} \quad \Longrightarrow \quad z(t) = z(t_0) - \frac{1}{Q_n} \int_{t_0}^{t} \eta(\tau)\, I(\tau)\, d\tau. \tag{2.50}$$

Here Q_n must be in ampere-seconds (A s) if t is in seconds, that is $Q_n[\text{A s}] = 3600\, Q_n[\text{Ah}]$. The factor η is the coulombic efficiency: $\eta = 1$ during discharge and $\eta = \eta_C \leq 1$ during charge,

since only that fraction of the charge entering is stored. The block diagram of Fig. 2.27 shows the implementation. For a sample period T_s the rectangular (Euler) discrete form is

$$z_{k+1} = z_k - \frac{\eta_k T_s}{3600\, Q_n} I_k, \qquad Q_n \text{ in Ah},\ T_s \text{ in s}. \tag{2.51}$$

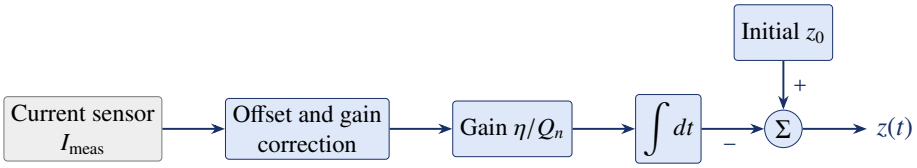


Figure 2.27: Coulomb counting block diagram.

Assumptions and limitations. The method assumes a known initial SoC, known capacity and an accurate sensor. It is open-loop: errors accumulate without bound. A constant current offset ΔI produces an SoC error that grows linearly,

$$\Delta z(t) = \frac{\Delta I\, t}{3600\, Q_n}. \tag{2.52}$$

For example, a 0.5 A offset on a 180 Ah pack gives an error of 0.28% per hour in operation. Remedies include periodic reset at full charge or using OCV at rest [16]. Its advantages are simplicity and low computation, and for these reasons it remains the backbone of most commercial estimators. Algorithm 2 gives the steps.

Algorithm 2 Coulomb counting-based SoC estimation

Require: z_0, Q_n (Ah), η_C, T_s, offset I_{off}
Ensure: z_k
$z \leftarrow z_0$
loop
  Read I_{meas}; $I \leftarrow I_{\text{meas}} - I_{\text{off}}$
  if $I < 0$ **then** $\eta \leftarrow \eta_C$
  else $\eta \leftarrow 1$
  end if
  $z \leftarrow z - \eta I T_s/(3600\, Q_n)$
  Limit z to $[0, 1]$
  if cell at full-charge condition **then** $z \leftarrow 1$
  end if
end loop

Example 8: Battery SoC calculation using Coulomb counting

Problem. A 60 Ah module starts at SoC 90% and experiences the profile below. Find the SoC after each stage.

Given. $z_0 = 0.90$, $Q_n = 60$ Ah. Stage 1: 60 A for 10 min. Stage 2: 120 A for 5 min. Stage 3: regeneration at 60 A for 2 min with $\eta_C = 0.98$. Stage 4: 45 A for 20 min.

Required. z after each stage.

Equations. (2.50) in the form $\Delta z = -\eta I \Delta t / Q_n$ with Δt in hours.

Solution.

$$z_1 = 0.90 - \frac{60(10/60)}{60} = 0.7333$$
$$z_2 = 0.7333 - \frac{120(5/60)}{60} = 0.5667$$
$$z_3 = 0.5667 + \frac{0.98 \times 60(2/60)}{60} = 0.5993$$
$$z_4 = 0.5993 - \frac{45(20/60)}{60} = 0.3493$$

Answer. SoC is 73.3%, 56.7%, 59.9% and finally 34.9%.

Interpretation. Net charge removed is 33.04 Ah. Regeneration recovered only 1.96 Ah of the 2 Ah that entered, the remainder being lost to side reactions.

2.7.1.4 Open-Circuit Voltage Method

After a rest, a cell's terminal voltage equals its OCV, which is a function of SoC (Fig. 2.28). Hence one can invert the stored curve:

$$\hat{z} = \mathrm{OCV}^{-1}(V_{\mathrm{rest}}, T). \tag{2.53}$$

Its *sensitivity* sets the accuracy: a voltage measurement error δV gives an SoC error $\delta z = \delta V / (dV_{oc}/dz)$. For NMC at mid-SoC the slope is about 0.5 V per unit SoC, so a 2 mV error gives 0.4% SoC. For LFP on its plateau the slope may be about 0.05 V per unit or less, giving at least 4% SoC. Figure 2.28 shows this contrast.

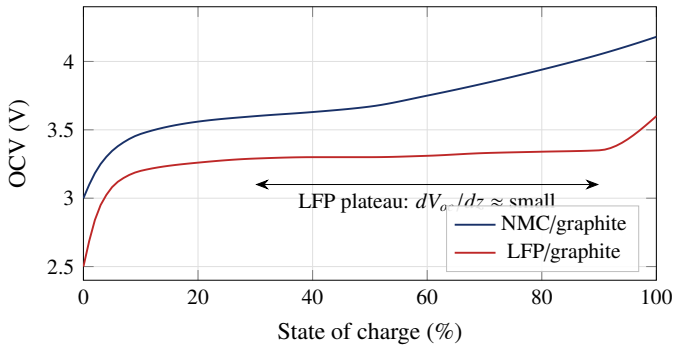


Figure 2.28: OCV versus SoC curves for NMC and LFP (illustrative).

Assumptions and limitations. The method needs the cell at equilibrium, which can take many minutes to hours, so it cannot run during driving. Hysteresis and temperature shift the curve. Its advantage is that it provides an absolute reference without drift, so it complements Coulomb counting. Algorithm 3 shows this standard combination.

Algorithm 3 OCV-based SoC estimation

Require: Table OCV(z, T), rest threshold I_{rest}, rest time t_{rest}
Ensure: Corrected z
$t_{\text{r}} \leftarrow 0$
loop
  Read V, I, T
  if $|I| < I_{\text{rest}}$ **then** $t_{\text{r}} \leftarrow t_{\text{r}} + T_s$
  else $t_{\text{r}} \leftarrow 0$
  end if
  if $t_{\text{r}} > t_{\text{rest}}$ **and** $|dV_{oc}/dz|$ is large at V **then**
    $z \leftarrow$ inverse look-up of OCV table at (V, T)
  end if
end loop

2.7.1.5 Model-Based and Equivalent Circuit-Based Estimation

A model-based estimator runs a battery model in parallel with the real battery. The difference between measured and predicted voltage is then used to correct the states. This is a closed-loop observer, and unlike Coulomb counting, it can recover from a wrong initial SoC. Using the equivalent circuit model of Section 2.11 keeps the computation feasible for a BMS microcontroller.

2.7.1.6 Kalman Filter

The Kalman filter (KF) is the optimal recursive estimator of the states of a *linear* system with Gaussian noise. It treats the problem as a weighted average of two uncertain sources: the model prediction, and the measurement. It sets the weight (the gain) according to their covariances. Since the battery output depends on the nonlinear OCV(z), the basic KF needs extending.

2.7.1.7 Extended Kalman Filter

The EKF linearises the nonlinear model at each step [17, 18].

State-space model. Take the 1-RC model of Section 2.11 with the states $\mathbf{x} = [z,\ V_1]^\top$, input $u = I$, and output $y = V_t$:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, u_k) + \mathbf{w}_k = \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} -\dfrac{\eta T_s}{3600\, Q_n} \\ R_1(1-a) \end{bmatrix} u_k + \mathbf{w}_k, \tag{2.54}$$

$$y_k = h(\mathbf{x}_k, u_k) + v_k = V_{oc}(z_k) - V_{1,k} - R_0 u_k + v_k, \tag{2.55}$$

with $a = e^{-T_s/(R_1 C_1)}$, process noise $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ and sensor noise $v \sim \mathcal{N}(0, R_v)$.

Jacobians. Linearisation gives

$$\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix}, \qquad \mathbf{C}_k = \frac{\partial h}{\partial \mathbf{x}}\bigg|_{\hat{\mathbf{x}}_k^-} = \begin{bmatrix} \dfrac{dV_{oc}}{dz}\bigg|_{\hat{z}_k^-} & -1 \end{bmatrix}. \tag{2.56}$$

Prediction (time update).

$$\hat{\mathbf{x}}_k^- = \mathbf{f}(\hat{\mathbf{x}}_{k-1}^+, u_{k-1}) \qquad \text{state prediction} \tag{2.57}$$

$$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}^+\mathbf{A}^\top + \mathbf{Q} \qquad \text{covariance prediction} \tag{2.58}$$

Correction (measurement update).

$$\hat{y}_k = h(\hat{\mathbf{x}}_k^-, u_k) \qquad \text{measurement prediction} \tag{2.59}$$

$$e_k = y_k - \hat{y}_k \qquad \text{innovation} \tag{2.60}$$

$$\mathbf{L}_k = \mathbf{P}_k^-\mathbf{C}_k^\top(\mathbf{C}_k\mathbf{P}_k^-\mathbf{C}_k^\top + R_v)^{-1} \qquad \text{Kalman gain} \tag{2.61}$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{L}_k e_k \qquad \text{state correction} \tag{2.62}$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{L}_k\mathbf{C}_k)\mathbf{P}_k^- \qquad \text{covariance correction} \tag{2.63}$$

The gain (2.61) is the heart of the method. A large model uncertainty $\mathbf{P}^-$ raises the gain, so the filter trusts the voltage; a noisy sensor R_v lowers it. Observe that the SoC correction depends upon dV_{oc}/dz: on the LFP plateau this slope is near zero, little correction occurs, and the filter reverts in effect to Coulomb counting. Figure 2.29 and Algorithm 4 set out the recursion.

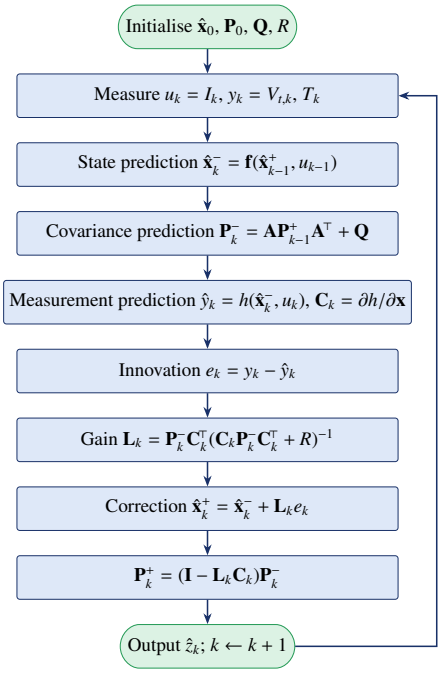


Figure 2.29: EKF-based SoC estimation flowchart.

Algorithm 4 EKF-based SoC estimation

Require: $\hat{\mathbf{x}}_0$, $\mathbf{P}_0$, $\mathbf{Q}$, R_v, parameters Q_n, R_0, R_1, C_1, OCV table
Ensure: $\hat{z}_k$ and its variance
$\hat{\mathbf{x}}^+ \leftarrow \hat{\mathbf{x}}_0$; $\mathbf{P}^+ \leftarrow \mathbf{P}_0$; $a \leftarrow e^{-T_s/(R_1C_1)}$
for $k = 1, 2, \ldots$ **do**
  Read $u_k = I_k$ and $y_k = V_{t,k}$
  Predict state and covariance using (2.57) and (2.58)
  Evaluate dV_{oc}/dz at $\hat{z}^-$ and form $\mathbf{C}_k$
  Compute $\hat{y}_k$, then e_k, and $\mathbf{L}_k$ using (2.59)–(2.61)
  $\hat{\mathbf{x}}^+ \leftarrow \hat{\mathbf{x}}^- + \mathbf{L}_k e_k$; $\mathbf{P}^+ \leftarrow (\mathbf{I} - \mathbf{L}_k\mathbf{C}_k)\mathbf{P}^-$
  Output $\hat{z}_k$ = first element of $\hat{\mathbf{x}}^+$
end for

Assumptions, advantages and disadvantages. EKF assumes a sufficiently accurate model, approximately Gaussian noise, and small linearisation error. It corrects initial errors and provides an error bound. On the other hand it requires tuning of $\mathbf{Q}$ and R_v and more computation, and its accuracy depends on the model parameters, which vary with temperature and age.

2.7.1.8 Unscented Kalman Filter

Rather than linearising, the UKF propagates a small set of deterministically chosen *sigma points* through the true nonlinear functions, and recovers the mean and covariance from them. It needs no Jacobians and copes better with strong nonlinearity, at the cost of more function evaluations per step.

2.7.1.9 Adaptive Estimation Techniques

Adaptive filters update the noise covariances or the model parameters online. Examples include adaptive EKF, dual EKF (one filter for states, another for parameters), and fusion with recursive least squares. They mitigate the parameter drift with ageing, but they increase the risk of instability, and need careful design.

2.7.1.10 Machine Learning-Based SoC Estimation

Data-driven methods learn the map from measured voltage, current and temperature to SoC, using neural networks such as feed-forward or long short-term memory networks, or support vector regression. They need no explicit electrochemical model, but require large, representative training data, generalise poorly outside it, and are harder to certify for safety. Hybrids that combine a learned model with a physics-based filter are an active research area [19].

2.7.1.11 Effect of Temperature and Ageing; Initial SoC and Sensor Errors

Temperature changes OCV slightly, and changes capacity and resistance significantly, so model parameters must be scheduled on temperature. Ageing reduces Q_n; if the SoC calculation keeps using the new-cell value, the gauge will read high. Initial SoC uncertainty arises from long rest with self-discharge and from the hysteresis of OCV, and it is corrected by a closed-loop estimator. Sensor errors have three forms: current offset, which causes drift by (2.52); gain

error, which scales the integrated charge; and voltage error, which matters most on flat OCV curves. Table 2.3 compares the methods.

Table 2.3: Comparison of SoC estimation methods.

Method	Computation	Strength	Main limitation
Coulomb counting	Very low	Simple, smooth	Drift; needs z_0 and Q_n
OCV look-up	Low	Absolute, no drift	Needs rest; weak on LFP
EKF / UKF	Medium	Closed loop; corrects z_0; gives error bound	Model and tuning dependent
Adaptive filter	Medium to high	Tracks parameter change	Stability, tuning effort
Machine learning	High to train	Captures nonlinearity	Data hungry; validation effort

2.7.2 State of Health

2.7.2.1 Definition

SoH is a figure of merit of the present battery relative to its condition when new. There is no single definition, so the basis must always be stated.

Capacity-based SoH. This form reflects energy and range:

$$\mathrm{SoH}_Q = \frac{Q_{\mathrm{act}}}{Q_0}, \tag{2.64}$$

where Q_{act} is the present capacity and Q_0 the beginning-of-life capacity. Automotive end of life is commonly taken as 70–80%.

Resistance-based SoH. This form reflects power capability. It is defined so that it equals 1 when new and 0 at the end-of-life resistance R_{EoL}, often $2R_0$:

$$\mathrm{SoH}_R = \frac{R_{\mathrm{EoL}} - R}{R_{\mathrm{EoL}} - R_0}. \tag{2.65}$$

Power-based SoH. Since $P_{\max} \propto 1/R$ from (2.34),

$$\mathrm{SoH}_P = \frac{P_{\max}}{P_{\max,0}} \approx \frac{R_0}{R}. \tag{2.66}$$

2.7.2.2 Battery Degradation Mechanisms

Degradation is usually grouped into three modes [11]:

- **Loss of lithium inventory (LLI):** SEI growth and plating consume cyclable lithium.
- **Loss of active material (LAM):** particle cracking, binder failure, and transition metal dissolution reduce the host sites.
- **Conductivity loss / impedance rise:** a thicker SEI, electrolyte decomposition and contact loss raise resistance.

2.7.2.3 Capacity Fade and Internal Resistance Growth

Relative capacity fade and resistance growth are defined as

$$\mathrm{CF} = \frac{Q_0 - Q_{\mathrm{act}}}{Q_0} = 1 - \mathrm{SoH}_Q, \qquad \mathrm{RG} = \frac{R - R_0}{R_0}. \tag{2.67}$$

Figures 2.30 and 2.31 show typical illustrative trends. The curve flattens with time when SEI growth is diffusion-limited, which gives the widely observed square-root-of-time trend. Note the dashed *knee* in Fig. 2.30: after long, near-linear ageing, some cells fall off rapidly, for instance due to lithium plating or loss of electrolyte. This knee makes end-of-life prediction by simple extrapolation unreliable.

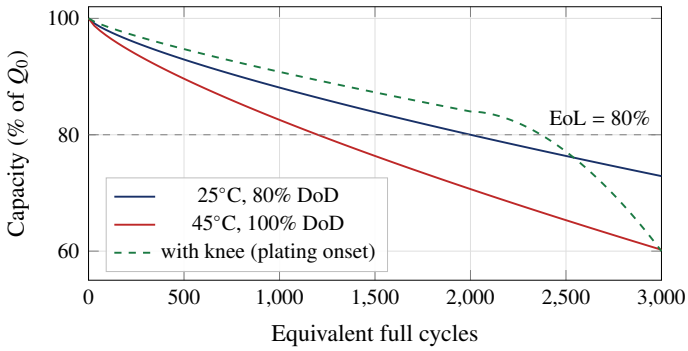


Figure 2.30: Capacity degradation versus cycle number (illustrative trends).

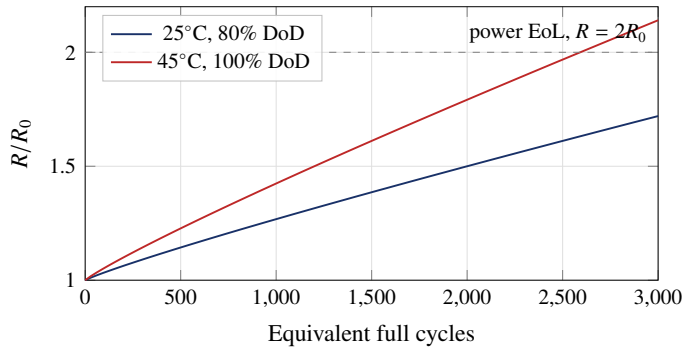


Figure 2.31: Internal resistance growth with ageing (illustrative trends).

2.7.2.4 Calendar Ageing and Cycle Ageing

Calendar ageing is often modelled empirically as

$$\mathrm{CF}_{\mathrm{cal}} = k_{\mathrm{cal}}(z, T)\ \sqrt{t}, \tag{2.68}$$

where k_{cal} rises with SoC and temperature following an Arrhenius form (2.24). Cycle ageing is commonly written as a function of throughput:

$$\mathrm{CF}_{\mathrm{cyc}} = k_{\mathrm{cyc}}(\mathrm{DoD}, C_{\mathrm{rate}}, T)\, N^{\beta}, \tag{2.69}$$

where N is the number of equivalent full cycles and β an empirical exponent, frequently 0.5–1.

2.7.2.5 Temperature, Depth-of-Discharge and C-Rate Effects

High temperature accelerates SEI growth and calendar ageing, while low temperature with charging favours plating. Larger DoD causes larger electrode volume changes per cycle. For a first estimate the total throughput is assumed fixed, which gives the **cycle life estimate**

$$N_{\mathrm{life}}(\mathrm{DoD}) \approx N_{\mathrm{ref}} \left(\frac{\mathrm{DoD}_{\mathrm{ref}}}{\mathrm{DoD}} \right)^{\gamma}, \tag{2.70}$$

where $\gamma \geq 1$ is obtained from test data. Higher C-rates add heat and mechanical stress, and so reduce life.

2.7.2.6 Battery Life Prediction and Remaining Useful Life

Over an interval the mean **degradation rate** per cycle is

$$r_d = \frac{\mathrm{SoH}(N_1) - \mathrm{SoH}(N_2)}{N_2 - N_1}. \tag{2.71}$$

If it stays constant, the **remaining useful life** is

$$\mathrm{RUL} = \frac{\mathrm{SoH}(N) - \mathrm{SoH}_{\mathrm{EoL}}}{r_d} \qquad [\text{cycles}]. \tag{2.72}$$

RUL is hence the prognostic counterpart of SoH: SoH describes the present, RUL extrapolates to the threshold. The extrapolation is only as good as the assumption of unchanged rate and operating conditions.

2.7.2.7 SoH Estimation Methods

Figure 2.32 shows the structure. On board, a true full discharge rarely occurs, so the BMS measures capacity from partial charges: between two points of known SoC z_1 and z_2,

$$Q_{\mathrm{act}} = \frac{\int_{t_1}^{t_2} \eta I \, dt}{z_2 - z_1}, \tag{2.73}$$

where a charge interval that follows rest (giving reliable OCV-derived z) is the natural choice. Resistance is estimated from a step in current, $R = \Delta V/\Delta I$. Algorithm 5 gives this capacity-based procedure.

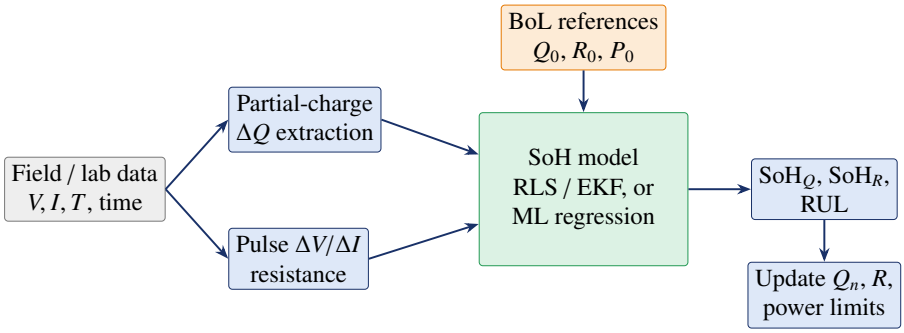


Figure 2.32: SoH estimation architecture.

Algorithm 5 Capacity-based SoH estimation

Require: Q_0, OCV table, $\Delta z_{\min}$, filter constant α
Ensure: SoH_Q, updated Q_n
After rest, set $z_1 \leftarrow \mathrm{OCV}^{-1}(V_{\mathrm{rest}})$; $\Delta q \leftarrow 0$
During the charging session: $\Delta q \leftarrow \Delta q + \eta |I| T_s/3600$
After the following rest, $z_2 \leftarrow \mathrm{OCV}^{-1}(V_{\mathrm{rest}})$
if $|z_2 - z_1| \geq \Delta z_{\min}$ **then**
    $Q_{\mathrm{est}} \leftarrow \Delta q/|z_2 - z_1|$
    $Q_n \leftarrow (1-\alpha)Q_n + \alpha Q_{\mathrm{est}}$
    $\mathrm{SoH}_Q \leftarrow Q_n/Q_0$
end if

Assumptions and limitations. The approach assumes accurate rest OCV and current; the larger the window, the smaller the error. Filtering with a low α avoids jumps, since capacity changes slowly, but it delays tracking.

2.7.2.8 Data-Driven SoH Estimation

Data-driven methods extract health indicators from routine data, such as voltage curve shape, incremental capacity peaks dQ/dV, or temperature rise, and regress SoH on them.

- **Regression** (linear, polynomial or Gaussian process) is transparent. A Gaussian process also gives a confidence interval.
- **Artificial neural networks (ANN)** capture nonlinear relations, whereas recurrent forms learn time dependence.
- **Support vector machines (SVM)** in regression form work with comparatively little data.
- **Ensembles and transfer learning** adapt models from one cell type to another.

A notable result is that cycle life can be predicted from early-cycle discharge voltage features [20]. All such models require data that span the expected conditions, and they cannot be trusted outside the training range.

Example 9: Capacity-based SoH calculation

Problem. A 60 Ah cell measures 52.8 Ah after 1500 equivalent full cycles, and its resistance has risen from 1.8 mΩ to 2.7 mΩ. The vehicle uses 300 equivalent cycles per year. Take EoL as 80% capacity or $R = 2R_0$.

Given. $Q_0 = 60$ Ah, $Q = 52.8$ Ah, $N = 1500$, $R_0 = 1.8$ mΩ, $R = 2.7$ mΩ (hypothetical test data).

Required. SoH_Q, capacity fade, SoH_R, RG, degradation rate, and RUL in cycles and years assuming linear fade.

Equations. (2.64), (2.65), (2.67), (2.71), (2.72).

Solution. $\mathrm{SoH}_Q = 52.8/60 = 0.88$. CF $= 0.12$. $R_{\mathrm{EoL}} = 3.6$ mΩ, so $\mathrm{SoH}_R = (3.6 - 2.7)/(3.6 - 1.8) = 0.50$. RG $= (2.7 - 1.8)/1.8 = 0.50$. $r_d = 0.12/1500 = 8.0 \times 10^{-5}$ per cycle. RUL $= (0.88 - 0.80)/(8.0 \times 10^{-5}) = 1000$ cycles, or $1000/300 = 3.3$ years.

Answer. $\mathrm{SoH}_Q = 88\%$, fade 12%, $\mathrm{SoH}_R = 50\%$, resistance growth 50%; RUL about 1000 cycles (3.3 years).

Interpretation. The two SoH values differ sharply. This battery has lost only 12% of capacity, yet it has used half of its resistance allowance, so a power-sensitive application would retire it first. Also, the RUL ignores a possible knee.

2.8 Governing Equations and Waveforms

This section assembles the fundamental equations and applies them to a single reference: a hypothetical 50 Ah NMC cell with $R_0 = 2$ mΩ, $R_1 = 1.5$ mΩ and $C_1 = 20$ kF. *All curves in this section are simulated from that model; they are illustrative, not measurements.*

2.8.1 Voltage, Current, Power and Energy Equations

The **battery voltage equation** of the dynamic model (derived in Section 2.11) is

$$V_t(t) = V_{oc}(z(t)) - I(t)R_0 - V_1(t). \tag{2.74}$$

Under discharge $V_t < V_{oc}$, and under charging, with $I < 0$, the IR_0 term adds, so $V_t > V_{oc}$. This one equation covers both cases. Treated as the **current equation**, at a commanded terminal voltage, it gives

$$I(t) = \frac{V_{oc}(z) - V_1 - V_t}{R_0}. \tag{2.75}$$

The **power equation** and the **energy equation** are

$$P(t) = V_t(t)I(t), \qquad E(t_1, t_2) = \int_{t_1}^{t_2} V_t(t)I(t)\,dt. \tag{2.76}$$

The sign of P gives the direction: positive is delivered to the load, negative absorbed.

2.8.2 OCV Characteristics and Polarisation Voltage

OCV(z) is the static part of the voltage, and it is shown in Fig. 2.33. The small separation between the rest curves after charge and after discharge is *hysteresis*. The polarisation voltage is the dynamic part, which is the state V_1 in (2.74), whose time constant gives the battery dynamic response.

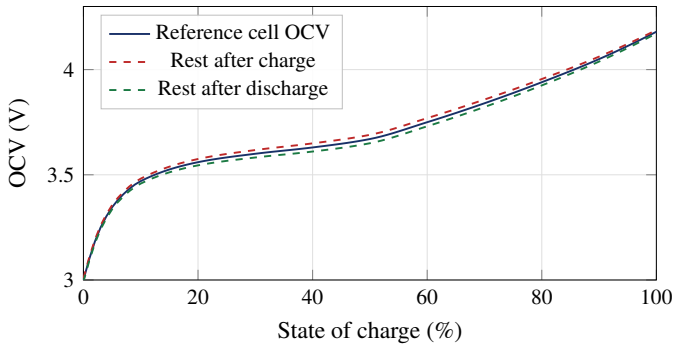


Figure 2.33: OCV versus SoC with a small hysteresis band (illustrative).

2.8.3 Constant-Current Discharge Response

Figure 2.34 shows terminal voltage against time at three constant rates. Higher current gives a lower starting voltage due to the IR drop, and a shorter run time. Note how the time at 2C is a little under half that of 1C: the cut-off is reached earlier. The energy delivered is the area under V_tI, and thus it reduces as the rate rises.

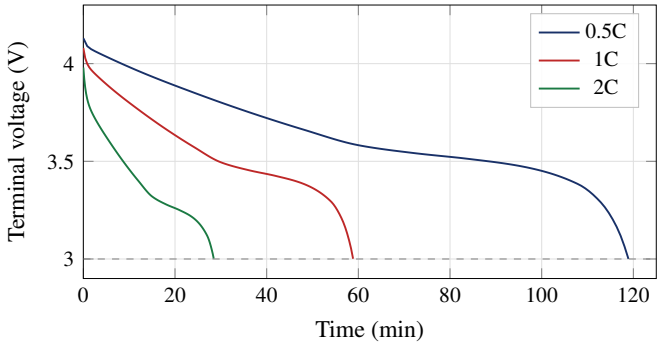


Figure 2.34: Battery voltage versus time during constant-current discharge (simulated).

2.8.4 Dynamic Operation: Current, Power, SoC and Voltage

A representative load is defined in Fig. 2.35: rest, 1C cruise, 2C acceleration, regenerative braking at 0.8C, light cruise, a 3C event, and rest. Negative current denotes regeneration.

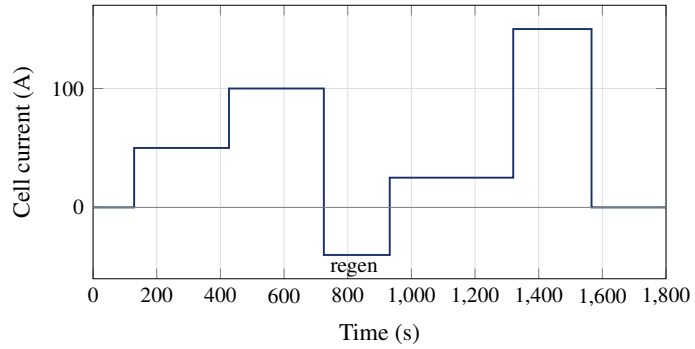


Figure 2.35: Battery current versus time for a dynamic profile (illustrative input to the simulation).

Figure 2.36 gives the corresponding power. It is close in shape to the current but it is not a scaled copy of it, because voltage falls as current rises. Figure 2.37 gives the SoC: it falls linearly in constant-current intervals with slope $-I/Q_n$, rises in regen, and is flat at rest. Figure 2.38 compares terminal voltage with OCV. Each step shows an instantaneous jump of $\Delta I\, R_0$, then a slower exponential settling with time constant $\tau_1 = R_1 C_1 = 30$ s. The difference between V_t and OCV during the 3C event is the voltage sag.

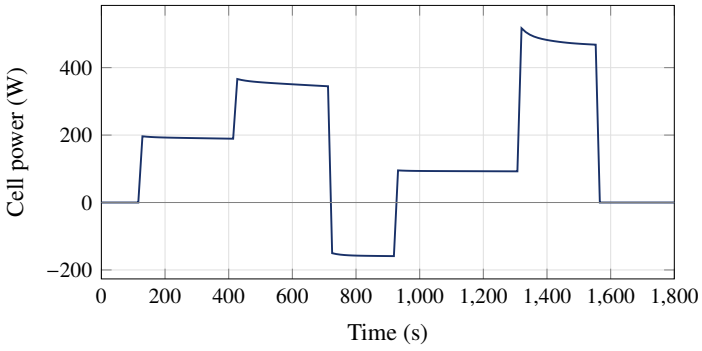


Figure 2.36: Battery power versus time (simulated).

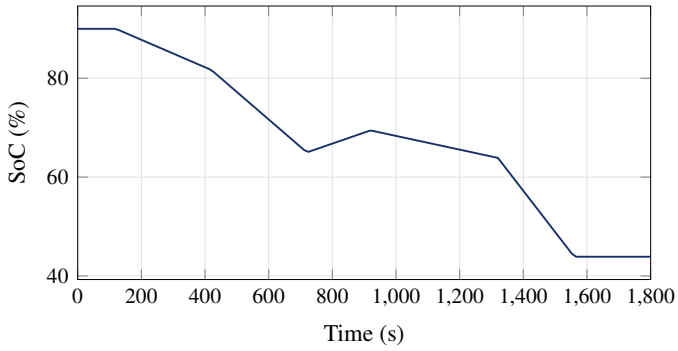


Figure 2.37: SoC versus time (simulated).

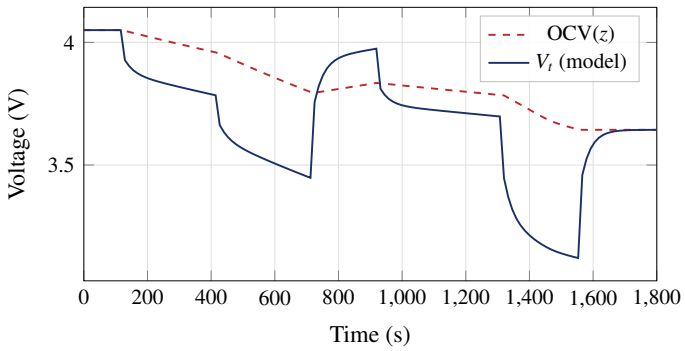


Figure 2.38: Battery voltage response during dynamic load conditions (simulated).

2.8.5 Battery Efficiency and Temperature Waveforms

The instantaneous efficiency from (2.29) and (2.30) dips during high-current intervals, as Fig. 2.39 shows, and it is 100% when idle only in the trivial sense that no power flows. Temperature follows (2.38): Fig. 2.40 shows that the cell integrates its losses, so it rises fastest in the high-current event and relaxes only slowly. It uses assumed values of 0.9 kg, 1000 J/kg K and $hA = 0.25$ W/K.

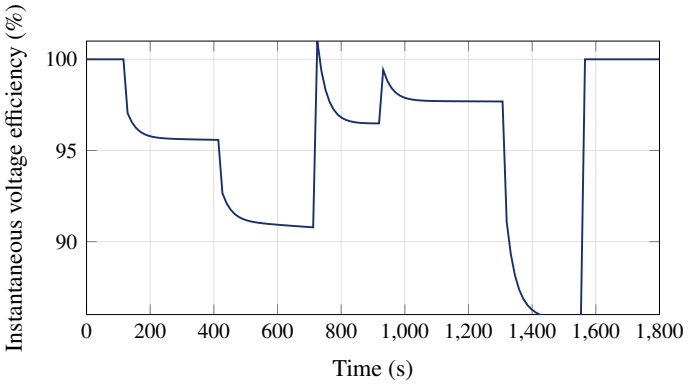


Figure 2.39: Instantaneous voltage efficiency during the dynamic profile (simulated).

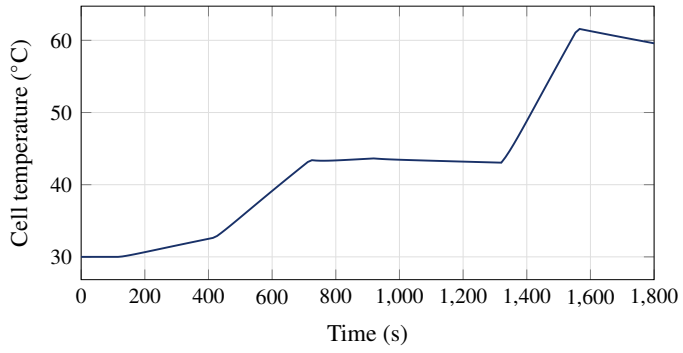


Figure 2.40: Cell temperature variation with time during the dynamic profile (simulated).

2.8.6 Pulse Current Response and Voltage Recovery

Figure 2.41 shows a 60 s, 2C pulse. At load application the voltage drops at once by IR_0 = 0.20 V, then falls exponentially towards an additional IR_1 = 0.15 V. Removal of the load reverses the same two features: the instantaneous rise of IR_0, and a slow *voltage recovery*. This response is the raw material of parameter identification (Section 2.11). A small permanent offset remains since the charge removed has lowered OCV slightly.

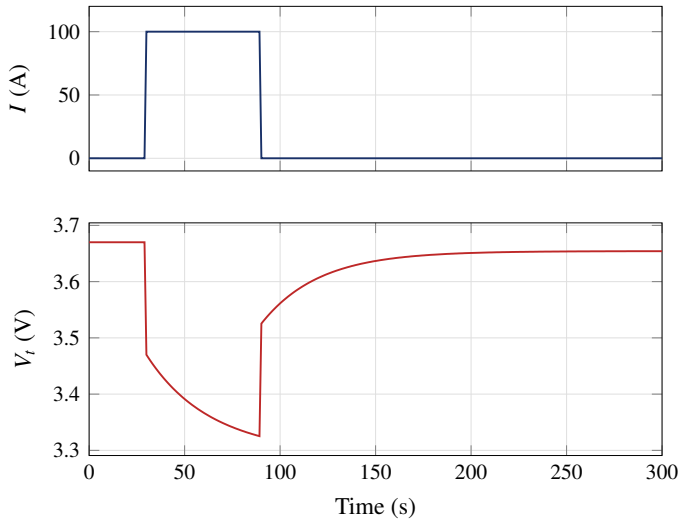


Figure 2.41: Pulse current response and voltage recovery after load removal (simulated).

2.8.7 Charging Voltage and Current Profiles

Figure 2.42 shows a 1C constant-current, constant-voltage charge from 10% SoC. During the first stage current is fixed and voltage climbs. When voltage reaches 4.2 V, the charger holds voltage, and current decays as the polarisation dissipates and OCV rises. Section 3.3 treats this in detail.

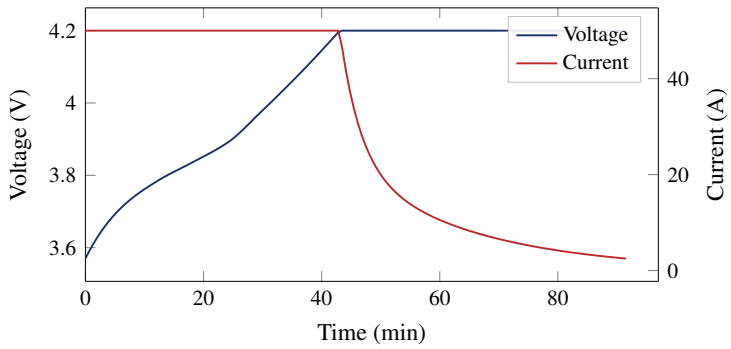


Figure 2.42: Charging voltage and current profiles (simulated).

Example 10: Battery energy estimation under a given operating profile

Problem. A hypothetical pack with mean OCV 360 V and R = 80 mΩ follows the profile in

Table 2.4. Assume that OCV is constant over the short profile. Find the energy delivered, the energy recovered, the net charge and the loss.

Given. See Table 2.4.

Required. V_t and energy for each segment; totals.

Equations. (2.15), (2.76) and (2.27).

Table 2.4: Operating profile and results for Example 10.

I (A)	t (min)	$V_t = 360 - 0.08I$ (V)	$P = V_t I$ (kW)	$E = Pt$ (kWh)
150	10	348.0	52.20	8.700
60	30	355.2	21.31	10.656
−80	5	366.4	−29.31	−2.443
0	10	360.0	0	0
220	4	342.4	75.33	5.022

Solution. Energy delivered = 8.700 + 10.656 + 5.022 = 24.38 kWh. Energy recovered = 2.44 kWh. Net = 21.94 kWh. Net charge = 25 + 30 − 6.67 + 0 + 14.67 = 63.0 Ah. Internal loss = $\sum I^2 R t$ = 0.745 kWh.

Answer. 24.38 kWh delivered, 2.44 kWh recovered, 21.94 kWh net, 63.0 Ah net, and a 0.745 kWh loss.

Interpretation. Regeneration returned 10% of the delivered energy. Note that the 220 A segment lasts only 4 minutes, yet its loss (0.258 kWh) is about a third of the total, because the loss scales with I^2.

2.9 Battery Parameters and Comparisons

Specifying a battery means agreeing on a set of parameters together with their test conditions. Table 2.5 lists them and their units; the following paragraphs explain those that are often misunderstood.

Table 2.5: Battery parameters and their units.

Parameter	Unit	Meaning
Nominal voltage	V	Representative mid-range voltage used for rating
Maximum voltage	V	Upper limit, set at end of charge
Minimum voltage	V	Lower cut-off on discharge
Open-circuit voltage	V	Rested, no-load equilibrium voltage
Terminal voltage	V	Voltage measured under load

Parameter	Unit	Meaning
Rated capacity	Ah	Charge at stated rate and temperature
Nominal energy	Wh, kWh	Nominal voltage × rated capacity
Usable energy	Wh, kWh	Energy available within the operating window
Specific energy	Wh/kg	Energy per unit mass
Energy density	Wh/L	Energy per unit volume
Specific power	W/kg	Power per unit mass, at stated duration
Power density	W/L	Power per unit volume
Internal resistance	mΩ	DC or AC value at stated SoC, T, pulse time
C-rate	h^{-1}	Current normalised to capacity
Charge efficiency	%, –	Energy stored divided by energy input
Discharge efficiency	%, –	Energy output divided by energy withdrawn from store
Coulombic efficiency	%, –	Charge out divided by charge in
Energy efficiency	%, –	Energy out divided by energy in
SoC	%, –	Remaining charge relative to full
SoH	%, –	Present versus initial condition
DoD	%, –	Charge removed relative to capacity; used window
Cycle life	cycles	Cycles to EoL under stated conditions
Calendar life	years	Life under storage conditions
Self-discharge rate	%/month	Open-circuit charge loss
Thermal characteristics	W/K, J/(kg K)	Heat capacity and conductance; entropic coefficient
Maximum charging current	A, C	Continuous or peak limit; may depend on T and SoC
Maximum discharging current	A, C	Continuous or peak limit
Operating temperature range	°C	Separate limits apply to charge and discharge

Voltage parameters. Nominal voltage is a label, not a measurement. Its operating range is bounded by V_{min} and V_{max}, and these bound the required inverter and charger range.

Internal resistance. Without its pulse duration, a resistance figure has no meaning: a 10 ms value is close to R_0, whereas an 18 s value also includes polarisation.

Charge and discharge limits. Datasheets give separate continuous and peak limits. Charge limits are lower than discharge limits and fall sharply at low temperature.

DoD. DoD $= 1 - z$ if measured from full charge. Note that in sizing, DoD means the permitted window.

Thermal characteristics. These cover specific heat, the conductance to coolant, and the temperature range. Lithium-ion cells should typically not be charged below about 0°C without a reduced current, although the actual limit is cell-specific.

2.9.1 Comparison Tables

Tables 2.7–2.9 give comparisons between technology choices. Table 2.2 has already given the comparison between lithium-ion chemistries, and it is not repeated here. All values are representative, drawn from [3, 4, 21].

Table 2.6: Lithium-ion versus lead-acid (representative, cell level).

Parameter	Lithium-ion	Lead-acid
Nominal cell voltage	3.2–3.7 V	2.0 V
Specific energy	110–300 Wh/kg	30–50 Wh/kg
Round-trip energy efficiency	90–97%	70–85%
Usable DoD	80–100%	About 50% for good life
Cycle life	1000–6000	200–800
Maintenance and management	Mandatory BMS	Simple charger; float tolerant
Initial cost per kWh	Higher	Lower
Recycling infrastructure	Developing	Mature

Table 2.6 shows that the higher purchase cost of lithium-ion is frequently recovered through its longer life and higher usable fraction. This is what has driven the shift of Indian e-rickshaws from lead-acid to lithium-ion.

Table 2.8 shows that no format dominates: the choice reflects the pack integrator's strategy for cooling, structure and joining.

2.10 BEV Battery Sizing

Section 2.6 sized a pack from energy alone. A complete design must also satisfy power, thermal, voltage, mass and volume constraints. The procedure is captured in Fig. 2.43: two paths, energy and power, join at the selection of cell, and the pack is then checked and iterated.

Table 2.7: LFP versus NMC (representative, cell level).

Parameter	LFP	NMC
Nominal voltage	3.2 V	3.6–3.7 V
Upper voltage	3.60–3.65 V	4.2 V (typ.)
Specific energy	110–190 Wh/kg	180–280 Wh/kg
Cycle life	2000–6000	1000–2500
Thermal stability	Very good	Moderate; falls as Ni rises
OCV slope	Very flat; SoC harder	Sloped; SoC easier
Low-temperature behaviour	Weaker	Better
Raw material	No Ni, Co	Uses Ni, Co, Mn
Advantages	Safety, life, cost	Range, energy density
Limitations	Pack mass and volume	Cost, supply, thermal care

Table 2.8: Cylindrical, prismatic and pouch cell formats.

Parameter	Cylindrical	Prismatic	Pouch
Housing	Steel can	Aluminium or steel case	Laminate foil
Unit capacity	Small to medium	Large	Medium to large
Packing efficiency	Lower (voids)	High	High
Cooling	Base or side, limited area	Face cooling	Large face area
Swelling	Contained by can	Managed by case and end plates	Needs external compression
Mechanical strength	High	High	Low
Interconnect count	Very high	Low	Low
Manufacturing	Mature, fast	Moderate	Stacking sensitive

Table 2.9: High-energy versus high-power batteries.

Parameter	High-energy	High-power
Electrodes	Thick, heavy loading	Thin, large area
Specific energy	High	Moderate
Internal resistance	Higher (per unit area)	Low
Typical continuous rate	0.3–2C	5C and above
Application	BEV	HEV, performance, buffers
Selection metric	Wh/kg, Wh/L	W/kg, charge acceptance

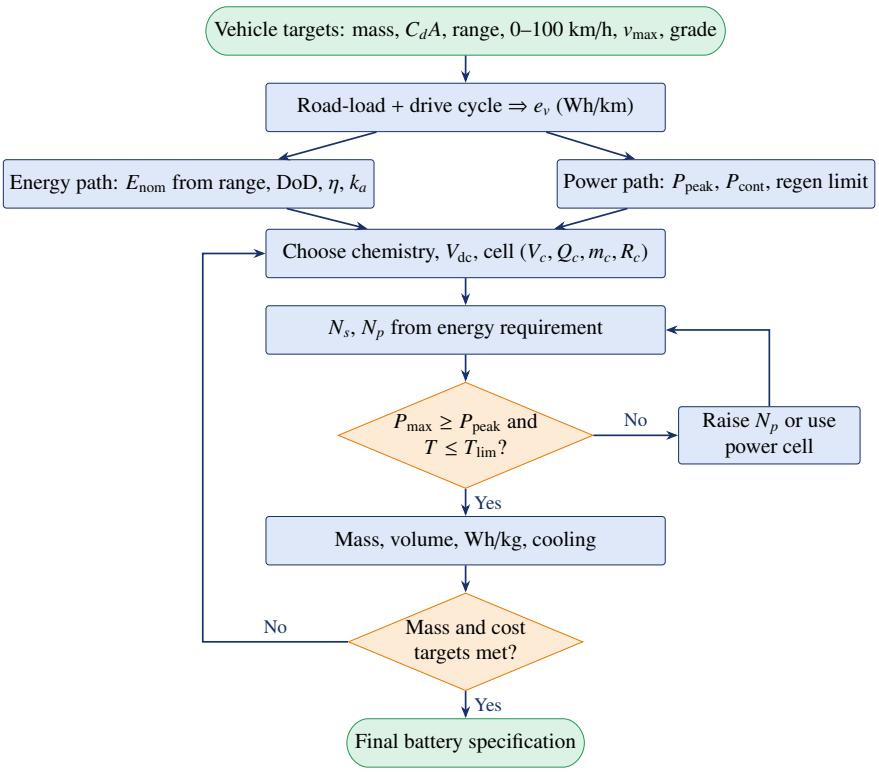


Figure 2.43: Complete battery sizing flowchart.

2.10.1 Vehicle-Level Energy Requirement

The tractive force at the wheels is

$$F_t = mg\,C_{rr}\cos\alpha + \tfrac{1}{2}\rho C_d A v^2 + mg\sin\alpha + \delta m a, \tag{2.77}$$

where m is mass (kg), $g = 9.81$ m/s^2, C_{rr} the rolling coefficient, α the road gradient angle, ρ the air density (kg/m^3), C_d the drag coefficient, A the frontal area (m^2), v the speed (m/s), δ the rotational inertia factor, and a the acceleration (m/s^2). The wheel power is $P_w = F_t v$. Fuller treatments of vehicle dynamics and drivetrain energy are given in [22, 23].

2.10.2 Drive-Cycle-Based and Range-Based Sizing

Drive-cycle-based sizing evaluates (2.77) at every sample of a speed trace of total distance D (for example WLTC or the Indian MIDC), using a regeneration fraction η_{reg} for negative power:

$$e_v = \frac{1}{D}\sum_{k}\left(\frac{\max(P_{w,k},0)}{\eta_{\text{dt}}} + \eta_{\text{reg}}\min(P_{w,k},0) + P_{\text{aux}}\right)\Delta t. \tag{2.78}$$

Range-based sizing then multiplies e_v by the design distance using (2.40) and (2.42).

2.10.3 Power-Based Sizing

Peak power and acceleration. To reach speed v_f in time t_a at about constant power, the energy argument gives

$$P_a \approx \frac{\delta m v_f^2}{2t_a}, \tag{2.79}$$

to which the road load is added. For a constant-power run the mean road-load power is roughly

$$\bar{P}_{\mathrm{rl}} \approx \tfrac{2}{3} m g C_{rr} v_f + \tfrac{2}{5} \cdot \tfrac{1}{2} \rho C_d A v_f^3. \tag{2.80}$$

Continuous power. This is the sustained demand, such as cruising at maximum speed, $P_{\mathrm{cont}} = \frac{1}{2}\rho C_d A v_{\max}^3 + m g C_{rr} v_{\max}$. It governs thermal design.

Gradeability. Holding speed v_g on a slope of $\tan\alpha$ requires, from (2.77),

$$P_g = \left(m g C_{rr} \cos\alpha + m g \sin\alpha + \tfrac{1}{2}\rho C_d A v_g^2 \right) v_g. \tag{2.81}$$

Battery-side peak. These wheel powers are converted to battery terminals by

$$P_{\mathrm{bat}} = \frac{P_w}{\eta_{\mathrm{dt}}} + P_{\mathrm{aux}}. \tag{2.82}$$

Regenerative braking constraints. The pack must also absorb the regen power, subject to (2.37) and the cell's charge rate limit. A small pack is often limited by charge acceptance, so its friction brakes do more work.

2.10.4 Selection of Voltage, Chemistry and Cell

The voltage is chosen as described in Section 2.6. Chemistry and cell selection is guided by Table 2.2. The power-to-energy ratio of the vehicle sets the required C-rate; if the pack sized for energy delivers the peak power below the cell's rated C-rate, the energy path governs, as it does for nearly all long-range BEVs.

2.10.5 Pack Mass, Volume and Figures of Merit

Pack mass is the cell mass divided by the *cell-to-pack mass fraction* f_m, and the volume follows from an assumed packing density:

$$m_{\mathrm{pack}} = \frac{N_{\mathrm{cells}} m_c}{f_m}, \qquad \mathcal{V}_{\mathrm{pack}} = \frac{E_{\mathrm{pack}}}{e_{v,\mathrm{pack}}}. \tag{2.83}$$

The *energy-to-weight ratio* and *power-to-weight ratio* are $E_{\mathrm{pack}}/m_{\mathrm{pack}}$ and $P_{\mathrm{peak}}/m_{\mathrm{pack}}$. Note the mass-compounding effect: added battery mass raises e_v, which requires more battery, so the calculation should be iterated.

2.10.6 Thermal Considerations, Ageing Margin, Safety Factor and Pack Efficiency

Heat generated at peak and continuous load follows from (2.27). The cooling system must hold cell temperature at the continuous load, and the pack's thermal mass absorbs short peaks. The ageing margin k_a assures end-of-life range, and power should be checked at end-of-life resistance, low SoC, and low temperature. Pack-level efficiency includes busbars and contactors: $\eta_{\text{pack}} = \eta_{\text{dis}}\,(1 - I^2 R_{\text{bus}}/P)$.

Algorithm 6 formalises the procedure.

Algorithm 6 BEV battery pack sizing

Require: Vehicle data, range d, acceleration, $v_{\max}$, grade, regen target, cell candidates
Ensure: Pack specification
repeat
  Compute e_v from (2.78); compute E_{nom} from (2.42)
  Compute P_{peak} and P_{cont} from (2.79)–(2.82)
  Select V_{dc} and cell; compute N_s and N_p
  $I_{\max} \leftarrow$ solution of $P_{\text{peak}} = (V_{\text{pack}} - IR_{\text{pack}})I$
  if $I_{\max} > C_{\max} Q_{\text{pack}}$ **or** $V_{\text{pack}} - I_{\max} R_{\text{pack}} < V_{\min}$ **then**
    Increase N_p or choose a higher-power cell
  end if
  Compute m_{pack}; update vehicle mass
until mass converged, and thermal, cost and packaging limits met
return pack specification

Example 11: Complete BEV battery pack sizing

Problem. Size the battery of a hypothetical passenger BEV. Every value below is a design assumption, not vehicle or manufacturer data.

Given. $m = 1850$ kg, $C_d = 0.26$, $A = 2.35$ m^2, $C_{rr} = 0.009$, $\rho = 1.2$ kg/m^3, $\delta = 1.05$; 0–100 km/h in 7.5 s; $v_{\max} = 170$ km/h; 30% grade at 20 km/h; $\eta_{\text{dt}} = 0.90$; $P_{\text{aux}} = 1.5$ kW; range 450 km; $e_v = 170$ Wh/km; DoD 0.92; $\eta_{\text{dis}} = 0.96$; $k_a = 1.10$; $V_{\text{dc}} \approx 380$ V. Cell: NMC cylindrical, 3.6 V, 5.0 Ah, 70 g, 20 mΩ, rated 3C peak; $f_m = 0.62$; pack packing density 260 Wh/L.

Required. All outputs listed in Table 2.11.

Equations. (2.42), (2.44)–(2.47), (2.79)–(2.83), (2.5).

Solution.

Step 1: energy. $E_{\text{use}} = 450 \times 170 = 76.5$ kWh, and $E_{\text{nom}} = 1.10 \times 76.5/(0.92 \times 0.96) = 95.28$ kWh.

Step 2: peak power. With $v_f = 27.78$ m/s,

$$P_a = \frac{1.05 \times 1850 \times 27.78^2}{2 \times 7.5} = 99.9 \text{ kW}, \qquad \bar{P}_{\text{rl}} = 3.02 + 3.14 = 6.17 \text{ kW},$$

so $P_w = 106.1$ kW and $P_{\text{peak}} = 106.1/0.90 + 1.5 = 119.4$ kW at the battery.

Step 3: continuous power. At 47.22 m/s, $P_w = 46.3$ kW, so $P_{\text{cont}} = 46.3/0.90 + 1.5 = 53.0$ kW.

Step 4: gradeability. At $\tan\alpha = 0.30$ ($\alpha = 16.7°$) and 5.56 m/s, (2.81) gives $P_g = 29.9$ kW, or 33.2 kW before auxiliaries. This is below P_{peak} and does not govern.

Step 5: configuration. $N_s = \lceil 380/3.6 \rceil = 106$, $V_{\text{pack}} = 381.6$ V, $N_p = \lceil 95\,280/(381.6 \times 5) \rceil = \lceil 49.9 \rceil = 50$. Thus $N_{\text{cells}} = 5300$, $Q_{\text{pack}} = 250$ Ah, and $E_{\text{pack}} = 95.4$ kWh.

Step 6: current check. $R_{\text{pack}} = (106/50) \times 0.020 = 42.4$ mΩ. Using (2.5) with $P = 119.4$ kW gives $I_{\max} = 324.5$ A and $V_t = 367.8$ V. The rate is $324.5/250 = 1.30$C, below the 3C limit. Peak loss is $I^2R = 4.47$ kW. Continuous current is about 139 A, for a continuous heat load of about 0.82 kW. A 60 kW regen event corresponds to 0.63C.

Step 7: mass and volume. Cell mass $= 5300 \times 0.070 = 371$ kg; $m_{\text{pack}} = 371/0.62 = 598$ kg; $\mathcal{V}_{\text{pack}} = 95\,400/260 = 367$ L; energy-to-weight = 159 Wh/kg; power-to-weight = 199.5 W/kg.

Step 8: range. BoL $d = 95\,400 \times 0.92 \times 0.96/170 = 496$ km; EoL $d = 496/1.10 = 451$ km, which is above the target.

Energy-based versus power-based sizing. If the pack were sized for power alone, using the 3C limit at about 3.3 V under load, each cell would give about 49.5 W, so at least $119\,400/49.5 = 2412$ cells would be needed, or $N_p = 23$ for 2438 cells. That pack holds only 43.9 kWh, and gives only about 228 km. Table 2.10 shows the comparison: energy governs by a factor of more than two.

Table 2.10: Energy-based and power-based sizing for Example 11.

Quantity	Energy-based	Power-based
N_s / N_p	106 / 50	106 / 23
Cells	5300	2438
Pack energy (kWh)	95.4	43.9
Rate at peak power	1.30C	about 3C
Range when new (km)	496	228

Answer. See Table 2.11.

Table 2.11: Final battery specification for Example 11.

Output	Value	Output	Value
Required energy	95.28 kWh	Pack voltage	381.6 V
Peak battery power	119.4 kW	Pack capacity	250 Ah
Continuous power	53.0 kW	Pack energy	95.4 kWh
Chemistry	NMC/graphite	Maximum current	324.5 A
Cell voltage	3.6 V	Pack mass	about 598 kg
Cell capacity	5.0 Ah	Energy density	159 Wh/kg
Series cells	106	Expected range, new	496 km
Parallel strings	50	Expected range, EoL	451 km
Total cells	5300	Volume	about 367 L

Interpretation. The pack is 32% of the assumed vehicle mass, which illustrates why range is expensive. The design assumed that m and e_v already include this pack. Were they not, the iteration of Algorithm 6 would continue. The power check must also be repeated at low SoC, in cold conditions, and at end of life, when R may double.

2.11 Modelling of Battery: RC Equivalent Network Model

2.11.1 Introduction and Purpose of Battery Models

A battery model predicts terminal voltage, and internal states, from the applied current. In EV engineering, models serve four purposes: vehicle-level simulation of range and performance; design of the BMS estimators of Section 2.7; pack design, including thermal and balancing studies; and virtual testing of control software in MATLAB/Simulink before hardware exists [24].

2.11.2 Electrochemical Models versus Equivalent Circuit Models

Electrochemical models, such as the Doyle–Fuller–Newman model, solve coupled partial differential equations for concentration and potential. They reveal internal states such as the anode potential, which is valuable for plating-free fast charging, but they require many hard-to-measure parameters and heavy computation. *Equivalent circuit models* (ECMs) instead reproduce the terminal behaviour using a voltage source, resistors and capacitors [25, 26]. ECMs have few parameters and can run on a BMS processor, so they dominate on-board use. Their drawback is that the parameters are empirical and vary with SoC, temperature and age.

2.11.3 Ideal Voltage Source Model

The simplest representation, shown in Fig. 2.44, is

$$V_t = V_{oc}. \tag{2.84}$$

It says the voltage is independent of current and of history, so the battery is lossless. It is valid only as a rough energy calculation.

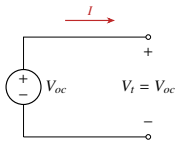


Figure 2.44: Ideal voltage source model.

2.11.4 Rint Model

Figure 2.45 adds SoC-dependent OCV and a series resistance. Kirchhoff's voltage law around the loop gives

$$V_t = V_{oc}(z) - IR_0, \tag{2.85}$$

together with the SoC dynamics of (2.50). Here R_0 may be scheduled on SoC, temperature and current direction.

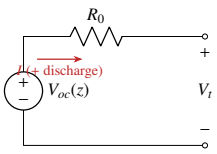


Figure 2.45: Rint battery model. Current is positive on discharge.

Limitations. The Rint model responds instantly. It cannot represent the slow sag during sustained current, or the recovery after the load is removed, as seen in Fig. 2.41. For vehicle-level energy studies it is often adequate, but it is not accurate enough for a good voltage-based SoC estimator.

2.11.5 Model Parameters of the RC Equivalent Circuit

Adding RC branches captures polarisation. Each element has a physical interpretation:

- **Open-circuit voltage source** $V_{oc}(z)$: the equilibrium voltage. Its dependence on z makes the model behave like a very large capacitor.
- **Ohmic resistance** R_0: electrolyte, foil, tab and contact resistance, responsible for the instantaneous step.
- **Polarisation resistance** R_1: the voltage drop from charge transfer and diffusion, once settled.
- **Polarisation capacitance** C_1: sets how quickly that drop develops, with $\tau_1 = R_1C_1$.

The *transient voltage response* is governed by these time constants, and the *steady-state response* is reached after about 5τ, when the capacitor current has decayed to zero, and $V_1 = IR_1$. Thereafter only OCV drifts. Together these yield the *dynamic battery behaviour*.

2.11.6 First-Order Thevenin Equivalent Circuit Model (1-RC)

2.11.6.1 Circuit and Governing Differential Equation

As illustrated in Fig. 2.46, the first-order Thevenin model consists of an ideal open-circuit voltage source, an ohmic resistance and one parallel RC polarisation branch. The resistance R_0 represents instantaneous ohmic losses, while the RC branch captures transient electrochemical polarisation.

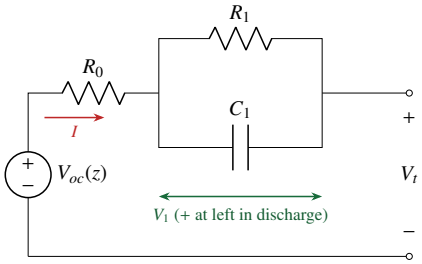


Figure 2.46: First-order Thevenin (1-RC) model.

With $I > 0$ on discharge, the current I divides between R_1 and C_1. Kirchhoff's current law at the RC node gives

$$I = \frac{V_1}{R_1} + C_1 \frac{dV_1}{dt}. \tag{2.86}$$

Rearranging gives the **polarisation voltage equation**

$$\frac{dV_1}{dt} = -\frac{V_1}{R_1 C_1} + \frac{I}{C_1}. \tag{2.87}$$

Kirchhoff's voltage law gives the **terminal voltage equation**

$$V_t = V_{oc}(z) - IR_0 - V_1. \tag{2.88}$$

Sign convention. The signs in (2.87) and (2.88) depend on the current convention. If a text defines charge current as positive, the same physical model is written $V_t = V_{oc} + IR_0 + V_1$ with V_1 redefined accordingly. Mixing the conventions is a common source of error in simulation; here the discharge-positive form is used throughout.

2.11.6.2 Dynamic Response Analysis

For constant I from an initial $V_1(0)$, solving the first-order linear equation (2.87) gives

$$V_1(t) = IR_1 + (V_1(0) - IR_1)e^{-t/\tau_1}, \qquad \tau_1 = R_1 C_1. \tag{2.89}$$

Starting from rest, $V_1(t) = IR_1(1 - e^{-t/\tau_1})$. When the load is removed at $t = 0$ with $I = 0$, the solution gives the *relaxation* $V_1(t) = V_1(0)e^{-t/\tau_1}$. The dimension of τ_1 is $\Omega \cdot \mathrm{F} = \mathrm{s}$.

2.11.6.3 State-Space Representation

Taking $\mathbf{x} = [z,\ V_1]^\top$ and $u = I$, the continuous-time model is

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 0 \\ 0 & -\dfrac{1}{R_1 C_1} \end{bmatrix} \mathbf{x} + \begin{bmatrix} -\dfrac{\eta}{Q_n} \\ \dfrac{1}{C_1} \end{bmatrix} u, \qquad y = V_{oc}(z) - V_1 - R_0 u, \tag{2.90}$$

where Q_n is in A s.

2.11.6.4 Discrete-Time Implementation

If the current is held constant across each sample T_s (zero-order hold), then the exact solution (2.89) gives

$$V_{1,k+1} = a\, V_{1,k} + R_1(1-a) I_k, \qquad a = e^{-T_s/\tau_1}, \tag{2.91}$$

which remains stable for every T_s, unlike the forward Euler form. This form is used in Algorithm 7.

Algorithm 7 1-RC battery model simulation

Require: Current profile I_k, z_0, Q_n (Ah), η_C, T_s, tables $V_{oc}(z)$, R_0, R_1, C_1
Ensure: $V_{t,k}$, z_k, $V_{1,k}$

$z \leftarrow z_0$; $\quad V_1 \leftarrow 0$
for $k = 0, 1, \ldots, N$ **do**
  Look up R_0, R_1, C_1 at (z, T); $\quad a \leftarrow e^{-T_s/(R_1 C_1)}$
  $V_{t,k} \leftarrow V_{oc}(z) - I_k R_0 - V_1$
  if $V_{t,k} < V_{\min}$ **or** $V_{t,k} > V_{\max}$ **then** flag limit
  end if
  if $I_k < 0$ **then** $\eta \leftarrow \eta_C$
  else $\eta \leftarrow 1$
  end if
  $z \leftarrow z - \eta I_k T_s / (3600\, Q_n)$
  $V_1 \leftarrow a V_1 + R_1(1-a) I_k$
end for

2.11.6.5 Parameter Identification from a Pulse Test

Figure 2.47 shows the response to a current pulse. Three features identify the parameters directly:

$$R_0 = \frac{\Delta V_0}{I}, \qquad R_1 = \frac{\Delta V_\infty}{I}, \qquad C_1 = \frac{\tau_1}{R_1}, \tag{2.92}$$

where ΔV_0 is the instantaneous step, ΔV_∞ is the slow further change, and τ_1 is the time to reach 63.2% of ΔV_∞. The recovery after the pulse, unaffected by the slow OCV movement during the load, is usually the cleaner part to fit.

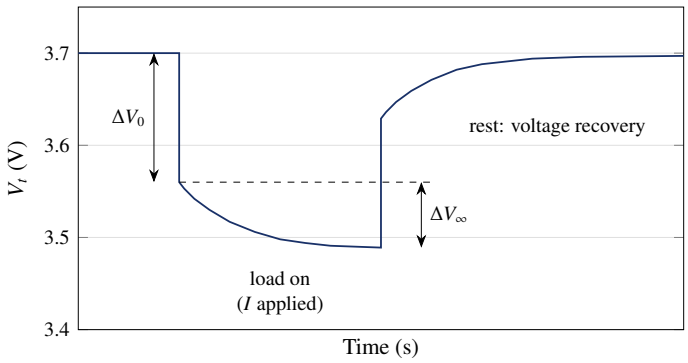


Figure 2.47: Battery pulse-test response, showing the instantaneous and polarisation voltage changes (schematic).

Example 12: 1-RC equivalent circuit model calculation

Problem. A hypothetical cell at rest experiences a constant 100 A discharge step. SoC change is neglected over the short interval.

Given. OCV = 3.70 V, $R_0 = 2$ mΩ, $R_1 = 1.5$ mΩ, $C_1 = 20\,000$ F, $I = 100$ A.

Required. Time constant; V_1 and V_t at $t = 0^+, 10, 30, 60, 150$ s; steady-state values; and recovery after the load is removed once settled.

Equations. (2.89) and (2.88).

Solution. $\tau_1 = 0.0015 \times 20\,000 = 30$ s, $IR_1 = 0.15$ V, $IR_0 = 0.20$ V. Hence $V_1(t) = 0.15(1 - e^{-t/30})$ and $V_t = 3.50 - V_1$.

t (s)	$e^{-t/30}$	V_1 (V)	V_t (V)
0^+	1.0000	0.0000	3.5000
10	0.7165	0.0425	3.4575
30	0.3679	0.0948	3.4052
60	0.1353	0.1297	3.3703
150	0.0067	0.1490	3.3510
∞	0	0.1500	3.3500

At load removal, V_t jumps by +0.20 V to 3.55 V, and $V_t(t) = 3.70 - 0.15e^{-t/30}$. After 30 s it is 3.645 V, and after 60 s it is 3.680 V.

Answer. $\tau_1 = 30$ s; transient as tabulated; steady-state $V_1 = 0.15$ V and $V_t = 3.35$ V.

Interpretation. Figure 2.48 plots this response. A Rint model with only R_0 would predict 3.50 V forever, an error of 150 mV. On a flat OCV curve, that would be a large SoC error, which is why a voltage-based estimator requires the RC branch. Using discrete step $T_s = 1$ s:

$a = e^{-1/30} = 0.96722$, and $V_{1,1} = (0.0015)(0.03278)(100) = 4.92$ mV, which agrees with the exact solution at $t = 1$ s.

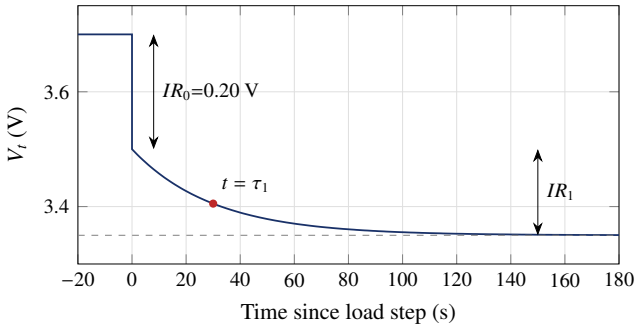


Figure 2.48: Transient response from Example 12.

2.11.7 Second-Order Thevenin Equivalent Circuit Model (2-RC)

A single time constant cannot fit both the fast charge-transfer dynamics (seconds) and slow diffusion (minutes). The 2-RC model of Fig. 2.49 provides two.

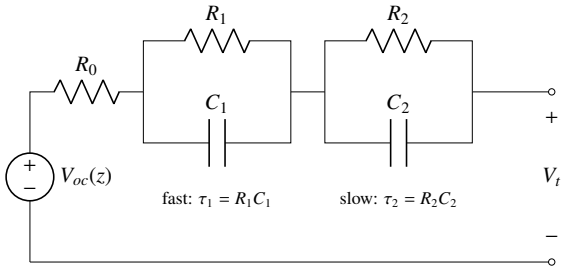


Figure 2.49: Second-order Thevenin (2-RC) model.

2.11.7.1 Governing Equations

Each branch obeys an equation of the same form as (2.87), and the voltages add:

$$\frac{dV_1}{dt} = -\frac{V_1}{R_1C_1} + \frac{I}{C_1}, \tag{2.93}$$

$$\frac{dV_2}{dt} = -\frac{V_2}{R_2C_2} + \frac{I}{C_2}, \tag{2.94}$$

$$V_t = V_{oc}(z) - IR_0 - V_1 - V_2. \tag{2.95}$$

2.11.7.2 Fast and Slow Polarisation Dynamics

The branch with the small time constant τ_1 represents charge-transfer and double-layer effects. The branch with the large τ_2 represents solid and electrolyte diffusion. Under a step I from rest,

$$V_t(t) = V_{oc} - IR_0 - IR_1(1 - e^{-t/\tau_1}) - IR_2(1 - e^{-t/\tau_2}). \tag{2.96}$$

2.11.7.3 Parameter Identification

Relaxation data after a pulse is fitted using a two-term exponential,

$$V_t(t) = V_\infty - A_1 e^{-t/\tau_1} - A_2 e^{-t/\tau_2}, \tag{2.97}$$

giving $R_1 = A_1/I$, $R_2 = A_2/I$, $C_1 = \tau_1/R_1$, $C_2 = \tau_2/R_2$. A well-separated pair of time constants, such as 20 s and 300 s, is needed for a meaningful fit.

2.11.7.4 Comparison with the 1-RC Model

Table 2.12 sets the options side by side. Comparative studies report that the gain from additional RC branches diminishes quickly [26].

Table 2.12: Comparison of equivalent circuit models.

Model	States	Parameters	Captures	Use
Ideal	0	1	Energy only	Rough estimate
Rint	1 (z)	2	Instantaneous drop	Vehicle energy studies
1-RC	2	4	One transient	Common BMS estimator
2-RC	3	6	Fast and slow transients	Accurate SoC, SoP
n-RC	$n+1$	$2n+2$	Spectrum of dynamics	Offline; risk of over-fitting

Algorithm 8 2-RC battery model simulation

Require: I_k, z_0, Q_n, T_s, tables $V_{oc}(z)$, R_0, R_1, C_1, R_2, C_2
Ensure: $V_{t,k}$, z_k
$z \leftarrow z_0$; $V_1 \leftarrow 0$; $V_2 \leftarrow 0$
for $k = 0, \ldots, N$ **do**
  $a_1 \leftarrow e^{-T_s/(R_1C_1)}$; $a_2 \leftarrow e^{-T_s/(R_2C_2)}$
  $V_{t,k} \leftarrow V_{oc}(z) - I_k R_0 - V_1 - V_2$
  $z \leftarrow z - \eta I_k T_s/(3600\, Q_n)$
  $V_1 \leftarrow a_1 V_1 + R_1(1-a_1) I_k$; $V_2 \leftarrow a_2 V_2 + R_2(1-a_2) I_k$
end for

Algorithm 8 extends Algorithm 7 by one state, with a further exact update.

Example 13: 2-RC equivalent circuit model calculation

Problem. A hypothetical cell at rest takes a 100 A discharge step. Neglect SoC change.

Given. OCV 3.70 V, $R_0 = 2$ mΩ, $R_1 = 1$ mΩ, $C_1 = 20\,000$ F, $R_2 = 2$ mΩ, $C_2 = 150\,000$ F.

Required. τ_1, τ_2, V_1, V_2 and V_t at 10, 60, 300 and 900 s; steady-state voltage.

Equations. (2.96).

Solution. $\tau_1 = 20$ s; $\tau_2 = 300$ s; $IR_0 = 0.20$ V; $IR_1 = 0.10$ V; $IR_2 = 0.20$ V.

t (s)	V_1 (V)	V_2 (V)	V_t (V)
10	0.0393	0.0066	3.4541
60	0.0950	0.0363	3.3687
300	0.1000	0.1264	3.2736
900	0.1000	0.1900	3.2100
∞	0.1000	0.2000	3.2000

Answer. $\tau_1 = 20$ s, $\tau_2 = 300$ s; V_t tends to 3.20 V.

Interpretation. Figure 2.50 shows the separation of scales. By 60 s the fast branch has essentially settled while the slow branch has barely begun, so a 1-RC fit to the early part would miss about 0.16 V of later sag. This matters for long uphill climbs and sustained fast charging.

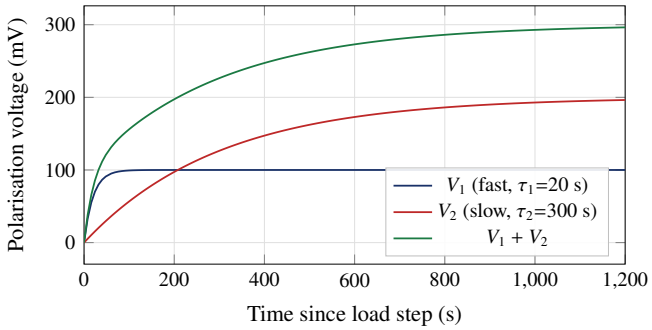


Figure 2.50: Fast and slow polarisation voltages from Example 13.

2.11.8 Higher-Order RC Models

With n branches, $V_t = V_{oc} - IR_0 - \sum_{i=1}^{n} V_i$. Beyond two branches the improvement in accuracy is modest, while the identification becomes ill-conditioned and the computational burden rises, particularly when n is repeated for each cell in the pack. An alternative, fractional-order element, is used for impedance fitting.

2.11.9 State-Space Battery Model

This subsection generalises the model for use in estimators. Figure 2.51 shows the architecture.

State vector. $\mathbf{x} = [z,\ V_1,\ V_2]^\top$.

Input vector. $u = I$, with T as a measured scheduling input.

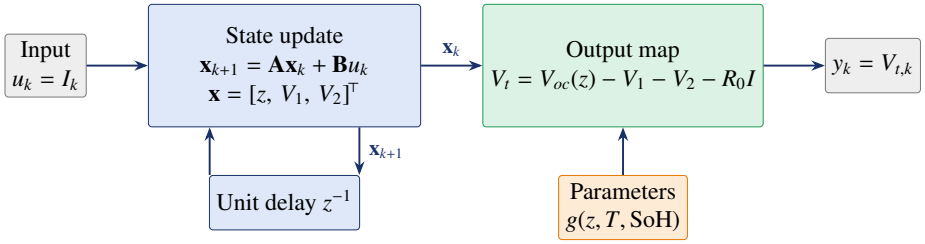


Figure 2.51: Battery model state-space architecture.

State transition model. From (2.50), (2.93) and (2.94):

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1/\tau_1 & 0 \\ 0 & 0 & -1/\tau_2 \end{bmatrix}}_{\mathbf{A}_c} \mathbf{x} + \underbrace{\begin{bmatrix} -\eta/Q_n \\ 1/C_1 \\ 1/C_2 \end{bmatrix}}_{\mathbf{B}_c} u. \tag{2.98}$$

Output and measurement equation. The measurement $y = V_t$ is nonlinear in z and direct in u:

$$y = V_{oc}(z) - V_1 - V_2 - R_0 u. \tag{2.99}$$

Discrete-time formulation. Exact zero-order-hold discretisation of the diagonal system gives

$$\mathbf{x}_{k+1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & a_1 & 0 \\ 0 & 0 & a_2 \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} -\eta T_s/Q_n \\ R_1(1-a_1) \\ R_2(1-a_2) \end{bmatrix} u_k, \quad a_i = e^{-T_s/\tau_i}, \tag{2.100}$$

$$y_k = V_{oc}(z_k) - V_{1,k} - V_{2,k} - R_0 u_k. \tag{2.101}$$

This is the form required by the EKF of Section 2.7, extended to three states. The linearised output row is $\mathbf{C}_k = [dV_{oc}/dz,\ -1,\ -1]$. Note that z is observable from voltage only where $dV_{oc}/dz \neq 0$.

2.11.10 Model Parameter Identification

2.11.10.1 Pulse Discharge Tests and HPPC Testing

A *pulse discharge test* applies a defined pulse at a known SoC and temperature. The *hybrid pulse power characterisation* (HPPC) test, from the U.S. DOE battery test manuals [27], repeats a discharge and regen pulse (10 s each in its classic form) at SoC steps of 10% with rest periods between. It yields a table of resistance against SoC, which leads to pulse power capability. An OCV test at very low current or with long rests provides the $V_{oc}(z)$ table.

2.11.10.2 Parameter Estimation Architecture

Figure 2.52 shows the general structure: the same current enters the cell and the model, and the voltage error drives an optimiser that adjusts the parameter vector $\boldsymbol{\theta}$ to minimise

$$J(\boldsymbol{\theta}) = \sum_{k=1}^{N} \left(V_{t,k} - \hat{V}_{t,k}(\boldsymbol{\theta})\right)^2. \tag{2.102}$$

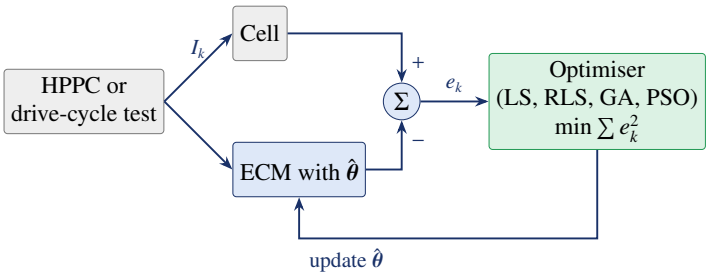


Figure 2.52: Parameter estimation architecture.

2.11.10.3 Least Squares Estimation

Let $y_k = V_{oc} - V_{t,k} = R_0 I_k + V_{1,k}$. Using (2.91), eliminate V_1:

$$y_k = a\, y_{k-1} + R_0\, I_k + \left[R_1(1-a) - aR_0\right] I_{k-1} = \boldsymbol{\phi}_k^{\top} \boldsymbol{\theta}, \tag{2.103}$$

with $\boldsymbol{\phi}_k = [y_{k-1},\ I_k,\ I_{k-1}]^{\top}$. This is linear in $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^{\top}$. Stacking N samples into $\mathbf{Y} = \boldsymbol{\Phi}\boldsymbol{\theta}$ gives the batch solution

$$\hat{\boldsymbol{\theta}} = (\boldsymbol{\Phi}^{\top}\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^{\top}\mathbf{Y}. \tag{2.104}$$

Physical values follow by $a = \theta_1$, $R_0 = \theta_2$, $R_1 = (\theta_3 + \theta_1\theta_2)/(1 - \theta_1)$, and $\tau_1 = -T_s/\ln a$. Least squares is fast and has a unique solution; it suits offline fits when the model is linear in the parameters.

2.11.10.4 Recursive Least Squares

Recursive least squares (RLS) updates the estimate at each sample, with a forgetting factor $\lambda \in (0.95, 1]$, which discounts old data:

$$\mathbf{K}_k = \frac{\mathbf{P}_{k-1}\boldsymbol{\phi}_k}{\lambda + \boldsymbol{\phi}_k^{\top}\mathbf{P}_{k-1}\boldsymbol{\phi}_k}, \tag{2.105}$$

$$\hat{\boldsymbol{\theta}}_k = \hat{\boldsymbol{\theta}}_{k-1} + \mathbf{K}_k(y_k - \boldsymbol{\phi}_k^{\top}\hat{\boldsymbol{\theta}}_{k-1}), \tag{2.106}$$

$$\mathbf{P}_k = \frac{1}{\lambda}(\mathbf{I} - \mathbf{K}_k\boldsymbol{\phi}_k^{\top})\mathbf{P}_{k-1}. \tag{2.107}$$

It is the natural choice for on-board tracking of resistance with ageing and temperature.

2.11.10.5 Genetic Algorithm and Particle Swarm Optimisation

A *genetic algorithm* evolves a population of parameter sets through selection, crossover and mutation, using (2.102) as fitness. *Particle swarm optimisation* moves candidates through the search space, guided by their own best and the swarm's best solutions. Both are derivative-free global methods, useful when the model is nonlinear in its parameters or has many local minima. They are computationally expensive, so they are used offline.

2.11.10.6 Optimisation-Based Parameter Estimation

Gradient-based nonlinear least squares, as in the MATLAB Parameter Estimator [24], refines all parameters over drive-cycle data, and is fast when a good initial guess exists. A practical workflow uses pulse fitting for the initial guess and optimisation for refinement.

2.11.11 Model Validation

A model must be validated against data *not* used for fitting, usually a dynamic profile at various temperatures. The normal metric is the root-mean-square voltage error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(V_{t,k} - \hat{V}_{t,k})^2}. \tag{2.108}$$

Values on the order of 10 mV or lower, excluding the extreme SoC ends, are typically taken as good for ECMs. Validation should cover the full SoC and temperature range, because a model tuned at 25°C and mid-SoC may fail in the cold, where accurate limits matter most.

2.12 Concept of Cell Balancing

2.12.1 Introduction and Need for Cell Balancing

Cells in a series string carry identical current, and yet they do not stay identical. Since every cell must remain inside its voltage limits, the pack stops charging when the first cell reaches $V_{\max}$, and stops discharging when the first reaches $V_{\min}$. *Cell balancing* is the process of equalising the cells so that the pack can use its full capacity.

2.12.2 Cell Voltage Imbalance, Capacity Imbalance and their Causes

Voltage imbalance denotes a spread of terminal voltages, which, at rest, reflects a spread in SoC. *Capacity imbalance* is a spread in the Ah capacity itself. The causes include:

- manufacturing tolerance in capacity, resistance and self-discharge;
- temperature gradients across the pack, since warmer cells age and self-discharge faster;
- unequal ageing and mismatch that grows over the service life;
- parasitic loads of monitoring electronics that differ between modules.

2.12.3 Effects of Imbalance on Pack Performance

Figure 2.53 shows the effect: at the left, cells differ in SoC at rest. At the right, charging has had to stop because C2 has hit $V_{\max}$, though C3 could still accept charge. The discharge is similarly terminated by C3. For a string with cell capacities Q_i and SoCs z_i, the usable pack capacity is

$$Q_{\text{pack}} = \min_i(z_i Q_i) + \min_i\left[(1 - z_i)Q_i\right]. \tag{2.109}$$

The first term is the charge that can still be withdrawn; the second is what can still be added. With identical cells at equal SoC this equals Q. Any spread reduces it. Over time an unbalanced pack also stresses the weakest cell the most, at the highest and lowest voltages, accelerating divergence.

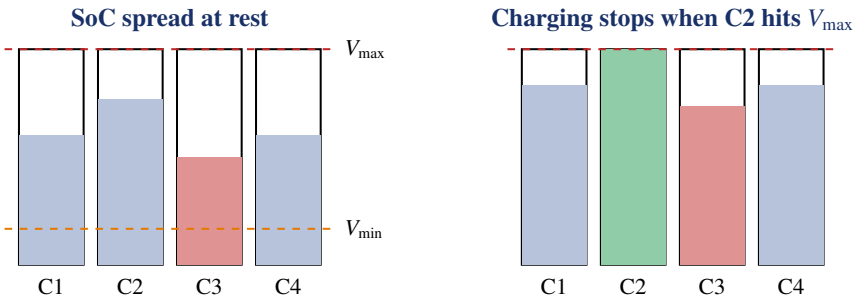


Figure 2.53: Series battery string imbalance diagram.

2.12.4 Passive Cell Balancing (Resistor-Based)

Passive balancing dissipates excess charge from the highest cells as heat. As Fig. 2.54 shows, a resistor R_b and a switch sit across each cell. When the BMS closes Sk, the cell discharges through its resistor, with

$$I_b = \frac{V_c}{R_b}, \qquad P_b = \frac{V_c^2}{R_b}, \qquad t_b = \frac{\Delta z\, Q_c}{I_b}. \tag{2.110}$$

The energy flow is one-way: from the cell into the resistor and then to the coolant. The circuit is cheap, simple and integrated in most AFE ICs, but it wastes energy, generates heat within the pack, and is limited to about 50–200 mA. Because of the low current, it runs for hours, typically near the top of charge or during parking. It corrects SoC differences, but it cannot help with discharge, as it cannot lift a weak cell.

2.12.5 Active Cell Balancing

Active balancing moves charge from high cells to low cells, with a converter, instead of burning it.

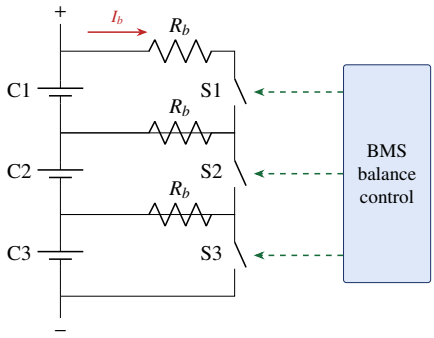


Figure 2.54: Passive balancing circuit.

2.12.5.1 Capacitor-Based and Switched-Capacitor Balancing

In Fig. 2.55, a flying capacitor C_f alternates between two neighbouring cells. In the first switching state (S_a closed), it charges from the higher-voltage cell. In the second (S_b closed), it gives charge to the lower one. The charge moved per cycle is $\Delta q = C_f \Delta V$, so the mean balancing current at switching frequency f_{sw} is

$$I_{\text{avg}} = f_{sw} C_f \Delta V. \tag{2.111}$$

No sensing logic is needed and the circuit is simple, yet the current falls as ΔV falls, so performance is poor on flat OCV curves, and energy passes only between neighbours.

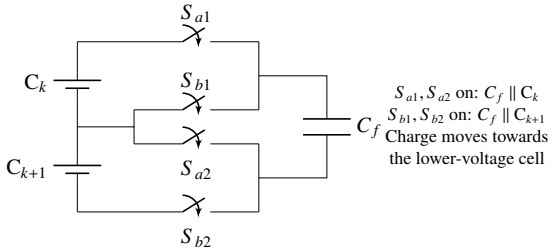


Figure 2.55: Active balancing circuit: switched-capacitor type between adjacent cells.

2.12.5.2 Inductor-Based Balancing

In Fig. 2.56, an inductor is connected first across the high cell by Q_1, storing energy $\frac{1}{2}LI_{pk}^2$, with peak current $I_{pk} = V_c D/(f_{sw} L)$ for duty cycle D. When Q_1 opens, the inductor current continues through Q_2 (or its diode) and flows into the neighbouring cell. It works independently of the voltage difference, and gives a higher current than a capacitor, at the cost of magnetic parts and more control.

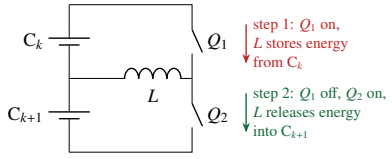


Figure 2.56: Active balancing circuit: inductor-based buck-boost shuttle.

2.12.5.3 Transformer-Based Balancing

Transformer-based circuits (Fig. 2.57) use a shared magnetic core. Energy may be taken from the pack and delivered to the lowest cell (pack-to-cell), or vice versa, so charge moves between any cells in one step, rather than through all neighbours. This gives fast balancing, but with high cost, size and control complexity.

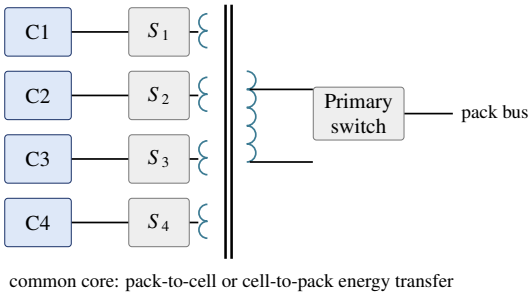


Figure 2.57: Active balancing circuit: multi-winding transformer (schematic).

2.12.6 Cell Balancing Algorithms

2.12.6.1 Voltage-Based and SoC-Based Balancing

Voltage-based balancing equalises measured voltages. It is simple, but voltage under load includes *IR* differences, and on flat curves the voltage hardly reveals SoC, so it is normally done at rest or at the top of charge. *SoC-based* balancing equalises the estimated SoC and can run at any time, so it works better for LFP, but it depends on the estimator. *Capacity-aware* methods go further by accounting for each cell's capacity using (2.109).

2.12.6.2 Balancing Thresholds, Current, Losses and Efficiency

The algorithm needs an on-threshold, an off-threshold to prevent chatter, a temperature guard, and an accurate voltage measurement: thresholds are usually around 5–20 mV, and depend on the AFE accuracy and chemistry. The balancing charge is proportional to current and time, so a large pack with low balancing current may need many hours. Balancing loss is the energy not transferred, and the efficiency of an active balancer is

$$\eta_{\text{bal}} = \frac{E_{\text{received}}}{E_{\text{removed}}}. \tag{2.112}$$

For a passive balancer it is by definition zero: all removed energy becomes heat.

Figure 2.58 shows how cell voltages converge under the algorithm, whereas Fig. 2.59 and Algorithm 9 give the control logic.

Table 2.13 shows that the cost of active balancing must be justified. For packs of well-matched cells, passive balancing suffices and is the overwhelming industrial choice. Active methods become attractive for large-format cells, in second-life packs with wide mismatch, and where capacity is at a premium.

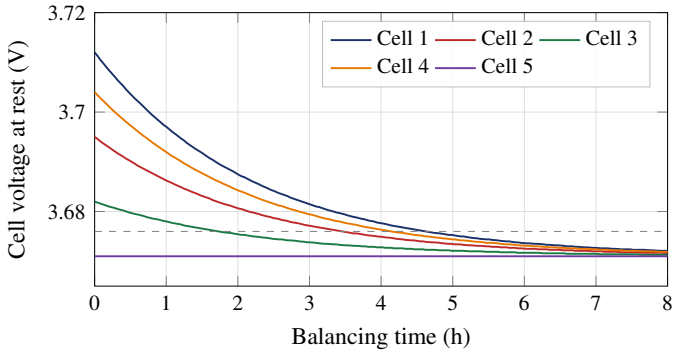


Figure 2.58: Cell voltage imbalance waveform during passive balancing at rest (illustrative): high cells are bled towards the lowest cell.

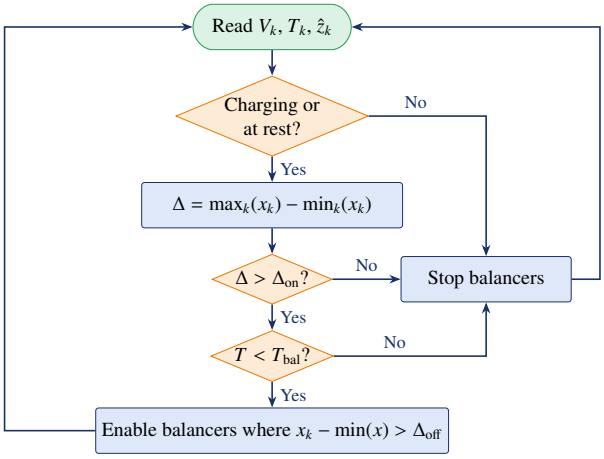


Figure 2.59: Cell balancing control flowchart.

Algorithm 9 Battery cell balancing

Require: Cell quantities x_k (voltage or SoC), thresholds $\Delta_{on} > \Delta_{off}$, T_{bal}
Ensure: Switch commands S_k

loop
  Read x_k, T_k for all cells
  if pack is charging near top, or at rest, **and** $\max_k T_k < T_{bal}$ **then**
    $x_{min} \leftarrow \min_k x_k$; $\Delta \leftarrow \max_k x_k - x_{min}$
    if $\Delta > \Delta_{on}$ **then**
      for all cells k **do**
        $S_k \leftarrow (x_k - x_{min} > \Delta_{off})$
      end for
    end if
  else
    $S_k \leftarrow$ off for all k
  end if
  Pause briefly with all S_k off, and remeasure (avoids sensing error from balance current)
end loop

Table 2.13: Comparison of passive and active balancing techniques (indicative).

Attribute	Passive (resistor)	Switched capacitor	Inductor	Transformer
Energy path	Cell to heat	Adjacent cells	Adjacent cells	Any cell or pack
Typical current	50–200 mA	Depends on ΔV	1–5 A	1–10 A
Efficiency	0% (dissipative)	Moderate to high	High	High
Cost and size	Lowest	Low to moderate	Moderate	Highest
Control	Simple	Simple	Moderate	Complex
Heat in pack	High	Low	Low	Low
Extends discharge	No	Limited	Yes	Yes

Example 14: Passive and active cell balancing calculations

Problem. One 60 Ah cell at 3.7 V is 2% above its neighbours in SoC. Compare (a) passive balancing using a 33 Ω resistor and (b) an active balancer at 2 A with $\eta_{\text{bal}} = 0.85$.

Given. $Q_c = 60$ Ah, $\Delta z = 0.02$, $V_c = 3.7$ V (assume constant).

Required. Charge to move, balancing current, time, resistor power, and energy lost in each.

Equations. (2.110), (2.112).

Solution. Charge to move: $\Delta q = 0.02 \times 60 = 1.2$ Ah; equivalent energy $= 1.2 \times 3.7 = 4.44$ Wh.

(a) $I_b = 3.7/33 = 0.112$ A; $t_b = 1.2/0.112 = 10.7$ h; $P_b = 3.7^2/33 = 0.415$ W; the entire 4.44 Wh is lost as heat.

(b) $t = 1.2/2 = 0.6$ h; energy delivered to the other cells $= 0.85 \times 4.44 = 3.77$ Wh; energy lost $= 0.67$ Wh.

Answer. Passive: 112 mA, 10.7 h, 0.415 W, 4.44 Wh lost. Active: 0.6 h, 0.67 Wh lost.

Interpretation. Active balancing is 18 times faster and recovers most of the energy, but 4.44 Wh is negligible compared with a 70 kWh pack. This is why low-cost passive balancing is usually sufficient. Its real limitation is time, which is why it needs to be scheduled over long charging or parking periods.

2.13 Basic Battery Management Systems (BMS)

2.13.1 Introduction, Objectives and Functions

The BMS is the embedded system that keeps the battery in its safe operating area, makes best use of it, and reports its state to the vehicle [19, 21]. Its objectives are safety, performance and longevity. Figure 2.60 shows the layered functions: hardware at the base, sensing and actuation

above it, with protection and communication, and application-level estimation and control at the top.

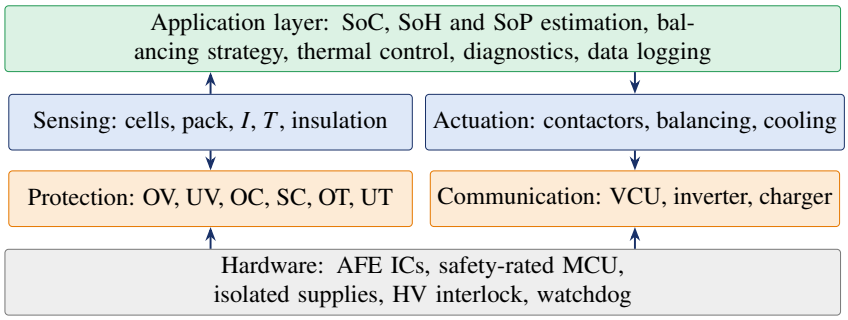


Figure 2.60: Complete BMS architecture by functional layer.

2.13.2 BMS Architecture

2.13.2.1 Centralized BMS

In a centralized BMS, one controller holds all measurement channels, and every cell group is wired to it (Fig. 2.61). It is compact and low in component cost, but the heavy, long sense harness is hard to route, raises failure points, and makes assembly and service difficult. It suits small packs.

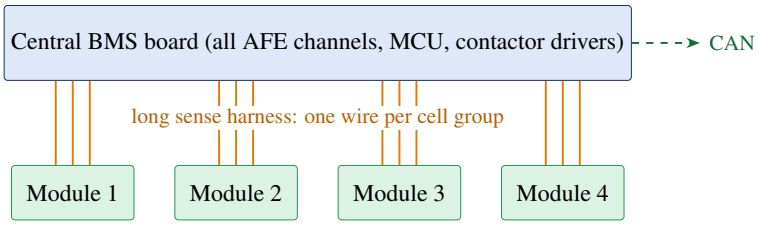


Figure 2.61: Centralized BMS architecture.

2.13.2.2 Distributed BMS

In a distributed (or master-slave) BMS, a cell monitoring unit (CMU) sits on each module, and communicates with a master over a digital link (Fig. 2.62). The sense wires are short, so the harness is light and noise is lower. The modules can be tested independently. It has more electronic parts, and needs a robust isolated link. This is the mainstream for passenger cars. Wireless BMS removes the link harness altogether.

2.13.2.3 Modular BMS

A modular BMS comprises several identical full-function units, of which one acts as the master. It is flexible and scalable, and suits heavy vehicles and stationary systems with many parallel packs, though it may duplicate functions and hence cost.

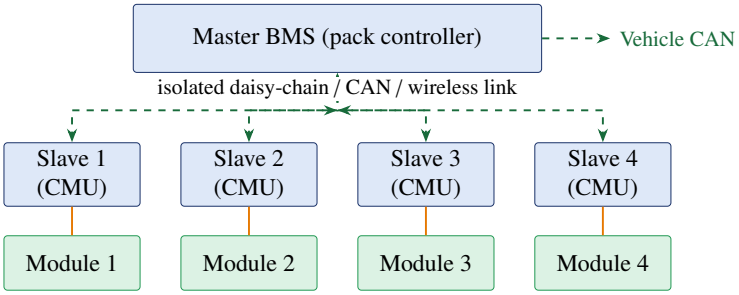


Figure 2.62: Distributed BMS architecture.

2.13.3 Monitoring Functions

2.13.3.1 Cell Voltage Monitoring

Every series group must be measured, since the limit applies to the worst cell. A dedicated analogue front-end (AFE) IC, as in Fig. 2.63, measures stacked cells, typically 12–18 per device, at the input of a precision ADC through an RC filter (R_f, C_f). Accuracy on the order of a few mV is needed, particularly for LFP. AFEs are daisy-chained through isolated links.

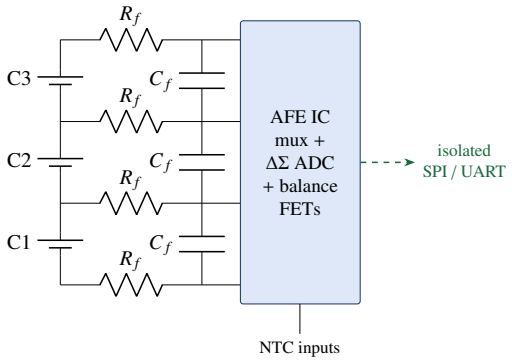


Figure 2.63: Cell monitoring circuit.

2.13.3.2 Pack Voltage, Current and Temperature Monitoring

Pack voltage is measured either side of the contactors, through high-impedance dividers, to verify contactor state and pre-charge. *Current* is measured by a shunt, which is accurate, or a Hall sensor, which is isolated; many packs use both for wide range and for diagnostic redundancy. It must be sampled simultaneously with voltage for the best estimation. *Temperature* is measured by NTC thermistors at selected positions, since it is impractical to sense every cell. Figure 2.64 shows how the measurements come together.

2.13.4 Estimation and Balancing

SoC, SoH and SoP estimation are as described in Section 2.7, and cell balancing as in Section 2.12. The BMS publishes dynamic current limits to the vehicle, using estimated SoP, so that the inverter can stay within limits without the contactors ever having to open.

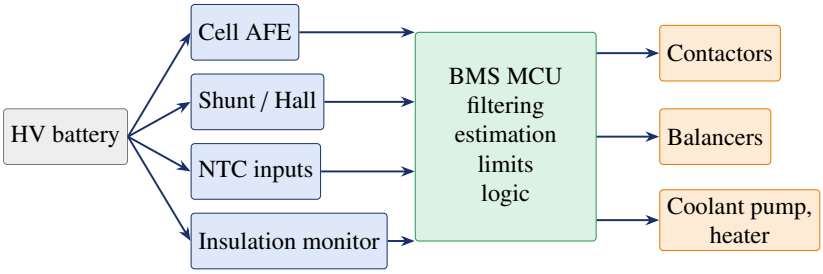


Figure 2.64: BMS sensing and control architecture.

2.13.5 Charge Control and Discharge Control

During charging, the BMS computes the permitted current and target voltage, sends them to the charger, and terminates charge. During driving it sets the discharge and regen current limits as functions of SoC, temperature and the voltage margin of the worst cell.

2.13.6 Contactor Control and Pre-Charge Circuit

Figure 2.65 shows the circuit. The inverter's DC-link capacitor C_{dc} presents almost a short circuit to a suddenly closed contactor, and the inrush would weld its contacts. So the sequence is: close K_{-}, close K_p so that the capacitor charges via R_p, then close K_{+} when the capacitor voltage is near the pack voltage, and finally open K_p. With an uncharged capacitor,

$$v_C(t) = V_{\mathrm{pack}}\left(1 - e^{-t/(R_p C_{\mathrm{dc}})}\right), \qquad i(t) = \frac{V_{\mathrm{pack}}}{R_p} e^{-t/(R_p C_{\mathrm{dc}})}. \tag{2.113}$$

For a 400 V pack, $R_p = 20\ \Omega$ and $C_{\mathrm{dc}} = 6$ mF, then $\tau = 0.12$ s, the peak current is 20 A, and 95% is reached at $3\tau = 0.36$ s (Fig. 2.66). The energy dissipated in R_p equals that stored in the capacitor, $\tfrac{1}{2} C_{\mathrm{dc}} V^2 = 480$ J, regardless of R_p, so the resistor is rated for pulse energy. If the capacitor voltage fails to rise as expected, a fault is flagged.

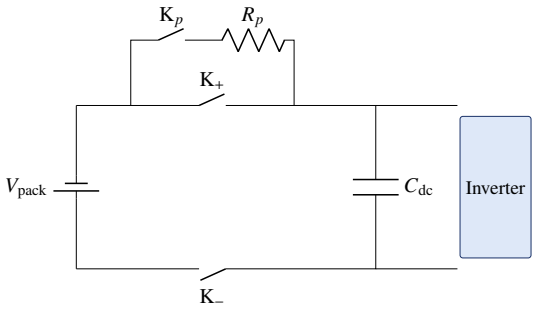


Figure 2.65: Contactors and pre-charge circuit.

2.13.7 Protection Functions

Table 2.14 summarises the protection functions. The action is graded: first a warning and derating, and only then disconnection, because opening the contactors while moving causes a loss of drive power.

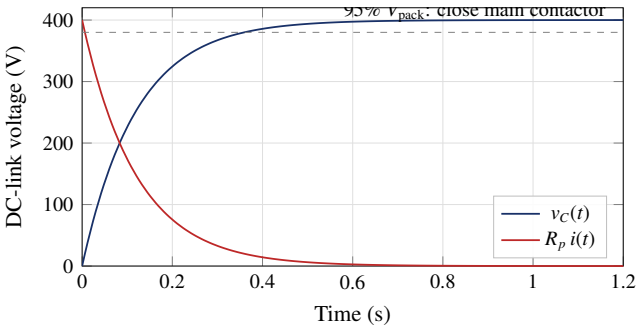


Figure 2.66: Pre-charge transient for the values in the text.

Table 2.14: BMS functions, measured parameters, protection limits and control actions. Limits are indicative for an NMC cell; *protection thresholds are chemistry- and manufacturer-dependent, not universal.*

Function	Measured	Indicative limit	Control action
Overvoltage	Cell voltage	About 4.2–4.25 V	Stop charge and regen; open contactor
Undervoltage	Cell voltage	About 2.5–3.0 V	Limit, then stop discharge
Overcurrent	Pack current	Time-dependent I–t limit	Derate; open contactor
Short circuit	Pack current	Far above peak rating	Fuse or pyro disconnect; open contactor
Overtemperature	Cell temperature	About 55–60°C (discharge)	Derate; maximum cooling; stop
Undertemperature	Cell temperature	Near 0°C (charge)	Reduce or block charge; preheat
Thermal runaway detection	dT/dt, dV/dt, gas, pressure	Rate or combined trigger	Alert driver; maximum cooling; open contactors
Insulation monitoring	R_{iso}	See text	Warn; inhibit closing
HV interlock	Loop continuity	Open circuit	Open contactors
Charge control	V, I, T, SoC	Cell-specific map	Set charger limits
Balancing	Cell V or SoC	5–20 mV	Enable balancers

Overvoltage and undervoltage protection act on the worst cell.

Overcurrent and short-circuit protection have different speeds: an overcurrent limit is managed by software through limits, but a short circuit far exceeds what a contactor can reliably break, so a fuse or pyrotechnic device clears it.

Over- and undertemperature protection consider that charge limits are stricter than discharge limits.

Thermal runaway detection. Because cell temperature is not measured everywhere, modern packs fuse several clues: rapid temperature rise, sudden voltage drop, pack pressure and gas sensors. The aim is to warn occupants early.

Insulation monitoring. The high-voltage bus floats with respect to the chassis, so a single insulation fault is not immediately dangerous, but a second would be. An insulation monitor measures the resistance between each rail and chassis. UN Regulation No. 100 sets minimum values relative to working voltage, namely 100 Ω/V for DC circuits and 500 Ω/V for AC circuits in its usual form [28], while the Indian AIS-038 is aligned with this regulation [29].

2.13.8 Communication Interfaces and CAN

Controller Area Network (CAN) is the usual link between BMS and the vehicle. It is a differential two-wire, multi-master bus with message priority and error detection. Classic CAN typically runs at up to 500 kbit/s on such buses, while CAN FD permits larger, faster payloads. Between AFEs, the common choices are isolated daisy chains and, increasingly, wireless links.

2.13.9 BMS Interaction with EV Control Systems

Figure 2.67 shows the network. The vehicle control unit (VCU) requests a drive or charge mode. The BMS answers with SoC, power limits and fault status. The inverter respects the current limits when converting the driver's torque request. During charging, the BMS talks to the on-board charger, or through the charge controller to a DC station, and requests target voltage and current. The thermal system receives coolant requests.

2.13.10 Battery Safety and Fault Diagnosis

Fault diagnosis must distinguish a genuine hazard from a sensor fault. The flowchart of Fig. 2.68 shows how: each check has a debounce time, and faults are classed by severity. A warning leads to derating and a diagnostic trouble code; a critical fault leads to contactor opening. BMS software is normally developed in line with the functional-safety approach of ISO 26262. Algorithm 10 gives the basic operating loop.

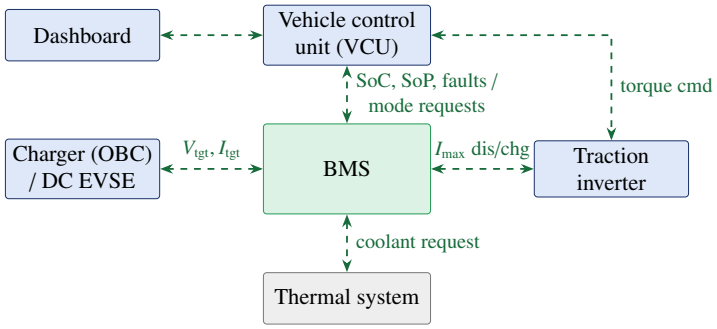


Figure 2.67: BMS interaction with inverter, charger and vehicle controller.

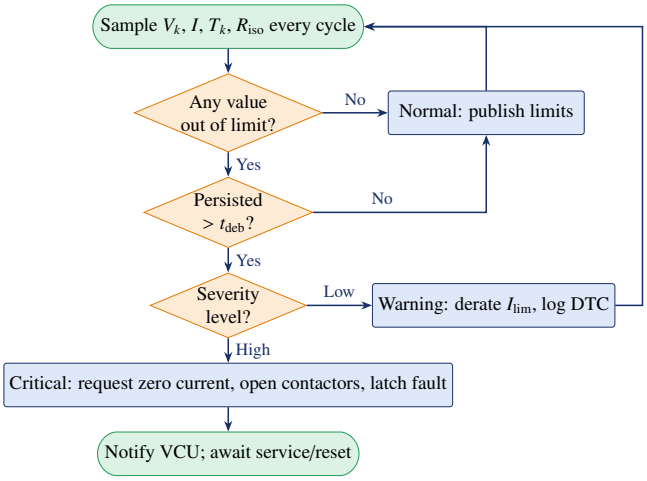


Figure 2.68: BMS protection and fault management flowchart.

Algorithm 10 Basic BMS monitoring and protection

Require: Limits $V_{\min}, V_{\max}, I_{\max}, T_{\min}, T_{\max}, R_{\text{iso,min}}$, debounce t_{deb}
Ensure: Contactor commands, current limits, status messages

Self-test; verify HVIL and R_{iso}; if faulted, keep contactors open
On request: close K$_-$; close K$_p$; wait for $v_C > 0.95V_{\text{pack}}$; close K$_+$; open K$_p$
loop
    Sample V_k, I, T_k, R_{iso}
    Update $\hat{z}$, SoH and SoP; compute $I_{\text{dis,lim}}$ and $I_{\text{chg,lim}}$
    if any limit exceeded **then** increment fault timer
    else clear fault timer
    end if
    if fault timer $> t_{\text{deb}}$ **then**
        if severity high **then**
            Request zero current; open contactors; latch fault
        else
            Derate limits; log diagnostic code
        end if
    end if
    Run balancing logic (Algorithm 9) and thermal control
    Transmit status and limits on CAN
end loop

Unit 2 (continued): Battery Chargers and Charging Systems

The energy that the battery delivers must be returned safely, efficiently and conveniently. Charging joins three systems that are designed by different parties: the grid, the charging equipment and the vehicle. Their interaction is governed by power electronics, control, communication and regulation, in that order of this chapter.

3.1 Basic Requirements for Charging Systems

3.1.1 Introduction and Purpose of EV Battery Chargers

A charger converts electrical power from the supply to the controlled DC voltage and current demanded by the battery. Its purposes are to restore energy in the required time, without harming the battery or the grid, and to protect the user. Note a matter of terminology: the equipment on the wall, the *electric vehicle supply equipment* (EVSE), is a charger only in the DC case. In AC charging, the EVSE merely delivers protected AC power, while the actual converter is in the vehicle.

3.1.2 Basic Charging Requirements

- **Charging power, voltage and current.** The charger must span the whole battery voltage window at the required current, $P = V_b I_b$. Both output limits matter: for a fixed current limit, a low-voltage pack charges at a lower power.
- **Charging efficiency.** This is the fraction of grid energy that reaches the battery, including AC-DC and DC-DC stages: $\eta_{\text{chg}} = E_{\text{bat}}/E_{\text{grid}}$.
- **Charging time.** From (3.13) below, time is set by energy and power, and, for chemistry reasons, by tapering at high SoC.
- **Battery chemistry compatibility.** Voltage limits and permitted rates are chemistry-specific, and the charger follows the BMS request.
- **Charging safety and isolation.** Grid and battery must be isolated in most designs by a high-frequency transformer, so that a single fault cannot put the chassis at a dangerous potential.
- **Power factor.** The charger must draw near-sinusoidal current in phase with the voltage, which requires an active power factor correction stage.

- **Harmonic distortion.** Rectifier currents contain harmonics, which load the distribution system, so total harmonic distortion must be kept low. For the current, it is defined by

$$\mathrm{THD}_I = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_1}, \tag{3.1}$$

where I_h is the rms value of harmonic h and I_1 the fundamental. The true power factor is then

$$\mathrm{PF} = \frac{\cos \varphi_1}{\sqrt{1 + \mathrm{THD}_I^2}}. \tag{3.2}$$

- **Bidirectional charging.** A charger with a reversible power stage can return energy, which is treated in Section 3.2.3.4.
- **Communication between EV and station.** Both parties must agree on limits, start and stop.
- **Battery temperature monitoring.** The BMS reports cell temperature, and the charging current is reduced accordingly, or cooling is applied.
- **Overvoltage and overcurrent protection.** These apply on both AC and DC sides, independent of the control loop.
- **Environmental conditions.** Enclosures must withstand heat, rain, dust, and voltage fluctuation. Batteries must be preheated in cold conditions or precooled for fast charging in high ambient temperature.

3.1.3 Onboard and Offboard Charging Systems

Figure 3.1 shows the overall charging architecture and the essential distinction. An *onboard charger* (OBC) travels in the vehicle, so its power is limited by mass, cost and cooling, and it uses AC from an EVSE. An *offboard charger* remains at the station, where mass and cooling are not as restricted, so it can feed DC at high power directly to the battery, bypassing the OBC.

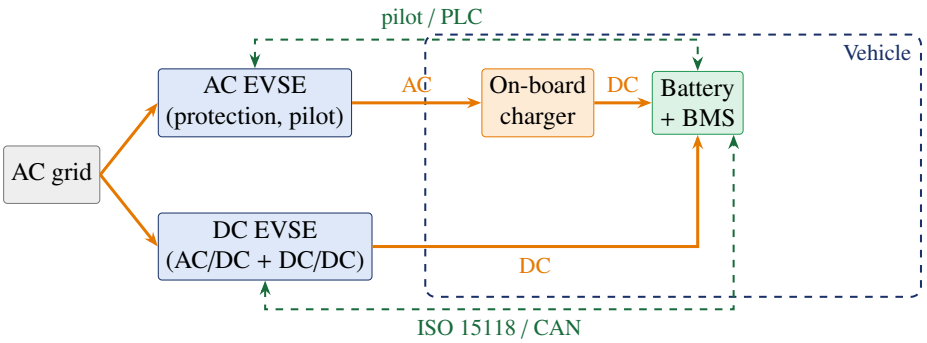


Figure 3.1: EV charging architecture: AC path through the onboard charger, DC path from an offboard charger.

3.2 Classification of Charging Architectures

3.2.1 Onboard Charger (OBC)

3.2.1.1 Working Principle

An OBC converts single- or three-phase AC to regulated DC. Figure 3.2 shows the block diagram.

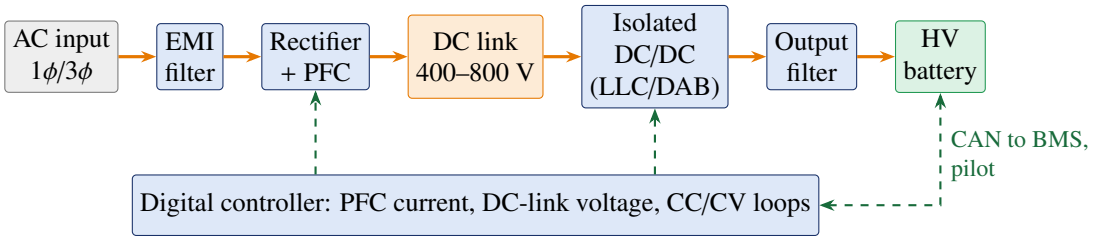


Figure 3.2: Onboard charger block diagram.

- **AC input and EMI filter.** The filter limits conducted emissions into the grid and protects against surges.
- **Rectifier and power factor correction (PFC).** A rectifier and boost stage shape the input current to follow the voltage, and regulate a DC link.
- **DC-DC conversion and isolation.** A high-frequency isolated converter, typically an LLC resonant converter or dual active bridge (DAB), adapts the DC-link voltage to the battery voltage, and provides galvanic isolation.
- **Battery-side DC output.** The filtered output feeds the battery through contactors.

Figure 3.3 shows these power conversion stages in a classical two-stage implementation. The boost PFC stage makes the input current sinusoidal; its controller has an inner current loop and an outer voltage loop. The H-bridge, resonant inductor and transformer give the isolation. The high frequency of the transformer makes it small enough to carry on board.

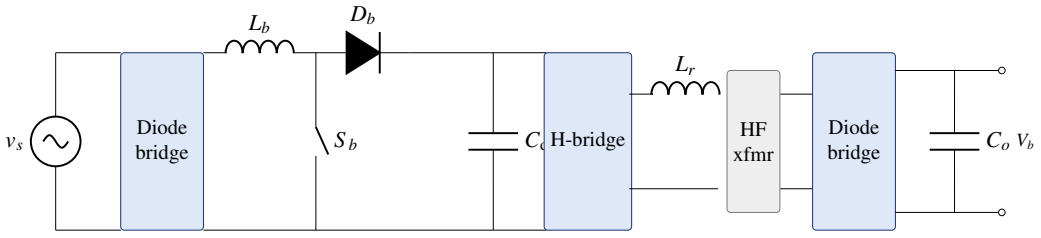


Figure 3.3: OBC power conversion stages: boost PFC followed by isolated DC-DC converter (simplified).

3.2.1.2 Single-Phase, Three-Phase and Bidirectional OBCs

Single-phase OBCs serve domestic connections and are commonly around 3.3 kW or 7.4 kW. The input power pulsates at twice line frequency, so the DC link needs a large capacitor. *Three-phase* OBCs, typically 11 kW or 22 kW, use a three-phase PFC stage and have a smoother power flow. *Bidirectional* OBCs replace diodes with active switches, typically using a totem-pole PFC and DAB, so power can flow from the battery to the AC side.

3.2.2 Offboard Charger

An offboard or DC charger, shown in Fig. 3.4, takes three-phase AC from a low-voltage or medium-voltage connection and converts it in two stages: an active front-end rectifier with PFC, then isolated DC-DC modules. For *high-power charging*, several modules are paralleled, and connections from the MV level may use a dedicated transformer. The *charging cable and connector* are often liquid-cooled at the highest currents, to keep the cable manageable. On the *battery-side interface*, the DC is connected straight to the pack through vehicle-side contactors. The charger controller exchanges limits with the BMS and, separately, connects to a back-end for authorisation and billing.

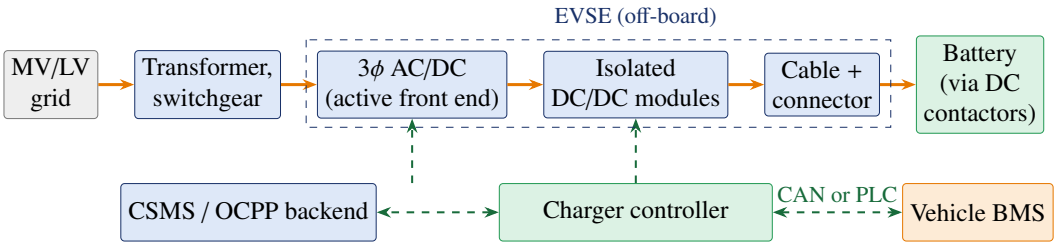


Figure 3.4: Offboard charger architecture, with power (orange) and communication and control (green) paths.

3.2.3 Other Charging Architectures

3.2.3.1 Conductive Charging

All the above are conductive charging, in which a metallic connection carries power. It is efficient and simple, and it is the worldwide norm.

3.2.3.2 Wireless Charging: Static and Dynamic

Wireless (inductive) charging transfers power between a ground coil and a vehicle coil through a resonant, high-frequency magnetic field. Compensation capacitors tune the coils so that a modest coupling across an air gap can transfer power. *Static* charging operates in a parking bay and offers convenience. *Dynamic* charging supplies power to a moving vehicle via coils embedded in the road, which might reduce the battery size, but it has high infrastructure cost and is still at the pilot stage. Efficiency depends strongly on alignment.

3.2.3.3 Battery Swapping

In battery swapping, a discharged pack is replaced by a charged one in minutes. It separates charging from the vehicle's parked time, and is attractive for two- and three-wheeler fleets in India. Packs may be charged off-peak. It needs standard packs, a stock of extra batteries, and a business model for ownership. IS 17896 addresses swap systems [30].

3.2.3.4 Bidirectional Charging: V2G, V2H, V2L and V2B

Bidirectional charging treats the vehicle battery as a distributed energy resource:

- **Vehicle-to-Grid (V2G):** energy is exported to the grid under aggregator or utility control, for peak support or frequency services.
- **Vehicle-to-Home (V2H):** the vehicle supplies a household, usually in a backup role, or to shift solar energy.
- **Vehicle-to-Building (V2B):** the principle of V2H is scaled for a commercial building, chiefly to shave peak demand.
- **Vehicle-to-Load (V2L):** the vehicle offers an AC socket to power tools or appliances.

The cost of V2G is the extra cycling of the battery, and the need for grid-code compliance, anti-islanding and metering, together with the high-level communication of ISO 15118-20 [31]. Table 3.1 compares the architectures.

Table 3.1: Comparison of charging architectures (qualitative).

Architecture	Typical power	Converter location	Strength	Limitation
Onboard AC	3–22 kW	Vehicle	Low infrastructure cost; home use	Slow; adds mass
Offboard DC	15–400 kW	Station	Fast charging	Cost; grid impact
Static wireless	3–11 kW	Both	Convenience	Efficiency; alignment; cost
Dynamic wireless	Pilot-stage values	Road and vehicle	Charging while moving	Infrastructure cost
Battery swapping	Minutes per swap	Swap station	Short downtime	Standardisation; spare stock
Bidirectional (V2X)	As per charger	Vehicle or station	Grid and home services	Added cycling, regulation

Example 15: EV onboard charger power and current calculation

Problem. (a) An 11 kW three-phase OBC on a 415 V supply has $\eta = 0.94$ and PF = 0.99. Find the input and output quantities at a 400 V battery. (b) A single-phase OBC draws 32 A

at 230 V, with the same efficiency and PF. Find its battery power. (c) Find the time to raise a 60 kWh battery from 20% to 80% with each.

Given. As above (hypothetical charger); battery current is taken at 400 V; battery charging losses are ignored in part (c).

Required. P_{in}, S, line current, battery current, and charging time.

Equations. $P_{\text{in}} = P_{\text{out}}/\eta$, $S = P_{\text{in}}/\text{PF}$, $I_L = S/(\sqrt{3}V_L)$, and $t = \Delta E/P_{\text{out}}$.

Solution. (a) $P_{\text{in}} = 11/0.94 = 11.70$ kW, $S = 11.70/0.99 = 11.82$ kVA, $I_L = 11\,820/(\sqrt{3} \times 415) = 16.4$ A, $I_b = 11\,000/400 = 27.5$ A.

(b) $S = 230 \times 32 = 7.36$ kVA, $P_{\text{out}} = 7.36 \times 0.99 \times 0.94 = 6.85$ kW, $I_b = 6849/400 = 17.1$ A.

(c) $\Delta E = 0.60 \times 60 = 36$ kWh; $t_a = 36/11 = 3.27$ h; $t_b = 36/6.85 = 5.26$ h.

Answer. (a) 11.70 kW, 11.82 kVA, 16.4 A per line, 27.5 A DC. (b) 6.85 kW, 17.1 A DC. (c) 3.27 h and 5.26 h.

Interpretation. "7.4 kW" is the input rating; the battery receives less. Three-phase charging at modest current gives more power, which shows why three-phase connections are preferred where available.

3.3 Charging Controls

3.3.1 Overview of Control

The charger controls either the output current or the output voltage. The BMS supplies a current limit and a voltage limit; the actual control mode is whichever limit is active. Using the discharge-positive convention, a charging current has $I < 0$. In this chapter it is more convenient to use the positive magnitude $I_c = -I$.

3.3.2 Constant Current (CC) Charging

The charger holds I_c at a set value I_{cc}. The SoC then rises linearly,

$$z(t) = z_0 + \frac{\eta_C I_{\text{cc}} t}{Q_n}, \tag{3.3}$$

where Q_n is in Ah with t in h. The terminal voltage is

$$V_t(t) = V_{oc}(z(t)) + I_{\text{cc}} R_0 + V_p(t), \tag{3.4}$$

in which $V_p = I_{\text{cc}} R_1 (1 - e^{-t/\tau_1})$ is the polarisation magnitude. Figure 3.5 illustrates this: the current is flat while voltage climbs after an initial step. Pure CC would drive the cell past $V_{\max}$, as the figure implies, so it needs voltage termination. Algorithm 11 sets out the logic.

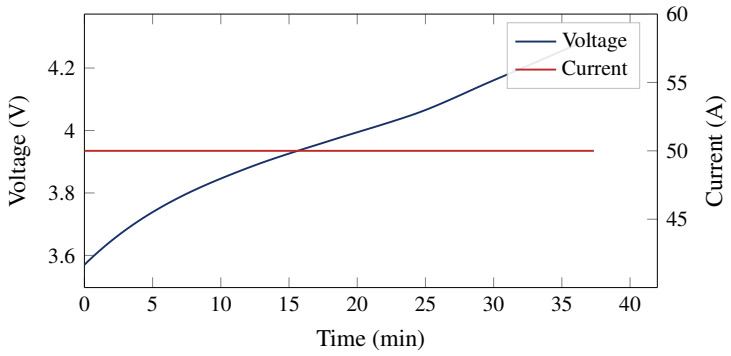


Figure 3.5: CC charging current and voltage (simulated, 50 Ah cell, 1C).

Algorithm 11 CC charging control

Require: I_{cc}, V_{max}, target SoC z_{tgt}, gains K_p, K_i
Ensure: Duty command d
1: Pre-checks: BMS ready, T in range
2: **while** $V_t < V_{max}$ **and** $\hat{z} < z_{tgt}$ **do**
3: $e \leftarrow I_{cc} - I_{c,meas}$; $\xi \leftarrow \xi + eT_s$
4: $d \leftarrow K_p e + K_i \xi$ (limit d with anti-windup)
5: **end while**
6: Stop or hand over to voltage control

3.3.3 Constant Voltage (CV) Charging

The charger holds the terminal voltage at V_{cv}. The current follows from (2.75):

$$I_c(t) = \frac{V_{cv} - V_{oc}(z) - V_p(t)}{R_0}. \tag{3.5}$$

Since OCV and polarisation rise as charge enters, I_c decays naturally toward zero. If OCV is linearised as $V_{oc} \approx V_0 + k_z z$, an effective capacitor $C_{eq} = Q_n/k_z$ results, and the taper is approximately exponential:

$$I_c(t) \approx I_0 e^{-t/\tau_{cv}}, \qquad \tau_{cv} \approx R_{ch} C_{eq}. \tag{3.6}$$

Here R_{ch} denotes the effective charging resistance. The charge accepted in time T is then

$$\Delta Q = \int_0^T I_c \, dt = I_0 \tau_{cv} (1 - e^{-T/\tau_{cv}}). \tag{3.7}$$

Applying CV directly to a discharged cell would draw a very large inrush, since the only limit is R_0, as shown by the current limit plateau in Fig. 3.6. So pure CV is never used on its own for a lithium-ion EV battery, except with a current clamp.

3.3.4 Constant Current–Constant Voltage (CC-CV) Charging

CC-CV is the standard method for lithium-ion, because it combines the fast charge of CC with the safe voltage limit of CV. Figure 3.7 shows the profile: phase 1 at constant current, then

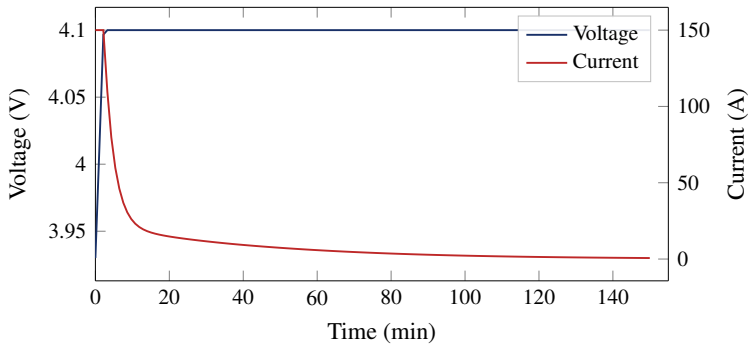


Figure 3.6: CV charging current and voltage at 4.1 V with a 3C current clamp (simulated).

Algorithm 12 CV charging control

Require: V_{cv}, I_{lim}, cut-off I_{cut}, gains K_{pv}, K_{iv}
Ensure: Current reference I^*
1: **repeat**
2: $e_v \leftarrow V_{\text{cv}} - V_t$; $\xi_v \leftarrow \xi_v + e_v T_s$
3: $I^* \leftarrow \min\left(K_{pv} e_v + K_{iv}\xi_v,\ I_{\text{lim}}\right)$
4: Pass I^* to inner current loop
5: **until** $I_{c,\text{meas}} < I_{\text{cut}}$
6: Open output; report end of charge

phase 2 at constant voltage.

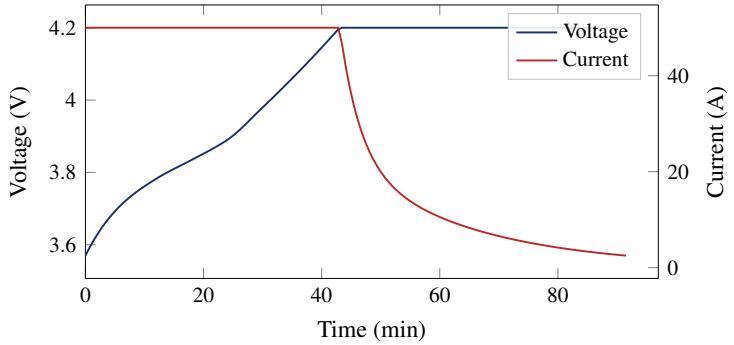


Figure 3.7: CC-CV charging profile (simulated, 50 Ah cell, 1C to 4.2 V, cut-off C/20).

Transition from CC to CV. The switch occurs at the instant V_t reaches $V_{\max}$. At that time the OCV is still below $V_{\max}$ by the overpotential $I_{\text{cc}}(R_0 + R_1)$. Therefore the switch-over SoC z_{sw} satisfies

$$V_{oc}(z_{\text{sw}}) + I_{\text{cc}} R_0 + V_p = V_{\max}. \tag{3.8}$$

This has an important consequence. Higher current, higher resistance (from ageing or cold), and hence greater overpotential, all mean that the CV stage starts at lower SoC, which is why a fast charge spends a large part of its final stage tapering. During the CV stage, the charger is effectively a voltage source, and the cell's overpotential drains away.

Control structure. A single cascaded loop handles both phases: the outer voltage regulator produces a current reference, which is clamped at I_{cc}. While $V_t < V_{\max}$, the outer loop saturates and the system is in CC; when the voltage reaches the limit, it comes out of saturation, and the transition is automatic and smooth. Thus, the **current regulation** and **voltage regulation** are nested:

$$I^* = \min\Big(I_{\mathrm{cc}},\ K_{pv}(V_{\max} - V_t) + K_{iv}\int (V_{\max} - V_t)\,dt\Big), \tag{3.9}$$

$$d = K_{pi}(I^* - I_c) + K_{ii}\int (I^* - I_c)\,dt, \tag{3.10}$$

where d is the converter duty or phase command.

Charging time estimation. The CC phase lasts

$$t_{\mathrm{cc}} = \frac{(z_{\mathrm{sw}} - z_0)Q_n}{\eta_C I_{\mathrm{cc}}}, \tag{3.11}$$

and, from (3.6), the CV phase, terminated at I_{cut}, lasts

$$t_{\mathrm{cv}} = \tau_{\mathrm{cv}} \ln \frac{I_{\mathrm{cc}}}{I_{\mathrm{cut}}}, \tag{3.12}$$

so that the overall time is $t_{\mathrm{cc}} + t_{\mathrm{cv}}$. When only a rough figure is needed for a constant-power charger,

$$t \approx \frac{(z_f - z_0)E_{\mathrm{nom}}}{\eta_{\mathrm{chg}} P_{\mathrm{chg}}}. \tag{3.13}$$

Termination criteria. Charging ends on any of: current below cut-off, typically C/20 to C/10; a target SoC; a timer; or a temperature or voltage fault.

Algorithm 13 CC-CV charging control

Require: I_{cc}, $V_{\max}$, I_{cut}, $t_{\max}$, z_{tgt}
Ensure: Charged battery; end-of-charge status
Verify connection, isolation and BMS readiness; ramp current
mode $\leftarrow$ CC
repeat
    Read V_t (highest cell voltage scaled), I_c, T, $\hat{z}$
    if mode = CC **and** $V_t \geq V_{\max}$ **then** mode $\leftarrow$ CV
    end if
    if mode = CC **then** $I^* \leftarrow I_{\mathrm{cc}}$
    else $I^* \leftarrow$ output of voltage regulator (3.9)
    end if
    Run current regulator (3.10)
until $I_c < I_{\mathrm{cut}}$ **or** $\hat{z} \geq z_{\mathrm{tgt}}$ **or** $t > t_{\max}$ **or** fault
Ramp down current; open contactors

Figure 3.8 shows SoC during the same charge. It rises linearly during CC, and it then bends over in CV: for the simulated case, around 80% of capacity is back in under half of the total

charge time. Figure 3.9 shows power, which rises during CC as the voltage climbs, peaks at the transition, and then falls with current. This is why a charger's advertised peak power is seen for only part of the session, and why a manufacturer's "10–80%" time is more informative.

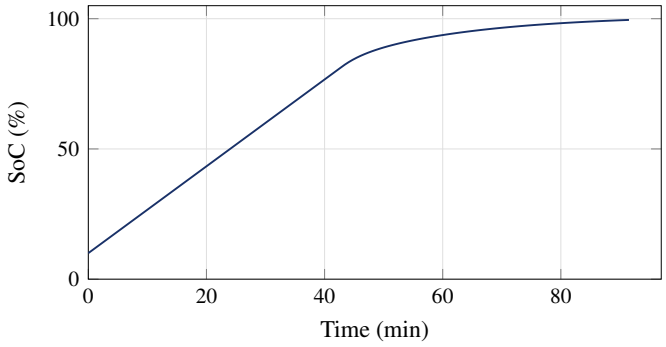


Figure 3.8: Battery SoC during charging (simulated).

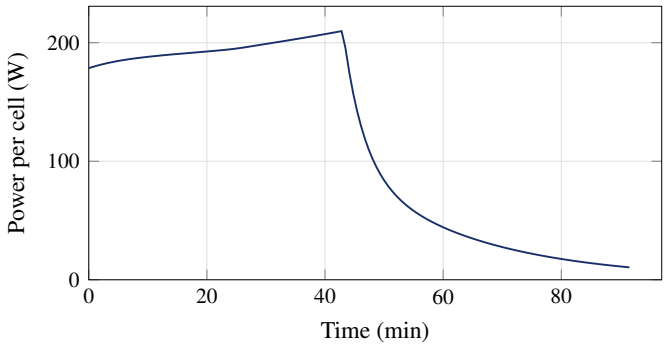


Figure 3.9: Charging power versus time (simulated).

Figure 3.10 shows temperature from (2.38). The rise follows the heating $I_c^2 R$ which is greatest in CC, and falls in the CV stage, when loss declines. Stronger cooling reduces the peak, which is the reason fast chargers need active thermal management.

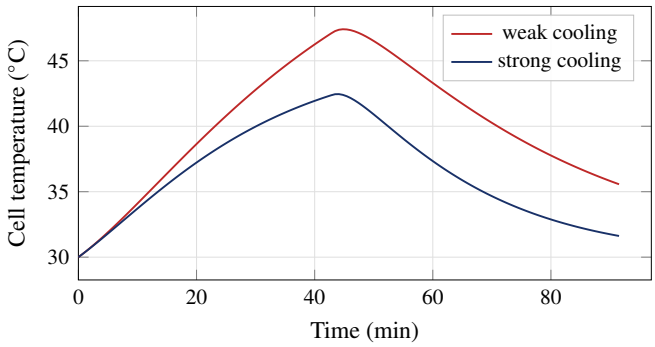


Figure 3.10: Battery temperature during charging with weak and strong cooling (simulated).

Example 16: CC-CV charging time calculation

Problem. A battery is charged from 20% SoC at 1C. The switch to CV occurs at 80% SoC.

The CV phase is cut off at C/20, by which time the battery is full. Assume exponential taper and $\eta_C = 1$.

Given. $z_0 = 0.20$, $z_{\text{sw}} = 0.80$, $I_{\text{cc}} = 1\text{C}$, $I_{\text{cut}} = 0.05\text{C}$.

Required. τ_{cv}, CC time, CV time, and total time.

Equations. (3.7), (3.11), (3.12).

Solution. CC: $t_{\text{cc}} = (0.80 - 0.20)Q/(1\ Q) = 0.60$ h. In CV, the remaining charge is $0.20\,Q$. With $e^{-T/\tau} = I_{\text{cut}}/I_{\text{cc}} = 0.05$, (3.7) gives

$$0.20\,Q = (1\,\text{h}^{-1}\ Q)\,\tau_{\text{cv}}(1 - 0.05) \quad \Rightarrow \quad \tau_{\text{cv}} = 0.2105\ \text{h}.$$

Then $t_{\text{cv}} = 0.2105 \ln 20 = 0.631$ h, and $t = 0.60 + 0.631 = 1.23$ h.

Answer. $\tau_{\text{cv}} = 0.2105$ h; CC 36 min; CV 37.8 min; total 1.23 h (about 74 min).

Interpretation. The last 20% of the charge takes slightly longer than the first 60%. This is why fast-charge stops are usually planned to about 80%. The result is independent of battery size, because the current is in C-rate.

3.3.5 Trickle Charging, Multi-Stage and Pulse Charging

Trickle charging. A small continuous current compensates for self-discharge. It suits lead-acid and NiMH float service. It is *not* used in the usual sense for lithium-ion cells, because they cannot absorb overcharge; however, a small current is used to recover deeply discharged cells to the safe voltage, which is often called pre-charge or trickle.

Multi-stage charging. A series of decreasing constant currents, sometimes called multi-stage constant current, is used, with each stage ending at a voltage limit. It approximates the battery's permissible current, with a higher rate at low SoC where plating risk is low.

Pulse charging. Current is applied in pulses separated by rests or brief discharges, aiming to allow concentration gradients to relax. Reported benefits vary [32], so it should be seen as an option to verify experimentally for each cell.

3.3.6 Fast Charging and Opportunity Charging

Fast charging at rates of about 1–3C or more is limited by lithium plating, heat, and the power available. Modern protocols use a stored map of allowed current against SoC and temperature [32]. *Opportunity charging* is a short top-up whenever the vehicle stops, such as a bus at a terminal. It keeps the battery at partial SoC, and permits a smaller pack, which suits LFP and LTO chemistries.

3.3.7 Temperature-Dependent and SoC-Dependent Charging

The current limit at any moment is the most restrictive of all the constraints:

$$I^{*} = \min\left(I_{\mathrm{cc}},\; I_{\mathrm{EVSE}},\; I_{\mathrm{lim}}(T),\; I_{\mathrm{lim}}(z),\; I_{v}\right), \tag{3.14}$$

where $I_{\mathrm{lim}}(T)$ is a temperature derating function, for example the trapezoid

$$I_{\mathrm{lim}}(T) = I_{\max} \times \begin{cases} 0 & T < T_1 \\ (T - T_1)/(T_2 - T_1) & T_1 \le T < T_2 \\ 1 & T_2 \le T < T_3 \\ (T_4 - T)/(T_4 - T_3) & T_3 \le T < T_4 \\ 0 & T \ge T_4 \end{cases} \tag{3.15}$$

The values $T_1 < T_2 < T_3 < T_4$ are cell-specific, and I_v is the output of the voltage loop. $I_{\mathrm{lim}}(z)$ tapers current towards high SoC. Algorithm 14 implements (3.14).

Algorithm 14 Basic temperature-dependent charging control

Require: $T_1, \ldots, T_4$, map $I_{\mathrm{lim}}(z)$, I_{EVSE}, $V_{\max}$
Ensure: I^{*}; thermal requests

loop
  Read $T_{\max}$, $T_{\min}$ of cells, V_t, $\hat{z}$
  if $T_{\min} < T_2$ **then** request heating
  end if
  if $T_{\max} > T_3$ **then** request cooling
  end if
  Compute $I_{\mathrm{lim}}(T)$ using $T_{\min}$ for low limit and $T_{\max}$ for high limit
  Compute I_v from the voltage regulator
  $I^{*} \leftarrow \min(I_{\mathrm{lim}}(T), I_{\mathrm{lim}}(z), I_{\mathrm{EVSE}}, I_v)$
  if $I^{*} = 0$ for longer than timeout **then** stop session
  end if
end loop

3.3.8 Battery Protection, Power Management and Efficiency During Charging

Battery protection remains the BMS's task: it can open its contactors if the charger fails to obey its limits. *Charging power management* shares a site's limited supply among several vehicles, with dynamic load management according to grid signals. *Charging efficiency* is

$$\eta_{\mathrm{chg}} = \frac{\int V_t I_c \, dt}{\int P_{\mathrm{grid}} \, dt} = \eta_{\mathrm{ACDC}}\, \eta_{\mathrm{DCDC}}, \tag{3.16}$$

and the chemical storage efficiency $\eta_{P,\mathrm{ch}}$ of (2.30) applies further on the way into the cells. Thus *charging power* at the battery, $P = V_t I_c$, always exceeds the power actually stored.

3.4 Current Regulations and Charging Power Standards

This section interprets the syllabus topic as the charging-current limits, electrical regulations and requirements governing EV charging equipment. A careful distinction between kinds of document is essential, and Table 3.2 provides the framework.

Table 3.2: Kinds of requirement that govern a charging installation.

Kind	Nature	Examples
Statutory requirement	Legally binding through an Act, rule or regulation	CEA safety regulations; Central Motor Vehicle Rules with AIS references
Policy guideline	Issued by a ministry to direct implementation	Ministry of Power EV charging guidelines
Technical standard	Consensus document; binding only when a law or contract adopts it	IEC 61851, IS 17017, ISO 15118
Connector specification	Defines interface geometry and ratings	IEC 62196 configurations, GB/T 20234
Manufacturer limit	Datasheet value for one product	Maximum OBC or cable current
Engineering practice	Accepted design habit, not a requirement	Derating margins, 80% target

Note on sources. IEC, ISO and BIS standards are copyrighted, and many clauses are not freely available. The values below come from public summaries of these documents as cited, and from the ratings commonly stated for them. Before any design or compliance decision, the current edition of the actual document must be consulted. Editions noted here were checked in November 2026.

3.4.1 Charging Current Limitations

3.4.1.1 AC Charging Current

The current in AC charging is limited by the *lowest* of four values: the installation's circuit rating, the EVSE rating, the cable rating and the vehicle's OBC rating. In Mode 3, the EVSE tells the vehicle its limit through the control pilot, and the OBC must not exceed it. IEC 61851-1 defines charging modes as summarised in Table 3.3 [33].

3.4.1.2 DC Charging Current and Voltage–Current Operating Limits

In DC charging, the station and vehicle exchange their maximum voltage, current and power, and the vehicle requests a current inside the overlap. The EVSE follows [34]. Its operating envelope has a constant-current boundary (limited by the cable and module current), a constant-power boundary (limited by supply), and a maximum voltage boundary. A 400 V car at a 500 A, 1000 V station is therefore limited to about 200 kW by its own voltage, and a station's nameplate figure is not the same as the power delivered.

Table 3.3: Charging modes of IEC 61851-1, as commonly summarised [33].

Mode	Supply	Arrangement	Typical maximum
1	AC	Non-dedicated socket, no in-cable protection	16 A
2	AC	Non-dedicated socket with in-cable control and protection device (IC-CPD)	32 A
3	AC	Dedicated EVSE with pilot control	63 A three-phase; 70 A single-phase
4	DC	Off-board charger with tethered cable	Depends on coupler; hundreds of amperes

3.4.2 Connector and Cable Ratings

A connector's rated current depends on its contact design and permitted temperature. Thus the rating in an IEC configuration is a maximum, while a product may be rated lower. Cables must be rated for the current at the operating temperature; high-power DC cables use liquid cooling, and the current is often limited by a temperature sensor at the contacts. IEC 62196-1 (2022 edition) sets general limits up to 690 V AC at 250 A, and 1500 V DC at 800 A [35]. The specific configurations are covered in Section 3.5.

3.4.3 Electrical Protection

3.4.3.1 Overcurrent Protection

Every charging point needs a dedicated final circuit with its own overcurrent device. The design load must be taken as continuous, as a charging session runs at full current for hours, and a diversity factor of unity generally applies unless an automatic load management system is installed [36, 37].

3.4.3.2 Residual Current Protection

A person's safety relies upon residual current devices (RCDs). In a charger, the rectifier can produce smooth DC fault currents, which can blind a conventional Type AC or Type A RCD by saturating its core. IEC 60364-7-722 therefore requires protection for each connection point by an RCD with a rated residual current of not more than 30 mA, of at least Type A, with additional measures against DC fault currents [38]. Measures commonly used are a Type B RCD, or a Type A RCD together with a residual direct current detecting device (RDC-DD) to IEC 62955 [39], which disconnects the supply on DC fault current. For Mode 2, IEC 62752 covers the IC-CPD [40]. In India, the National Electrical Code 2023 follows a similar approach [36], as summarised in [37].

3.4.3.3 Earthing, Grounding and Isolation Monitoring

The protective conductor bonds the vehicle body to the installation earth, and its continuity is continuously verified via the control pilot circuit. Indian practice refers to IS 3043 for earthing and IS 732 for wiring [37]. In isolated DC systems, an insulation monitoring device (IMD)

checks the DC bus before and during a session, as required by IEC 61851-23 [34]. Further, the DC coupler must be locked before energising, and it must stay locked while the voltage is above a safe limit.

3.4.4 Power Quality, Power Factor, Harmonics and Grid Connection

Large charging loads are non-linear, and they appear suddenly, which matters to the distribution company. In India, the requirements for connection flow from the CEA (Technical Standards for Connectivity) regulations and from the licensee's supply code, and they refer to harmonic limits of the IEEE 519 kind [41]. The Ministry of Power guidelines describe the connection process, referred to in the next subsection.

3.4.5 Communication, Safety and Charging Infrastructure Requirements

A compliant installation must provide, among other things: fire-retardant enclosures with suitable IP rating; a cable length limit; surge protection; mechanical protection from vehicles; emergency stop; and periodic inspection. Public chargers must also be networked to permit interoperable roaming through open protocols.

3.4.6 Indian Regulatory Framework

3.4.6.1 Ministry of Power Guidelines

The Ministry of Power issued the *Guidelines for Installation and Operation of Electric Vehicle Charging Infrastructure–2024* in September 2024 [42], with an amendment on 10 January 2025 [43]. Public summaries report the following provisions [44]:

- Setting up a charging station is a de-licensed activity.
- A single-part tariff is specified: not more than the average cost of supply (ACoS) until 31 March 2028, with 0.7 times ACoS in solar hours (9:00 to 16:00) and 1.3 times ACoS in other hours.
- Ceilings are set for service charges, which differ for AC and DC charging and the time of day. The notified tables should be consulted.
- Connection timelines for a new connection are 3 days in metropolitan areas, 7 days in other municipal areas, 15 days in rural areas, 30 days in rural hilly terrain, and 90 days where a line extension or substation is needed.
- Separate metering is required.
- The density targets for 2030 are reported as one station within a 1 km grid in urban areas and every 20 km along highways, with long-range points every 100 km for buses and trucks [45].

The 2025 amendment refines the definitions of charging infrastructure and of a charging station, supports round-the-clock operation, requires technology-agnostic land allotment bids, and revises the nodal agency provision [43]. These tariff and timeline provisions are *policy guidelines*, which take effect through state regulators.

3.4.6.2 Central Electricity Authority Regulations

The CEA (Measures relating to Safety and Electric Supply) Regulations, 2023 are *statutory*, and include specific safety provisions for charging stations, drawn from earlier amendments [46]. Summaries list requirements on overload protection, locking of DC connectors and fire prevention [37].

3.4.6.3 Bureau of Indian Standards

The BIS series IS 17017 corresponds to the IEC conductive charging standards, and is summarised in Table 3.4. These are technical standards, which become compulsory when adopted by a statute or tender.

Table 3.4: IS 17017 parts and related standards [30, 37].

Standard	Scope
IS 17017 (Part 1): 2018	General requirements for conductive charging
IS 17017 (Part 2/Secs 1, 2, 3): 2020	Plugs, socket-outlets, vehicle connectors and inlets; dimensions for AC and for DC/AC-DC couplers
IS 17017 (Part 2/Sec 6): 2021	DC couplers relying on electrical separation
IS 17017 (Part 21): 2019	EMC requirements
IS 17017 (Part 22/Sec 1): 2021	AC charge point for light EVs
IS 17017 (Part 23): 2021	DC EV supply equipment
IS 17017 (Part 24): 2021	Digital communication for DC charging control
IS 17017 (Part 25): 2021	DC EVSE with protection by electrical separation
IS 17017 (Part 30): 2025	Dual-gun DC EVSE
IS 17017 (Part 31): 2024	AC or DC EVSE with electrical separation for light EVs
IS 17896 (Parts 1, 2): 2022	Battery swap systems

3.4.6.4 Automotive Industry Standards

Vehicle-side requirements fall under the Central Motor Vehicle Rules through Automotive Industry Standards. AIS-038 (Rev. 2) covers electric power train safety of M and N category vehicles, and AIS-156 covers L category vehicles (two- and three-wheelers and quadricycles) [29, 47]. Amendments in 2022 added battery safety requirements, such as thermal propagation testing, in two phases. AIS-138 (Parts 1 and 2) covers EV conductive AC and DC charging systems.

3.4.7 International and Indian Standards in Context

IEC 61851 addresses the charging system; IEC 62196 the coupler; IEC 60364-7-722 the fixed installation; IEC 62752 and IEC 62955 the protective devices. The Indian IS 17017 series closely mirrors these. Table 3.11 at the end of the next section sets out the correspondence.

3.5 Charging Standards and Technologies

Standards turn three parties, car maker, charger maker and network operator, into one interoperable system. If a plug fits, the pilot signal is understood, and the back-end recognises the user, then any car can charge at any station. This section sets out the standards that provide this.

3.5.1 AC Charging Standards

3.5.1.1 Type 1 Connector

Type 1 (SAE J1772) is a five-pin, single-phase connector, widely used in North America and Japan. Under IEC 62196-2 it is rated to 32 A; SAE J1772 allows up to 80 A for North American use [35, 48].

3.5.1.2 Type 2 Connector

Type 2 is a seven-pin connector that handles single- and three-phase supply. It is the European norm and is adopted for AC charging of cars in India. It is rated up to 63 A three-phase and 70 A single-phase [35]. It gives up to 43.6 kW at 400 V, and about 45 kW at 415 V.

3.5.1.3 AC Charging Modes and Phases; Power Levels

AC modes were given in Table 3.3. The AC power available is

$$P_{1\phi} = V_{ph} I \,\mathrm{PF}, \qquad P_{3\phi} = \sqrt{3} V_L I \,\mathrm{PF}. \tag{3.17}$$

At unity power factor, 230 V at 16 A is 3.7 kW; 230 V at 32 A is 7.4 kW; 415 V at 16 A is 11.5 kW; and 415 V at 32 A is 23 kW.

Figure 3.11 shows the AC charging architecture. The EVSE contains only protection, metering, a contactor and the pilot circuit; conversion takes place in the vehicle's OBC.

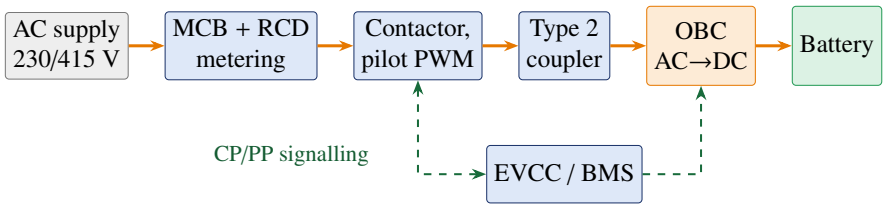


Figure 3.11: AC charging architecture.

Table 3.5: Type 1 versus Type 2 connectors [35, 48].

Attribute	Type 1	Type 2
Pins	5	7
Phases	Single	Single or three
Rating under IEC 62196-2	32 A	63 A three-phase; 70 A single-phase
Region	North America, Japan	Europe, India, others
Typical AC power	Up to 7.4 kW	Up to 22 kW common; 43 kW possible
DC companion	CCS1	CCS2

3.5.2 DC Charging Standards

3.5.2.1 CCS Combo 1 and Combo 2

The Combined Charging System (CCS) adds two DC pins below an AC connector, so one vehicle inlet serves both AC and DC. CCS1 (configuration EE in IEC 62196-3) extends Type 1; CCS2 (configuration FF) extends Type 2. They typically reach 500 A and 1000 V, about 500 kW, and communication uses power-line signalling over the control pilot wire [49]. CCS2 is the norm for Indian four-wheelers.

3.5.2.2 CHAdeMO

CHAdeMO originated in Japan (configuration AA). It uses CAN communication over dedicated pins, and supports bidirectional power flow in its established form. Its later versions go to high power, and it has joined GB/T development in the ChaoJi project [50].

3.5.2.3 GB/T

GB/T is the Chinese national system. GB/T 20234.2 covers AC; GB/T 20234.3 covers DC (configuration BB), and the 2023 revision raises the limits to 1500 V and 800 A, up to 1200 kW. Communication follows GB/T 27930 over CAN at 250 kbit/s [51].

3.5.2.4 Bharat AC-001 and Bharat DC-001

The Bharat chargers were specified in 2017 by a committee constituted by the Department of Heavy Industry for low-voltage, entry-level EVs [52]. *Bharat AC-001* has three 230 V, 15 A single-phase outlets with IEC 60309 industrial sockets, about 3.3 kW per output. *Bharat DC-001* has a DC output of 48–72 V, up to 200 A, and 10–15 kW, using a GB/T 20234.3 connector and CAN communication with the BMS [53, 54]. Being low-voltage specifications, they suited early commercial fleets, but have been overtaken as vehicle voltages rose.

3.5.2.5 Other Relevant Indian Charging Technologies

For light EVs, Indian standards have introduced compact AC/DC connectors, Type 6 and Type 7 in the IS 17017 (Part 2) series, which rely on electrical separation [37]. Dual-gun DC EVSE (IS 17017 Part 30) serves electric buses. The Megawatt Charging System (MCS) is emerging

for heavy trucks.

Figure 3.12 shows the DC fast-charging architecture, in which the charger is in the station, the OBC is bypassed, and the vehicle's communication controller (EVCC) sends requests to the supply equipment communication controller (SECC).

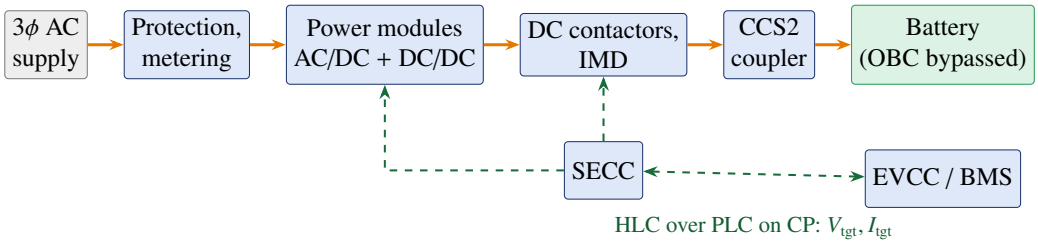


Figure 3.12: DC fast-charging architecture.

Table 3.6: CCS versus CHAdeMO versus GB/T: stated maxima from cited public sources. Installed chargers are often rated lower.

Attribute	CCS2 / CCS1	CHAdeMO	GB/T
IEC 62196-3 type	FF / EE	AA	BB
Voltage, current	1000 V, 500 A (typ.)	Up to 1000 V (recent versions)	Up to 1500 V, 800 A (2023)
Power	About 500 kW	High, ongoing via ChaoJi	Up to 1200 kW (2023)
EV communication	PLC on CP	CAN	CAN
Shared AC inlet	Yes	No	No
Region	Europe, India / N. America	Japan	China
V2G	Via ISO 15118-20	Established	Under development

3.5.3 Charging Communication Protocols

3.5.3.1 IEC 61851 Control Pilot

The control pilot (CP) is a simple but brilliant circuit, shown in Fig. 3.13. The EVSE drives a ±12 V signal through 1 kΩ. Resistors in the vehicle pull the positive level down, and the EVSE reads the result to know the state: state A at +12 V for no vehicle; state B at +9 V for connected; state C at +6 V for ready to charge (with S2 closed); state D at +3 V for ventilation required; state E at 0 V for error; and state F at −12 V for EVSE fault. These voltages come from the divider formed by the resistors of Fig. 3.13. For state C, the parallel of 2.74 kΩ and 1.3 kΩ is 882 Ω, and together with the diode drop it gives about +6 V.

The EVSE then applies a 1 kHz PWM signal, whose duty cycle D gives the available current. For the common range,

$$I_{\text{avail}} = 0.6\,D \quad [\text{A}], \qquad 10 \le D \le 85, \tag{3.18}$$

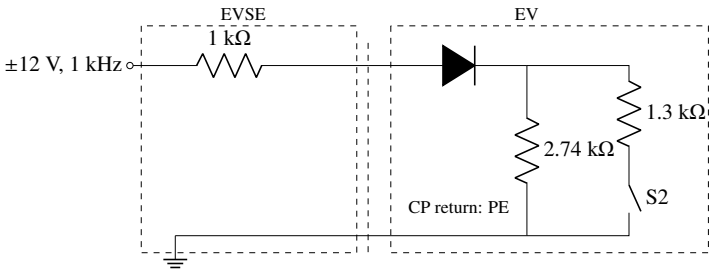


Figure 3.13: Control pilot circuit (simplified, values per the commonly published circuit of IEC 61851-1/SAE J1772).

with D in per cent, and $I_{\text{avail}} = 2.5(D-64)$ for $85 < D \leq 96$. So a 50% duty cycle means 30 A. A nominal 5% duty cycle signals that digital high-level communication is required. The proximity pilot (PP) separately identifies the cable's rating by a resistor, which prevents overloading of a low-rated cable.

3.5.3.2 ISO 15118

ISO 15118 defines high-level communication (HLC) between the EV and the supply equipment. ISO 15118-2 was the widely deployed AC and DC application protocol, and ISO 15118-20:2022 adds bidirectional power transfer, wireless charging and stronger security [31]. Data is carried over HomePlug Green PHY power-line communication on the CP wire [49].

3.5.3.3 DIN SPEC 70121

DIN SPEC 70121 preceded ISO 15118, and defines DC charging communication over the same physical layer. It was a practical interim specification, which remains widely used in the installed CCS base.

3.5.3.4 OCPP

The Open Charge Point Protocol (OCPP), from the Open Charge Alliance, links a charging station to its central management system, for authorisation, status, metering and remote control. OCPP 1.6 and 2.0.1 are widely deployed, and OCPP 2.1 has been published as IEC 63584-210:2025 [55]. OCPI is a separate protocol for roaming among networks.

3.5.3.5 EV–EVSE Communication, Authorisation, Plug and Charge, and Smart Charging

A DC session proceeds through a handshake: connection, compatibility check (limits), isolation test, pre-charge, energy transfer under the vehicle's current requests, and shut-down. *Charging authorisation* may be by RFID, an app, or a payment terminal. *Plug and Charge* uses digital certificates under ISO 15118, so that the vehicle identifies itself merely on plugging in. *Smart charging* schedules or modulates charging current according to tariff, grid and site constraints, using either the pilot duty cycle or OCPP profiles.

Table 3.7: Charging protocols and their functions.

Protocol	Link	Function
IEC 61851 CP / PP	EV – EVSE (signal)	Safety state, current limit, cable rating
DIN SPEC 70121	EV – DC EVSE (PLC)	DC charging control
ISO 15118-2, -20	EV – EVSE (PLC)	HLC, Plug and Charge, V2G (-20)
CAN (CHAdeMO, GB/T 27930)	EV – DC EVSE	DC charging control
OCPP	EVSE – back-end	Status, authorisation, tariffs, load control
OCPI	Back-end – back-end	Roaming between networks

3.5.4 Charging Technologies

Level 1 and Level 2 charging. The *Level* terminology is North American in origin (SAE J1772). Level 1 uses 120 V household supply, up to about 1.9 kW, whereas Level 2 uses 208–240 V, up to about 19.2 kW. IEC and Indian documents use *Modes* instead, although "Level" is often used loosely in India.

DC fast charging and ultra-fast charging. DC fast charging is commonly taken as about 25–150 kW, while "ultra-fast" or high-power charging means 150 kW and above, usually with 800 V vehicles for full use. These boundaries are industry conventions and not standard definitions.

Wireless charging, battery swapping, smart charging and bidirectional charging. These were explained in Sections 3.2 and in the communication subsection above. Together they extend charging beyond the plug, towards a vehicle that is an active part of the electricity system.

Table 3.8: AC versus DC charging.

Attribute	AC charging	DC charging
Converter location	Onboard	Offboard (station)
Power set by	OBC rating and supply	Station, cable and battery
Typical power	3–22 kW	15–400 kW
Equipment cost	Low	High
Typical use	Home, workplace	Public, highway, fleets
Communication	CP signal (HLC optional)	HLC mandatory
Grid impact	Low per point	High, needs planning

Table 3.9: Onboard versus offboard charging.

Attribute	Onboard	Offboard
Located in	Vehicle	Station
Constraint	Mass, volume, cooling	Site, grid connection, cost
Input	AC	Three-phase AC, often via a transformer
Output	Battery DC at low power	Battery DC at high power
Cost borne by	Vehicle buyer	Station operator
Upgrade path	Fixed for vehicle life	Station can be upgraded

Table 3.10: Charging technologies and typical power ranges (indicative conventions).

Technology	Typical power	Typical application
Level 1 / Mode 2 home socket	1.4–3.7 kW	2W, 3W, overnight car
Level 2 / Mode 3 AC	3.7–22 kW	Home, office, destination
DC fast	25–150 kW	Public, highway
Ultra-fast DC	150–400 kW and above	Highway, 800 V vehicles
Megawatt charging	Order of 1 MW and above	Heavy trucks, buses
Wireless (static)	3–11 kW	Parking
Battery swapping	Minutes per swap	2W, 3W fleets

Table 3.11: Indian charging standards and international counterparts [30, 37].

Subject	Indian document	International counterpart
General conductive charging	IS 17017 (Part 1)	IEC 61851-1
Plugs, sockets, couplers	IS 17017 (Part 2)	IEC 62196 series
DC EVSE	IS 17017 (Part 23)	IEC 61851-23
DC digital communication	IS 17017 (Part 24)	IEC 61851-24
Vehicle-grid communication	IS/ISO 15118	ISO 15118
Installation	NEC 2023; IS 732	IEC 60364-7-722
EV battery swap	IS 17896	IEC 62840
Vehicle HV safety	AIS-038 Rev. 2	UN R100
Light EV safety	AIS-156	UN R136

Tables 3.8 and 3.9 show the same fundamental choice from two angles. AC keeps the cost in the vehicle at low power and suits long parking, while DC moves the converter to the station and suits short stops. Table 3.10 ranks the technologies by power, indicating that the charger type should be chosen from the parking time and the battery size, and not simply from the highest power available. Table 3.11 shows that Indian standards closely follow the IEC structure, so a design based on IEC documents is largely transferable. However, the regional mix of connectors, the Indian grid conditions and the specific rules of Section 3.4.6 must be observed. Taken together, the connector standards ensure physical compatibility, the IEC 61851 family ensures electrical safety, ISO 15118 and CAN-based protocols ensure that vehicle and charger cooperate, and OCPP ensures that the station can be operated as part of a network; it is this layered interoperability that allows any compliant EV to charge safely at any compliant station.

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